Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
```

```python
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
```

```python
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```python
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : campaign

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv

- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

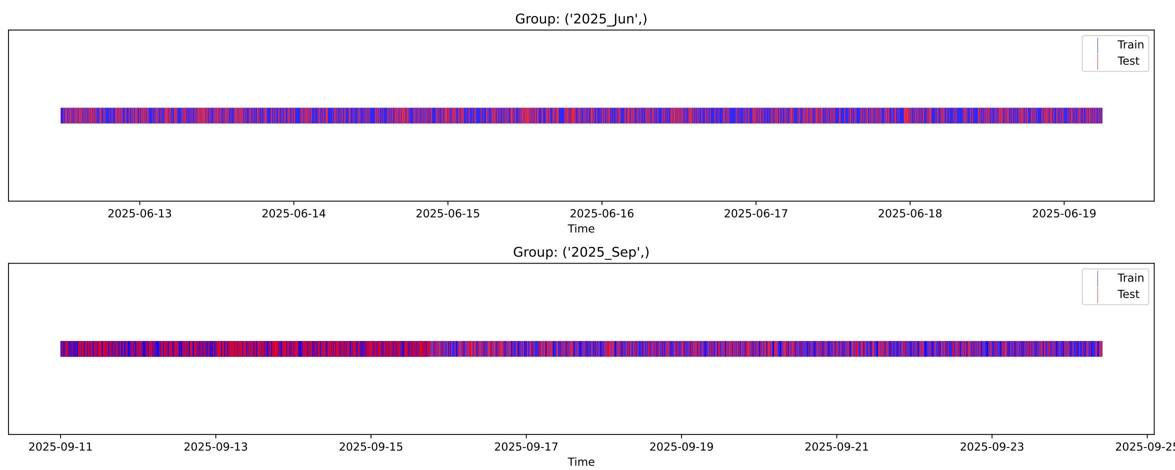
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

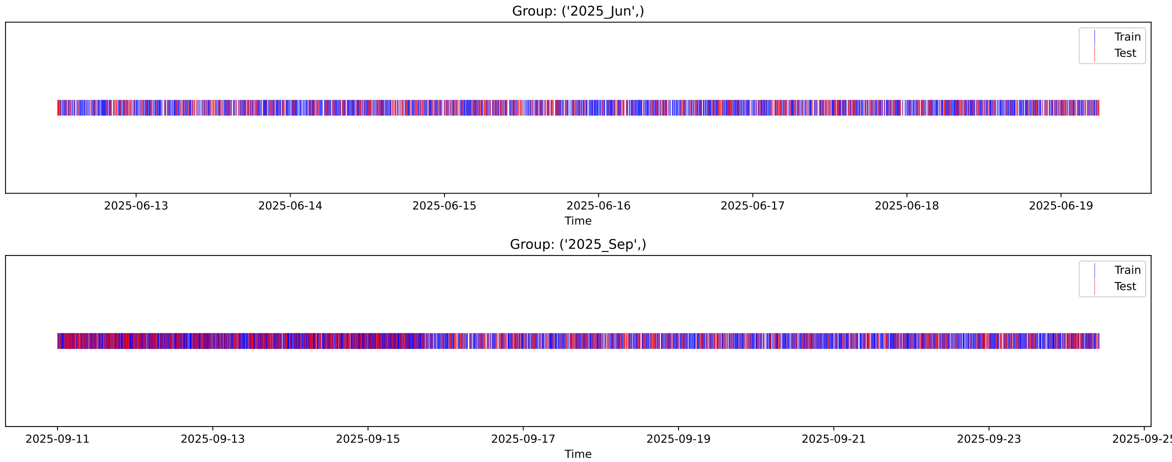
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

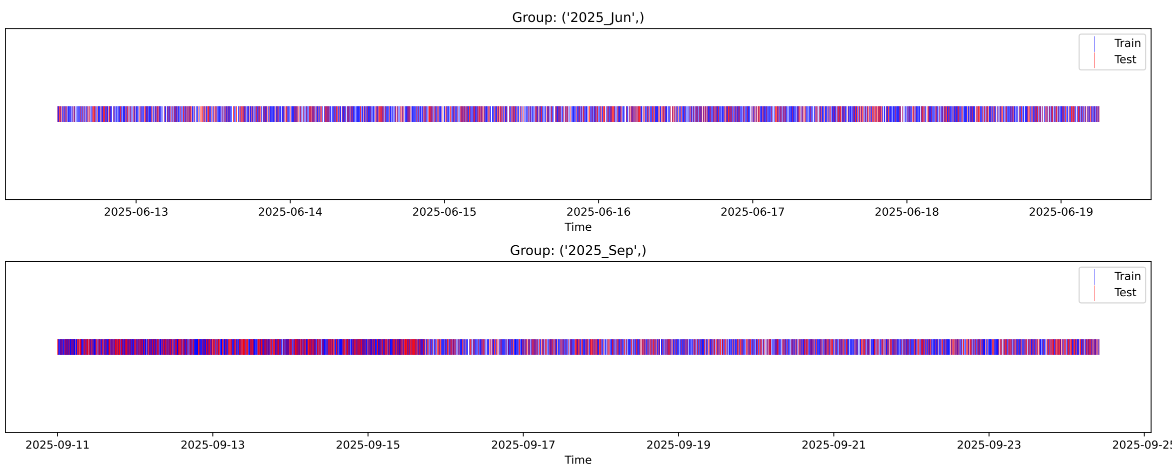
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



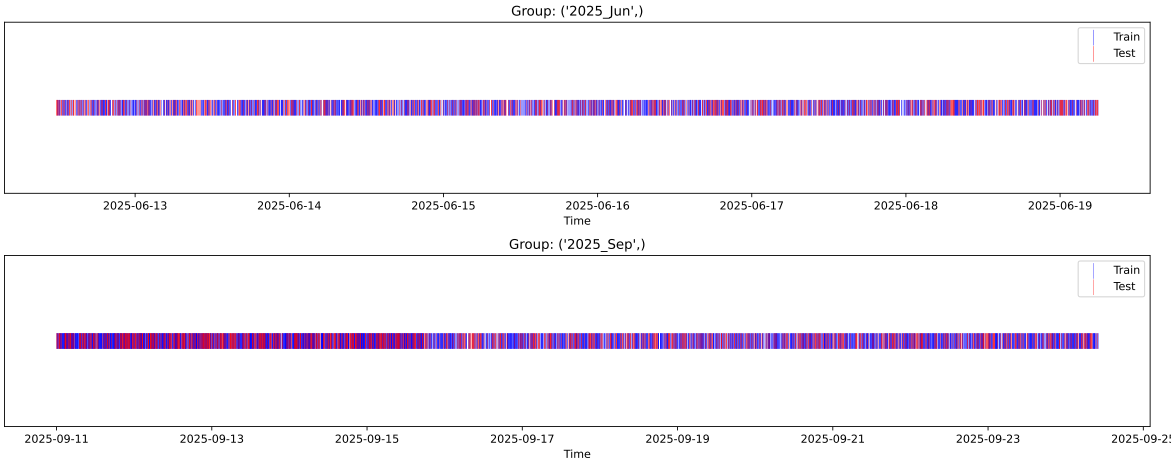
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



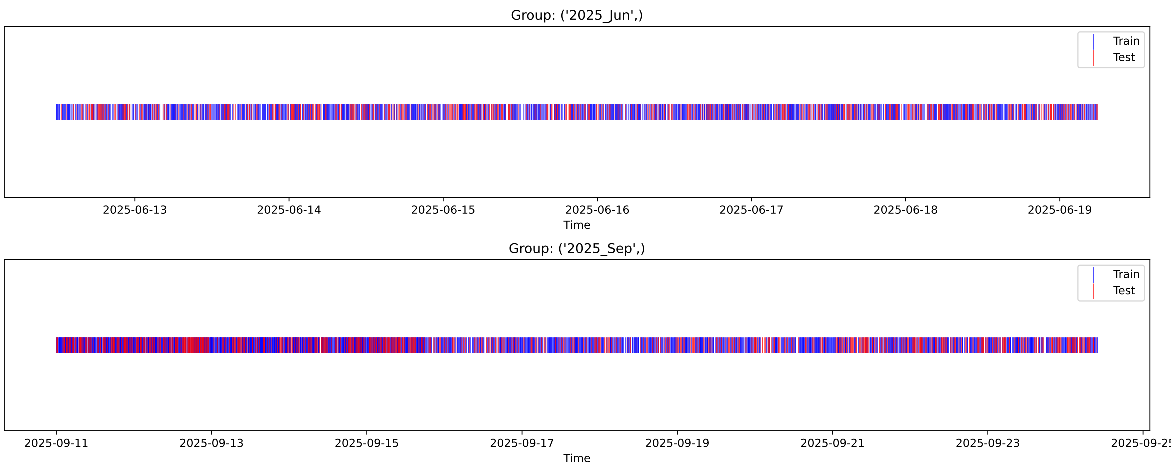
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



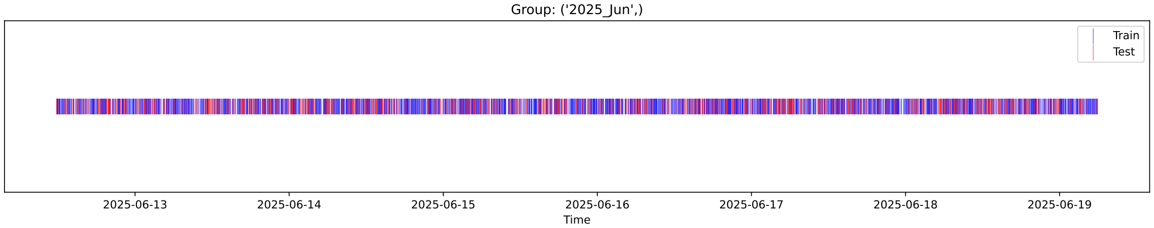
Fold 4/5

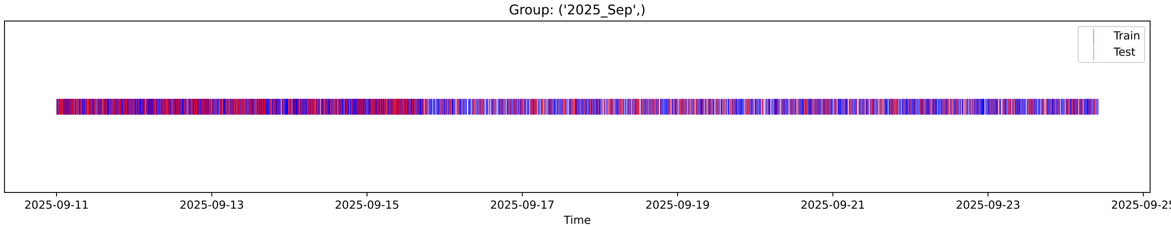
Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h





2.3 Model : Linear

2.3.1 Gridsearch — Group: 2025_Jun

Len(group) : 3172, **Len(training)** : 2221, **Len(test)** : 951

Distribution y_train vs y_test: y_train: mean=2.661, std=0.770, min=1.305, max=13.845 y_test: mean=2.621, std=0.704, min=1.253, max=5.627

Fold 0 — y_test: mean=2.68, std=0.87, y_train: mean=2.65, std=0.72

Fold 1 — y_test: mean=2.65, std=0.74, y_train: mean=2.66, std=0.79

Fold 2 — y_test: mean=2.62, std=0.73, y_train: mean=2.68, std=0.78

Fold 3 — y_test: mean=2.67, std=0.69, y_train: mean=2.66, std=0.80

Fold 4 — y_test: mean=2.66, std=0.75, y_train: mean=2.66, std=0.78

train scores CV: [0.23976731 0.2178682 0.21549938 0.22227879 0.21779003]

test scores CV: [0.19537288 0.24059213 0.2410925 0.22691821 0.24147471]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.2226	0.2291	0.0177	0.2502	0.2521	0.023	1.1005	0.9718

Coefficients:

coef_r_CO2 = -0.0303

coef_r_NH3 = -0.0589

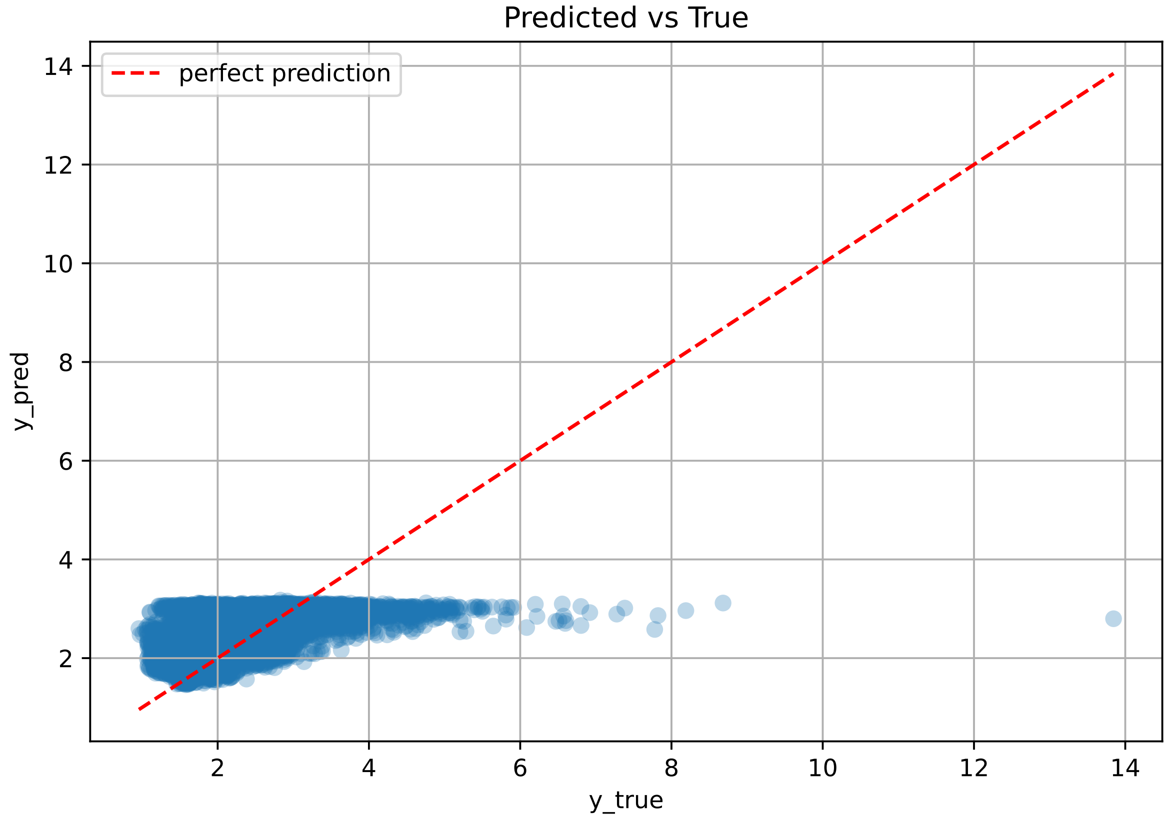
coef_r_THI = 1.1151

coef_r_temperature = -1.1856

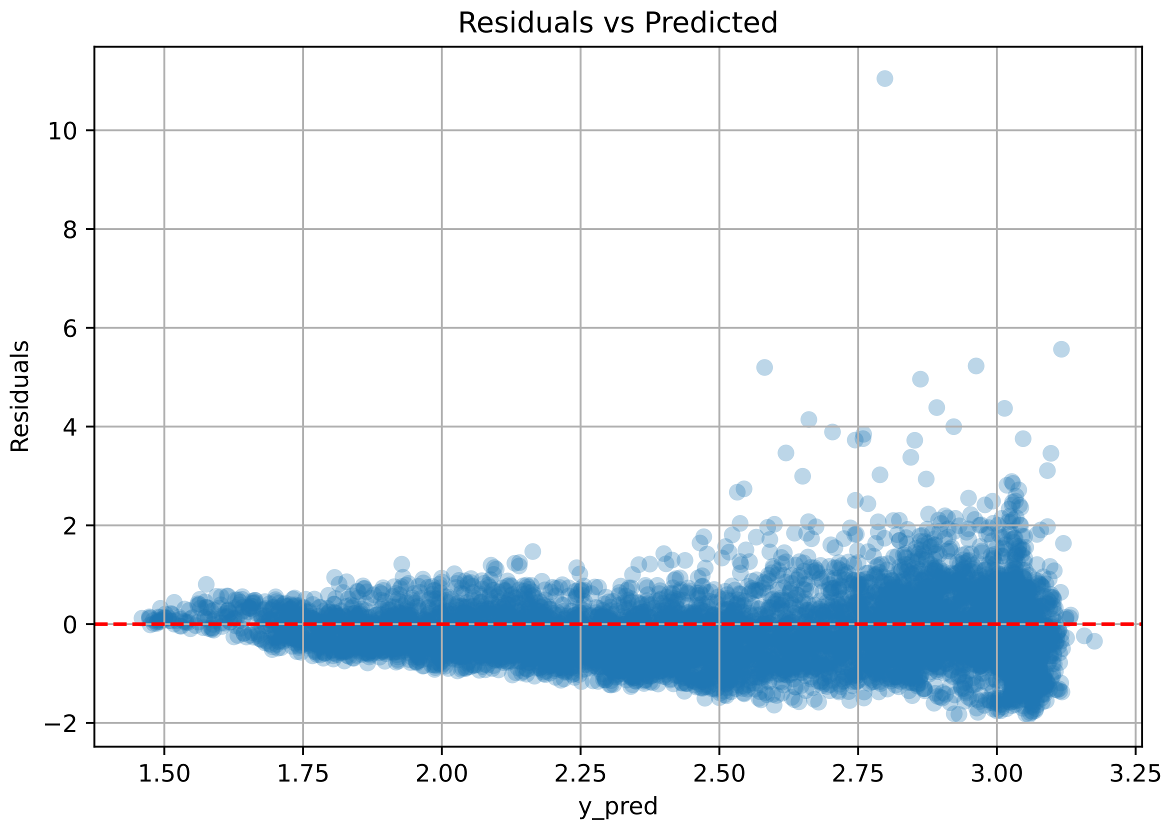
coef_r_umidity = -0.5032

intercept = 2.6606

2.3.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

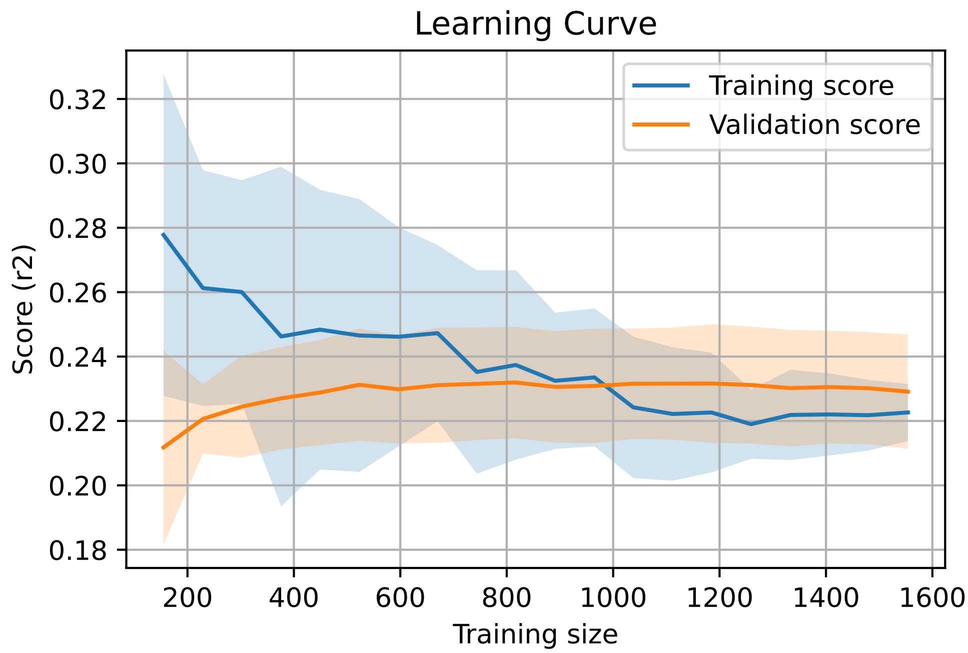
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 2025_Jun

Len(group) : 3172, **Len(training)** : 2221,

Len(training_real) : 1555, **Len(test_of_training)** : 666, **Len(test)** : 951



2.3.5 Gridsearch — Group: 2025_Sep

Len(group) : 4996, **Len(training)** : 3498, **Len(test)** : 1498

Distribution y_train vs y_test: y_train: mean=2.160, std=0.756, min=0.973, max=8.190 y_test: mean=2.160, std=0.777, min=0.959, max=7.779

Fold 0 — y_test: mean=2.15, std=0.74, y_train: mean=2.16, std=0.76

Fold 1 — y_test: mean=2.11, std=0.72, y_train: mean=2.18, std=0.77

Fold 2 — y_test: mean=2.15, std=0.75, y_train: mean=2.17, std=0.76

Fold 3 — y_test: mean=2.16, std=0.72, y_train: mean=2.16, std=0.77

Fold 4 — y_test: mean=2.15, std=0.75, y_train: mean=2.16, std=0.76

train scores CV: [0.38065158 0.37587093 0.35729885 0.36978918 0.37340349]

test scores CV: [0.36340016 0.37339033 0.42062432 0.39338444 0.38193969]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3714	0.3865	0.0197	0.3489	0.3494	-0.0371	0.9039	0.9608

Coefficients:

coef_r_CO2 = -0.0565

coef_r_NH3 = -0.0601

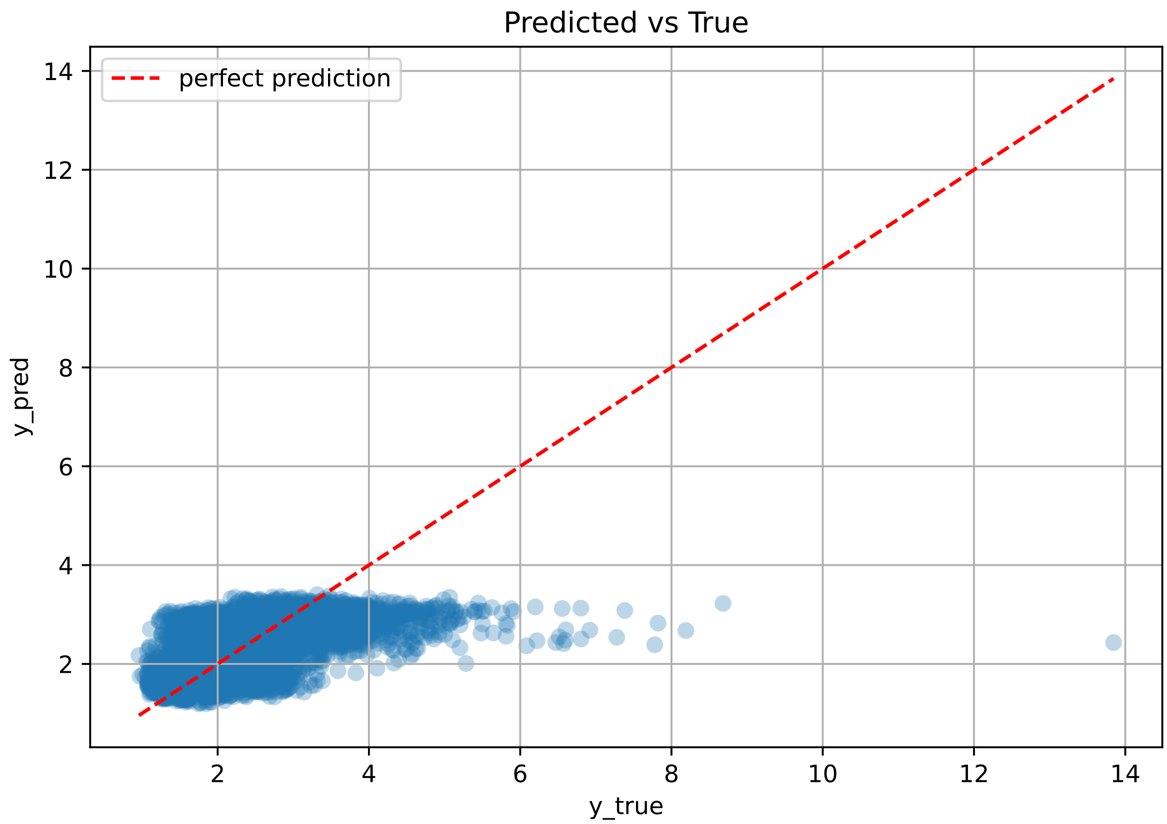
coef_r_THI = 0.8465

coef_r_temperature = -1.1296

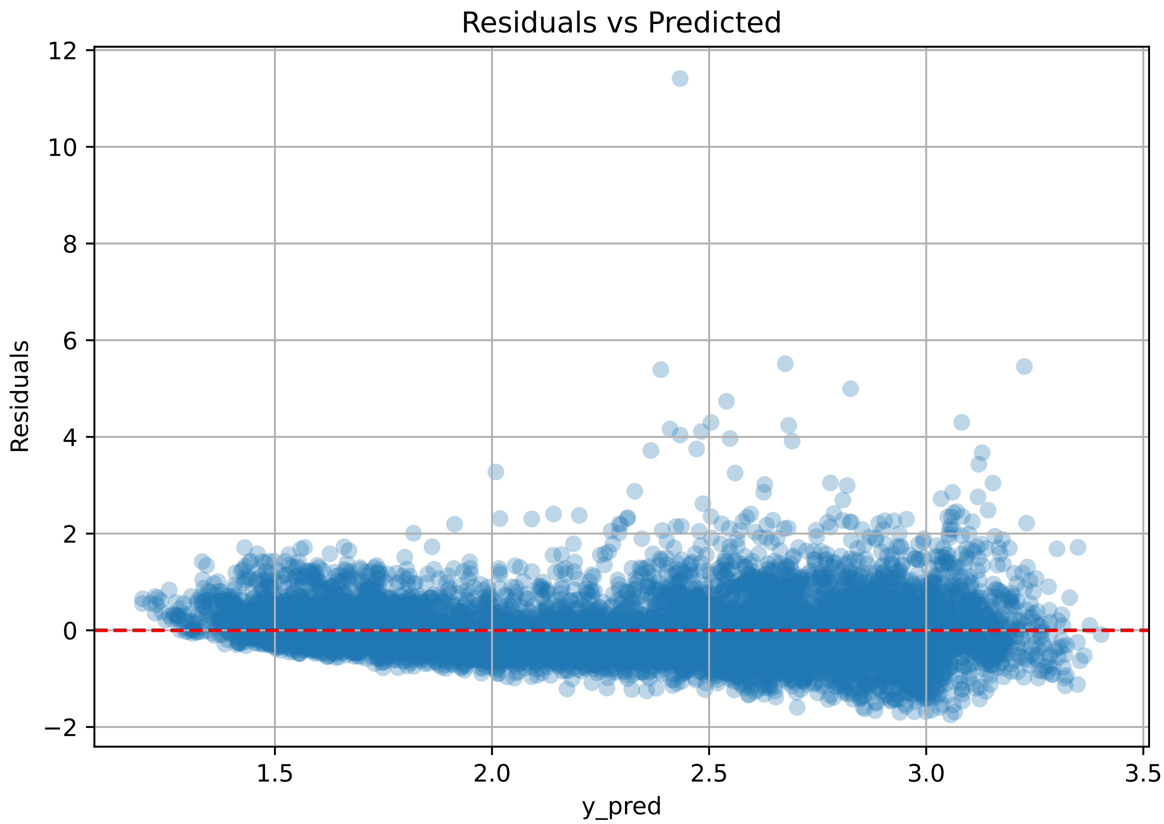
coef_r_umidity = -0.7945

intercept = 2.1596

2.3.6 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

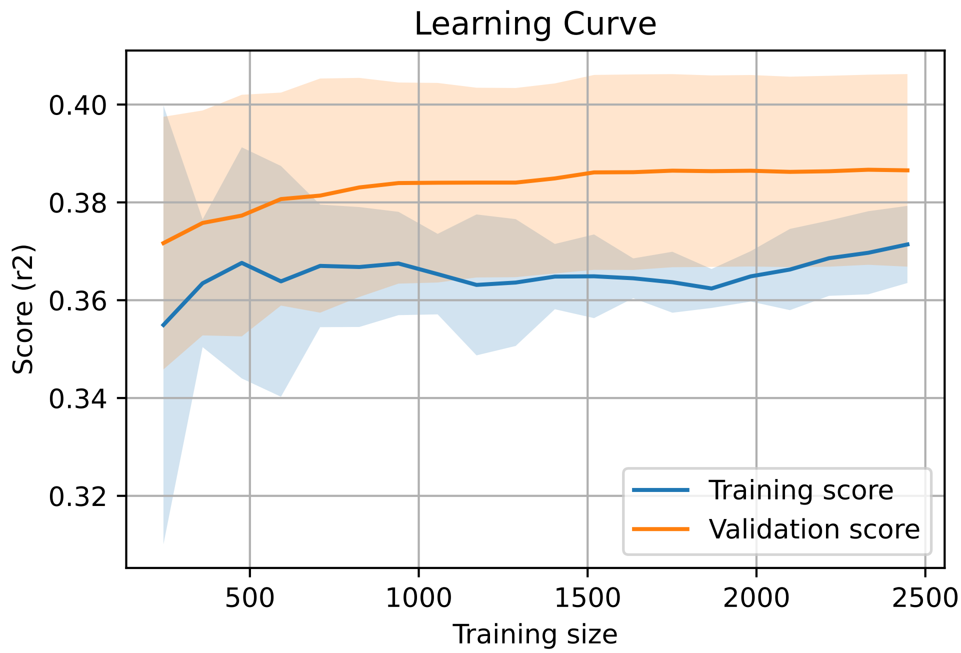
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2025_Sep

$\textbf{Len(group)} : 4996$, $\textbf{Len(training)} : 3498$,

$\textbf{Len(training\_real)} : 2449$, $\textbf{Len(test\_of\_training)} : 1049$, $\textbf{Len(test)} : 1498$



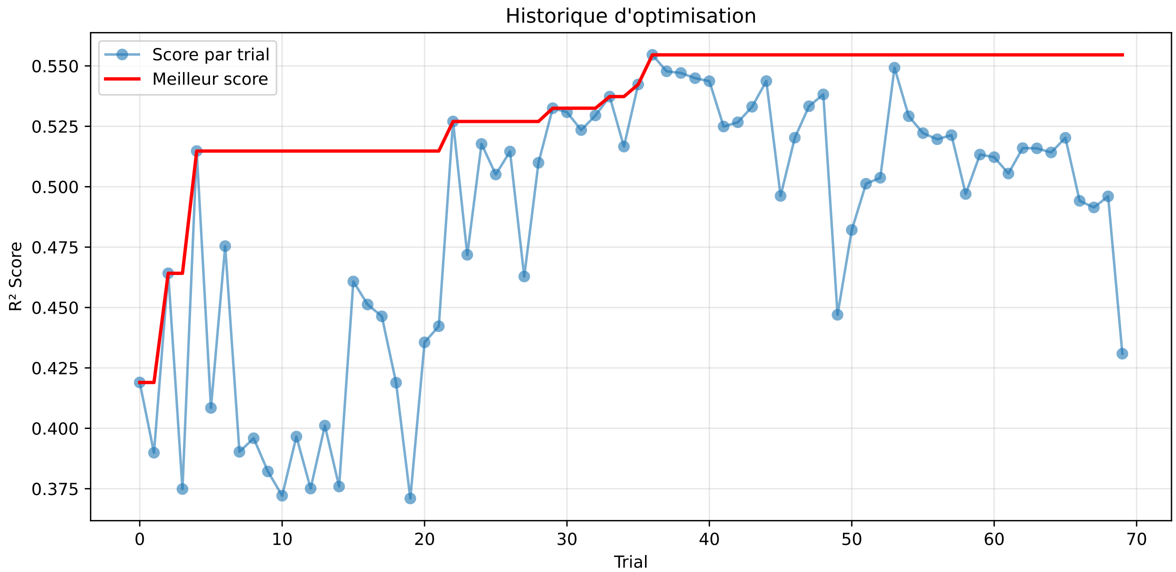
2.4 Model : LGBM

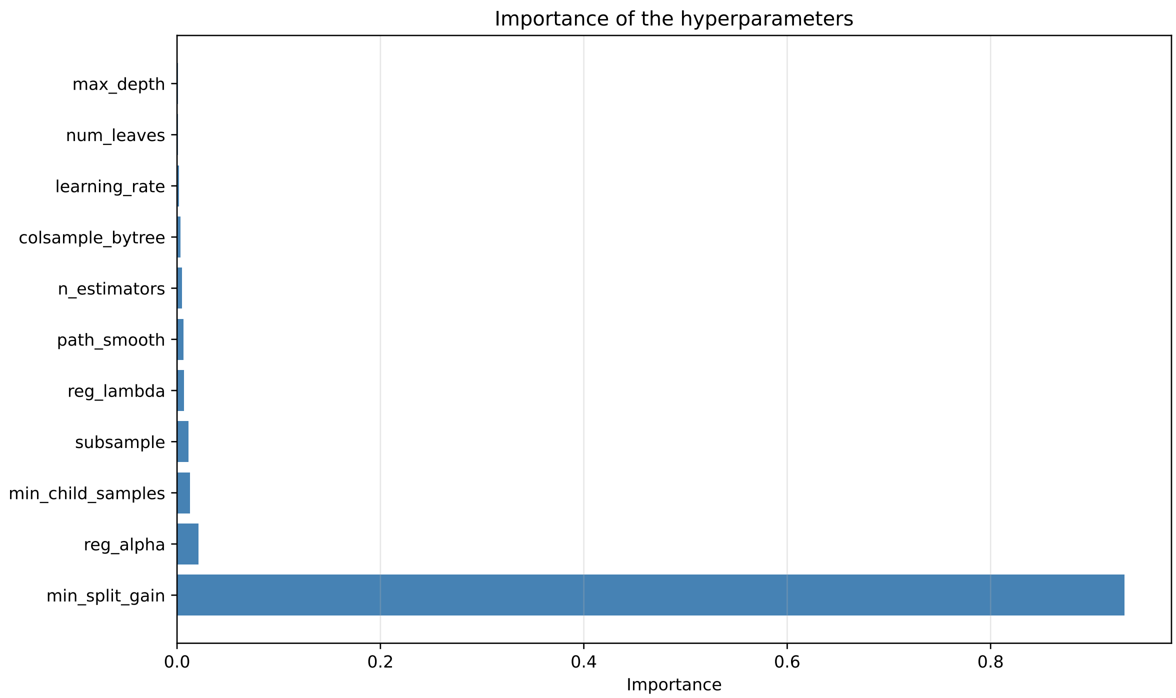
2.4.1 Gridsearch — Group: 2025_Jun

Len(group) : 3172, **Len(training)** : 2221, **Len(test)** : 951

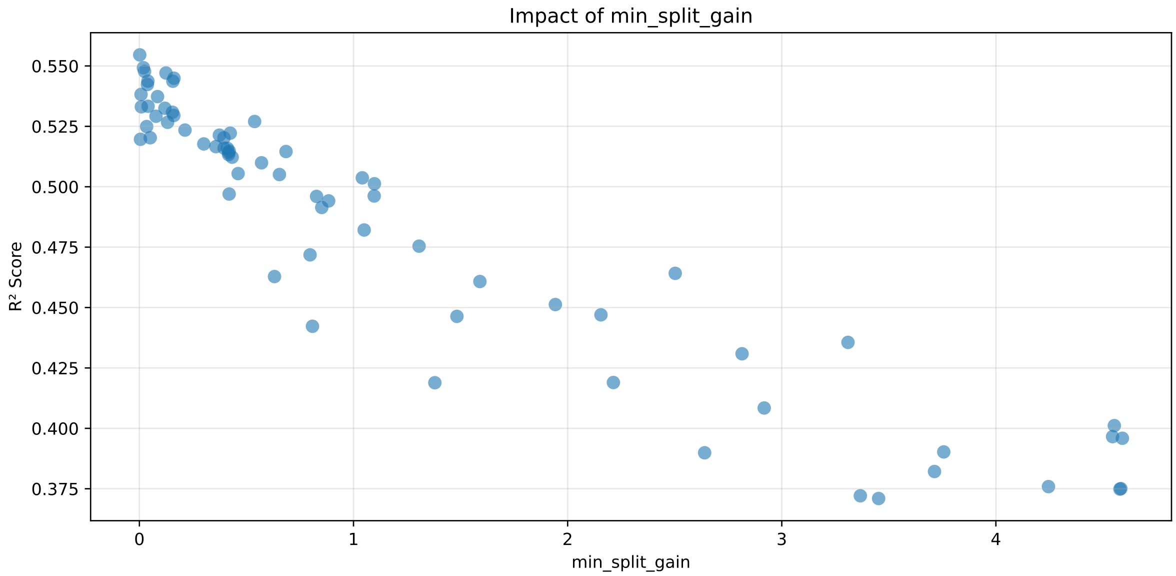
Distribution y_train vs y_test: y_train: mean=2.661, std=0.770, min=1.305, max=13.845 y_test: mean=2.621, std=0.704, min=1.253, max=5.627

==================== TOP Importance FEATURES ==================== feature importance r_CO2 158 r_NH3 116 r_umidity 109 r_temperature 99 r_THI 78





Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4557	0.4356	0.4816	0.0387	0.4356	0.5115	0.0759	1.1743	1.0463
2	0.4506	0.4309	0.4832	0.0363	0.4309	0.5291	0.0982	1.228	1.0458
3	0.4397	0.419	0.4669	0.0367	0.419	0.5132	0.0942	1.2249	1.0494
4	0.4275	0.4084	0.461	0.0375	0.4084	0.509	0.1005	1.2461	1.0468
5	0.4133	0.4011	0.4492	0.037	0.4011	0.4967	0.0955	1.2382	1.0304
6	0.4112	0.3966	0.4428	0.0359	0.3966	0.4902	0.0936	1.2361	1.0369
7	0.4143	0.3959	0.4475	0.036	0.3959	0.4876	0.0917	1.2317	1.0465
8	0.404	0.3902	0.4396	0.0365	0.3902	0.4884	0.0981	1.2514	1.0353
9	0.4076	0.3899	0.4387	0.0349	0.3899	0.4878	0.0979	1.2511	1.0455
10	0.3945	0.3821	0.4308	0.0347	0.3821	0.476	0.0939	1.2457	1.0323
---	---	---	---	---	---	---	---	---	---
70	0.7459	0.5546	0.5909	0.0223	-0.0194	0.6045	0.0499	1.0899	1.345

Hyperparameters:

Rank 1: {'reg___n_estimators': 300, 'reg___max_depth': 4, 'reg___learning_rate': 0.031839788985901285, 'reg___num_leaves': 58, 'reg___subsample': 0.5451476318070476, 'reg___colsample_bytree': 0.6340748170447167, 'reg___min_child_samples': 32, 'reg___reg_alpha': 1.4709816518504448, 'reg___reg_lambda': 0.7546165004255534, 'reg___min_split_gain': 3.309315830630476, 'reg___path_smooth': 3.6718640587038798}

Rank 2: {'reg___n_estimators': 600, 'reg___max_depth': 8, 'reg___learning_rate': 0.04966862793505803, 'reg___num_leaves': 39, 'reg___subsample': 0.5611308036269933, 'reg__colsample_bytree': 0.8693910689791702, 'reg__min_child_samples': 27, 'reg__reg_alpha': 2.032874054251555, 'reg__reg_lambda': 43.686084591474405, 'reg__min_split_gain': 2.8145769039147206, 'reg__path_smooth': 1.218092540562925}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.049829357145410716, 'reg__num_leaves': 34, 'reg__subsample': 0.7961921814968476, 'reg__colsample_bytree': 0.6468629774873205, 'reg__min_child_samples': 28, 'reg__reg_alpha': 6.015799640585944, 'reg__reg_lambda': 33.23205778998717, 'reg__min_split_gain': 2.2136668310504106, 'reg__path_smooth': 0.04156440398539463}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.06937793302751268, 'reg__num_leaves': 53, 'reg__subsample': 0.8479394088575971, 'reg__colsample_bytree': 0.7944337597246378, 'reg__min_child_samples': 25, 'reg__reg_alpha': 6.072498112882987, 'reg__reg_lambda': 27.985290400573238, 'reg__min_split_gain': 2.9180474080243495, 'reg__path_smooth': 1.116973887483581}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.0605390828345043, 'reg__num_leaves': 55, 'reg__subsample': 0.8493236067124508, 'reg__colsample_bytree': 0.708653082794882, 'reg__min_child_samples': 30, 'reg__reg_alpha': 2.5776332092732246, 'reg__reg_lambda': 18.929268317059776, 'reg__min_split_gain': 4.5531163979938984, 'reg__path_smooth': 3.8115385934141526}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.05267671630977974, 'reg__num_leaves': 44, 'reg__subsample': 0.6683617537516958, 'reg__colsample_bytree': 0.6619255347766608, 'reg__min_child_samples': 41, 'reg__reg_alpha': 2.369954653659546, 'reg__reg_lambda': 8.094995016829836, 'reg__min_split_gain': 4.544935519362395, 'reg__path_smooth': 4.484693300403765}

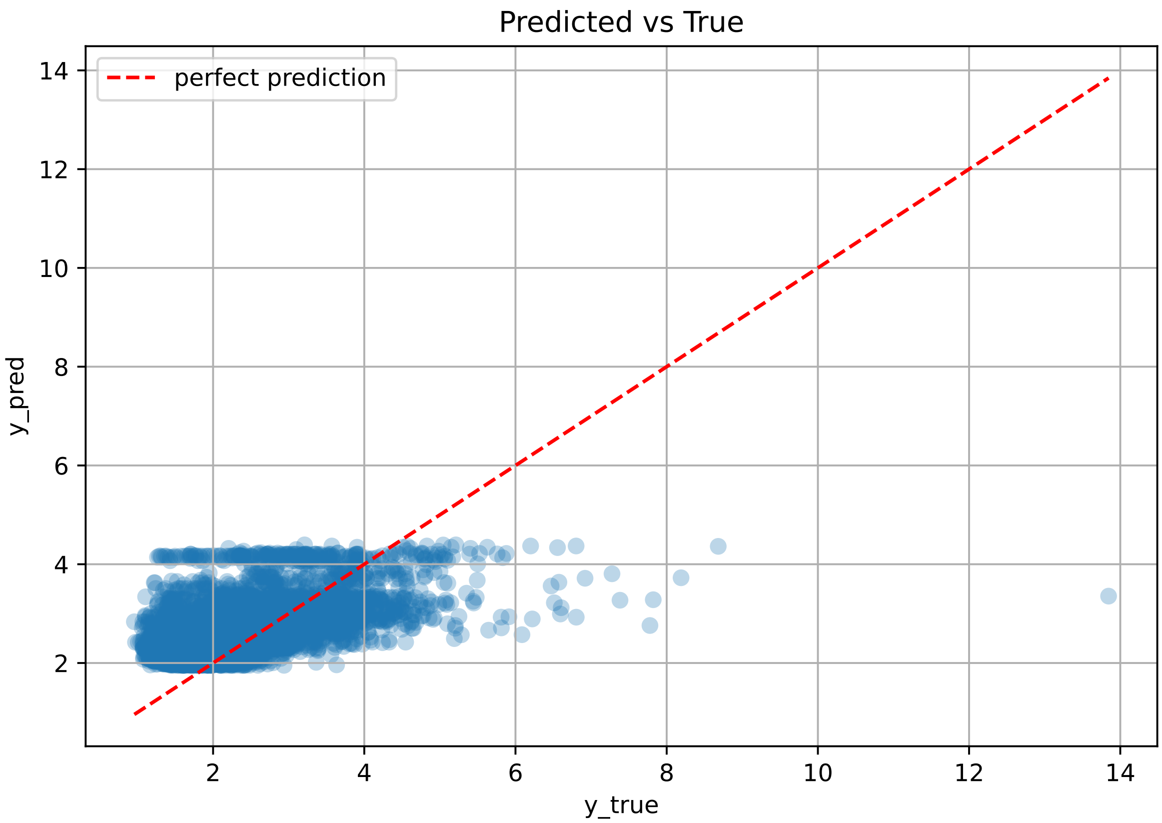
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.0627974098313729, 'reg__num_leaves': 35, 'reg__subsample': 0.6958018780471172, 'reg__colsample_bytree': 0.5716323771677476, 'reg__min_child_samples': 11, 'reg__reg_alpha': 0.10881843065661978, 'reg__reg_lambda': 48.10451448684149, 'reg__min_split_gain': 4.5906117947240705, 'reg__path_smooth': 2.175567414645794}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.06882971344446394, 'reg__num_leaves': 55, 'reg__subsample': 0.7634106455762952, 'reg__colsample_bytree': 0.7985660632020777, 'reg__min_child_samples': 26, 'reg__reg_alpha': 6.941405599994407, 'reg__reg_lambda': 25.79939121173011, 'reg__min_split_gain': 3.7565342238535235, 'reg__path_smooth': 1.9251830724307113}

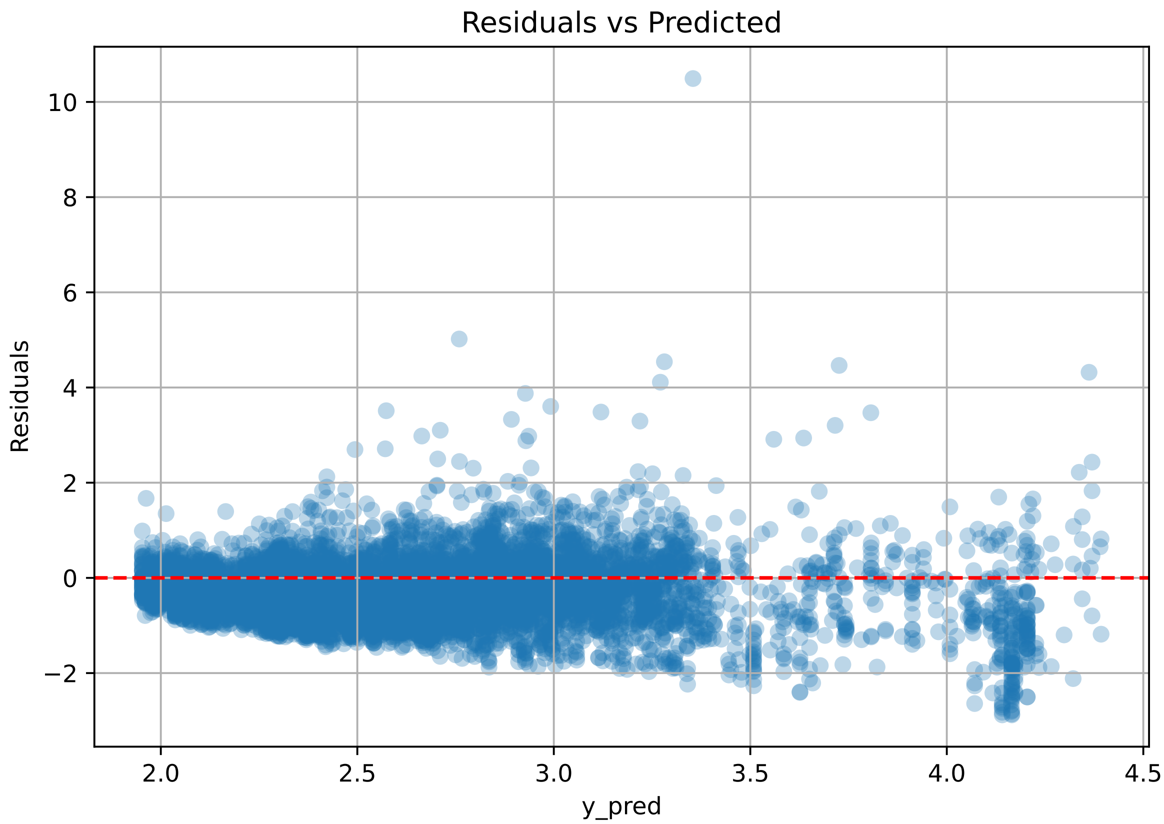
Rank 9: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.017251263168547377, 'reg__num_leaves': 28, 'reg__subsample': 0.7501815425819205, 'reg__colsample_bytree': 0.5058909213857211, 'reg__min_child_samples': 49, 'reg__reg_alpha': 8.736345130655437, 'reg__reg_lambda': 16.79783427272553, 'reg__min_split_gain': 2.6398207027726697, 'reg__path_smooth': 1.3086819026910745}

Rank 10: {’reg___n_estimators’: 200, ’reg___max_depth’: 7, ’reg___learning_rate’: 0.014808363748672265, ’reg___num_leaves’: 63, ’reg___subsample’: 0.6797777810645527, ’reg___colsample_bytree’: 0.6796957507849686, ’reg___min_child_samples’: 35, ’reg___reg_alpha’: 2.7312375064578243, ’reg___reg_lambda’: 49.8000656452134, ’reg___min_split_gain’: 3.7132878790077477, ’reg___path_smooth’: 4.780728513589757}

2.4.2 Scatter



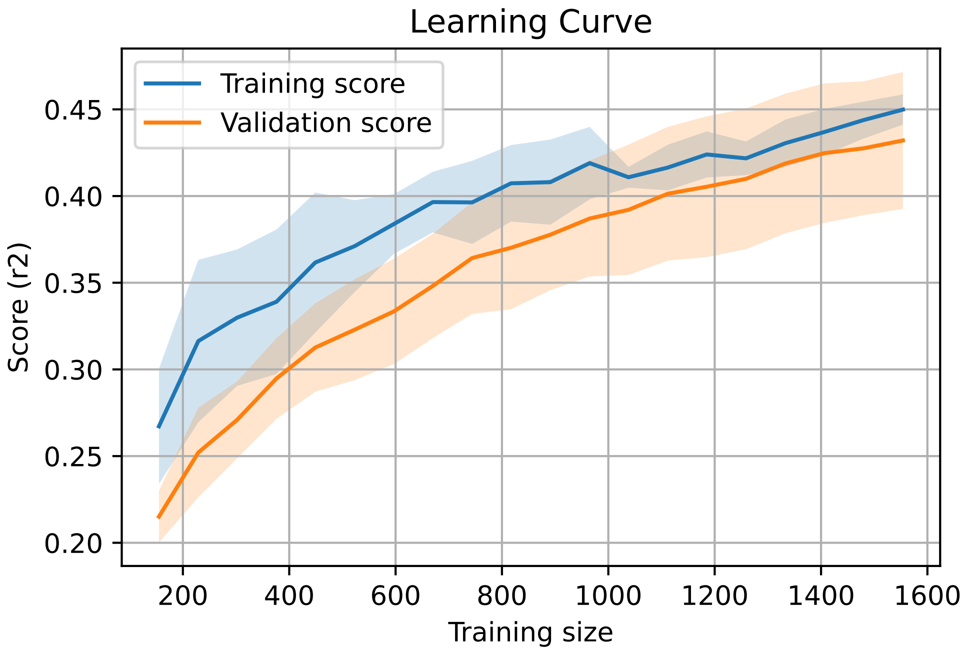
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 2025_Jun

$\mathbf{Len(group)}$: 3172, $\mathbf{Len(training)}$: 2221,

$\mathbf{Len(training\_real)}$: 1555, $\mathbf{Len(test\_of\_training)}$: 666, $\mathbf{Len(test)}$: 951

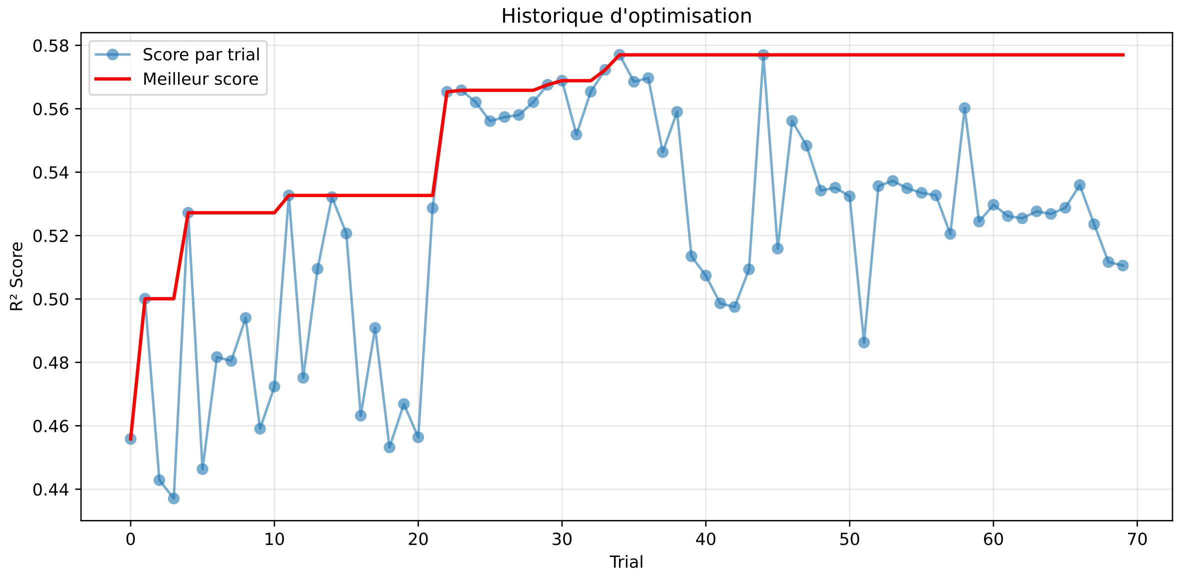


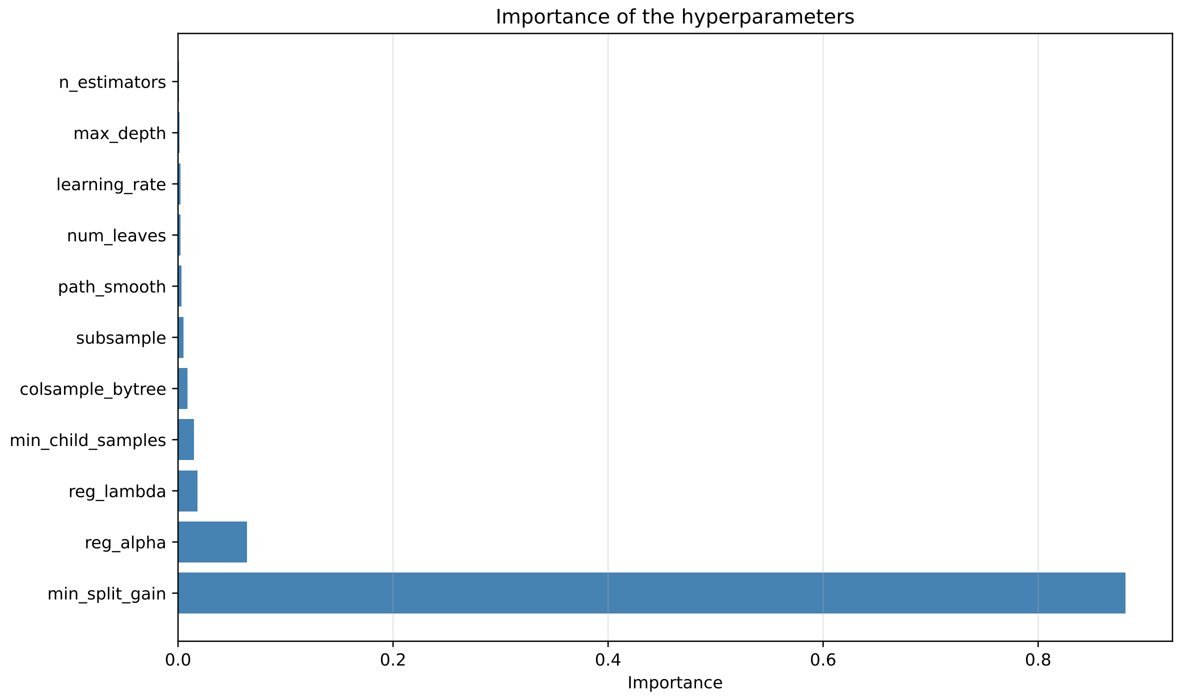
2.4.5 Gridsearch — Group: 2025_Sep

Len(group) : 4996, **Len(training)** : 3498, **Len(test)** : 1498

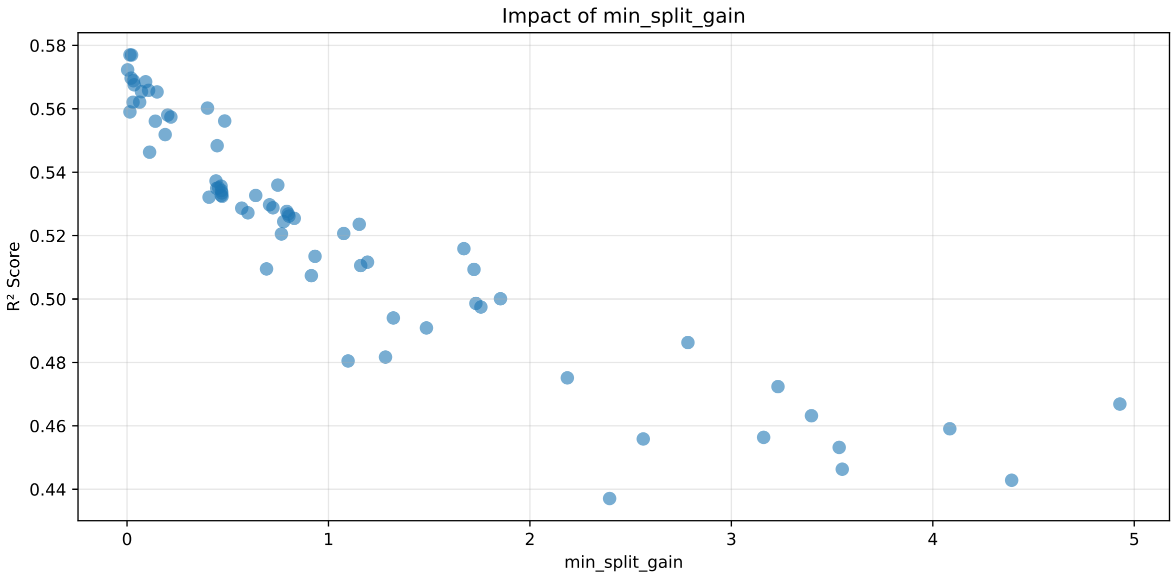
Distribution y_train vs y_test: y_train: mean=2.160, std=0.756, min=0.973, max=8.190 y_test: mean=2.160, std=0.777, min=0.959, max=7.779

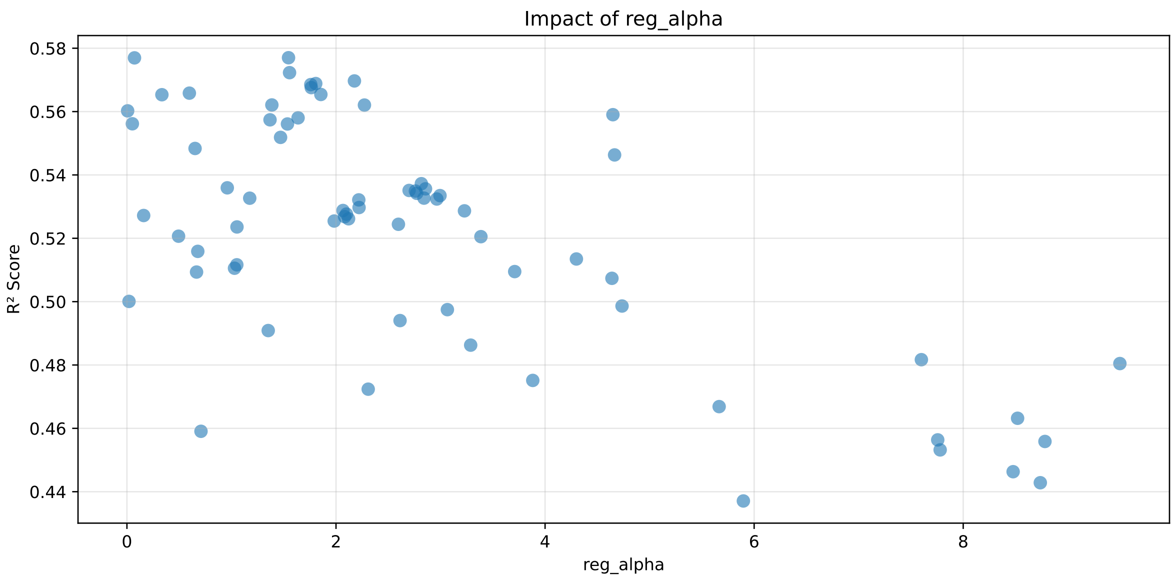
==================== TOP Importance FEATURES ==================== feature importance r_umidity 386 r_CO2 369 r_NH3 211 r_THI 196 r_temperature 133





Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



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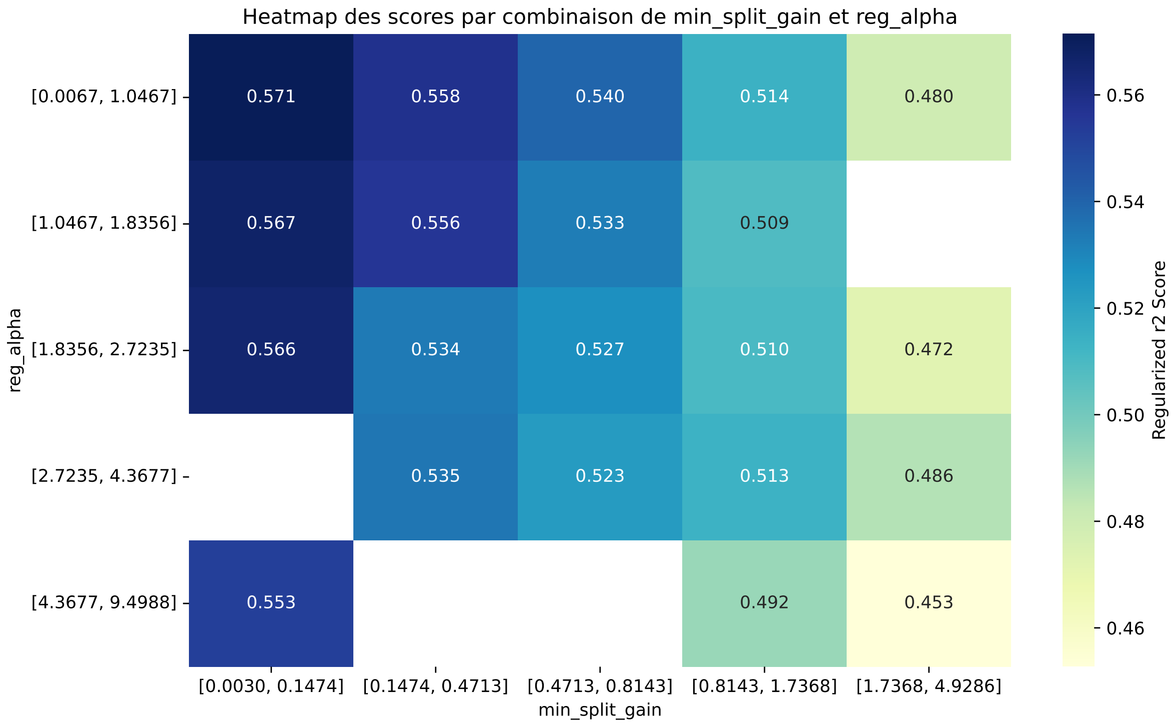
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4687	0.4371	0.4262	0.0211	0.3421	0.4412	0.0041	1.0093	1.0724
2	0.5055	0.4632	0.454	0.0256	0.3361	0.4721	0.0089	1.0193	1.0914
3	0.4979	0.4564	0.4504	0.0239	0.3319	0.4674	0.011	1.0242	1.0909
4	0.4804	0.4428	0.4361	0.0221	0.3302	0.4503	0.0075	1.0169	1.0847
5	0.5128	0.4669	0.4591	0.024	0.3291	0.4778	0.0109	1.0233	1.0983
6	0.4861	0.4463	0.4408	0.0225	0.327	0.455	0.0087	1.0194	1.0891
7	0.4993	0.4559	0.4496	0.0235	0.3256	0.4683	0.0124	1.0272	1.0952
8	0.5043	0.4591	0.4531	0.0238	0.3233	0.4688	0.0098	1.0213	1.0986
9	0.5354	0.4817	0.476	0.0247	0.3204	0.4897	0.008	1.0166	1.1116
10	0.4979	0.4532	0.4465	0.0237	0.3191	0.4686	0.0154	1.0339	1.0986
---	---	---	---	---	---	---	---	---	---
70	0.8402	0.577	0.578	0.0253	-0.2127	0.5961	0.0191	1.0331	1.4562

Hyperparameters:

Rank 1: {'reg___n_estimators': 200, 'reg___max_depth': 8, 'reg___learning_rate': 0.010930823294261214, 'reg___num_leaves': 37, 'reg___subsample': 0.6884473536398243, 'reg___colsample_bytree': 0.5093716637154961, 'reg___min_child_samples': 49, 'reg___reg_alpha': 5.897009493998189, 'reg___reg_lambda': 38.02395093985286, 'reg___min_split_gain': 2.395611316664679, 'reg___path_smooth': 2.9707406998230557}

Rank 2: {’reg___n_estimators’: 200, ’reg___max_depth’: 8, ’reg___learning_rate’: 0.07097222402747878, ’reg___num_leaves’: 37, ’reg___subsample’: 0.6170094756843711, ’reg___colsample_bytree’: 0.8900987359421166, ’reg___min_child_samples’: 37, ’reg___reg_alpha’: 8.520259165732284, ’reg___reg_lambda’: 18.936447979876338, ’reg___min_split_gain’: 3.397578174884514, ’reg___path_smooth’: 0.7411013389662124}

Rank 3: {’reg___n_estimators’: 700, ’reg___max_depth’: 8, ’reg___learning_rate’: 0.02058396742222129, ’reg___num_leaves’: 23, ’reg___subsample’: 0.5398334472923733, ’reg___colsample_bytree’: 0.5212631847735343, ’reg___min_child_samples’: 26, ’reg___reg_alpha’: 7.756301303582857, ’reg___reg_lambda’: 31.036924559927215, ’reg___min_split_gain’: 3.1601886098841856, ’reg___path_smooth’: 2.713063725887853}

Rank 4: {’reg___n_estimators’: 500, ’reg___max_depth’: 8, ’reg___learning_rate’: 0.0533742599168416, ’reg___num_leaves’: 28, ’reg___subsample’: 0.7539314983263365, ’reg___colsample_bytree’: 0.5988821250174826, ’reg___min_child_samples’: 11, ’reg___reg_alpha’: 8.737593556583935, ’reg___reg_lambda’: 33.89657874391073, ’reg___min_split_gain’: 4.391347949216105, ’reg___path_smooth’: 0.006002953500694463}

Rank 5: {’reg___n_estimators’: 700, ’reg___max_depth’: 6, ’reg___learning_rate’: 0.08104605367808392, ’reg___num_leaves’: 30, ’reg___subsample’: 0.7838280094217764, ’reg___colsample_bytree’: 0.7983389856316241, ’reg___min_child_samples’: 31, ’reg___reg_alpha’: 5.665243244326309, ’reg___reg_lambda’: 0.657525520104002, ’reg___min_split_gain’: 4.928618347468942, ’reg___path_smooth’: 1.0527052814646731}

Rank 6: {’reg___n_estimators’: 400, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.039932063781541086, ’reg___num_leaves’: 58, ’reg___subsample’: 0.5500866268034613, ’reg___colsample_bytree’: 0.6342086725701562, ’reg___min_child_samples’: 26, ’reg___reg_alpha’: 8.477623981199754, ’reg___reg_lambda’: 40.69808015474949, ’reg___min_split_gain’: 3.550465989252345, ’reg___path_smooth’: 4.016648226574189}

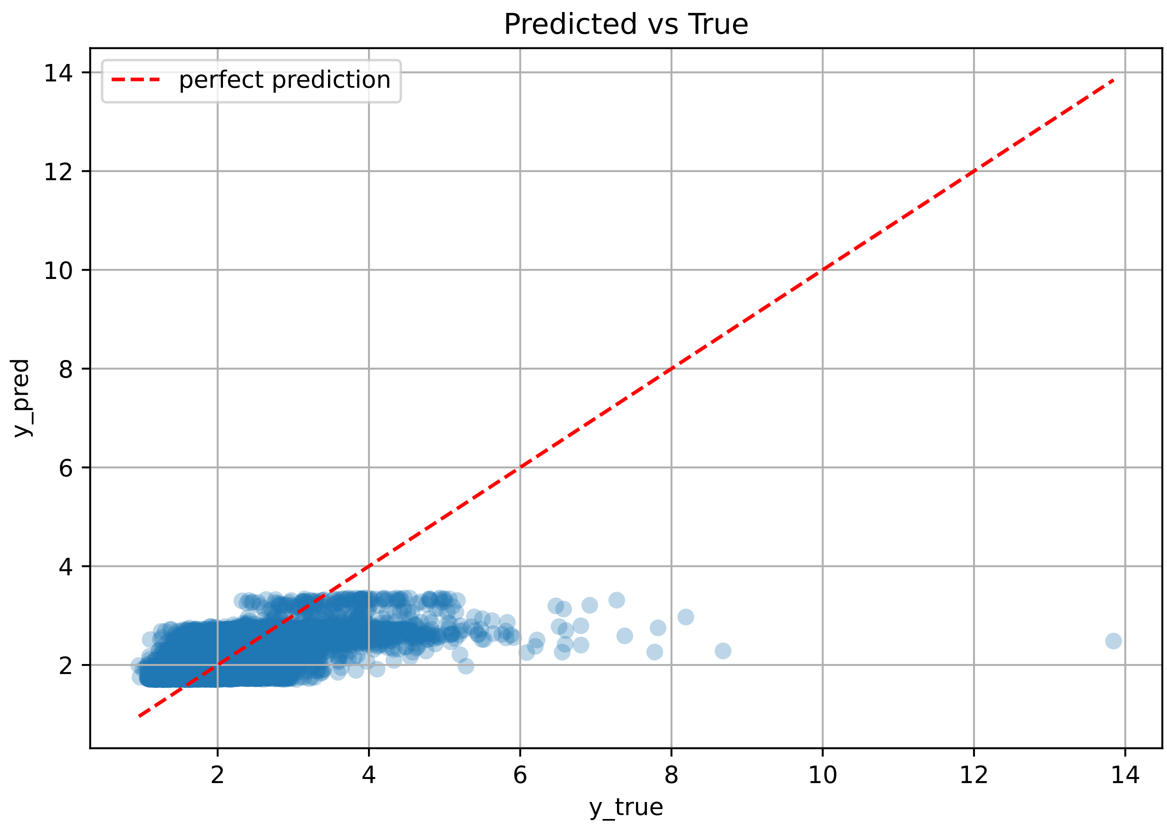
Rank 7: {’reg___n_estimators’: 800, ’reg___max_depth’: 6, ’reg___learning_rate’: 0.01755028096404159, ’reg___num_leaves’: 44, ’reg___subsample’: 0.5889215975033879, ’reg___colsample_bytree’: 0.5292524152178825, ’reg___min_child_samples’: 46, ’reg___reg_alpha’: 8.782093477246088, ’reg___reg_lambda’: 32.88299146079089, ’reg___min_split_gain’: 2.562623188987952, ’reg___path_smooth’: 1.8964652890459888}

Rank 8: {’reg___n_estimators’: 500, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.014888689985922698, ’reg___num_leaves’: 15, ’reg___subsample’: 0.7713378870382155, ’reg___colsample_bytree’: 0.6003981380204584, ’reg___min_child_samples’: 39, ’reg___reg_alpha’: 0.708401843474219, ’reg___reg_lambda’: 35.450297061503875, ’reg___min_split_gain’: 4.084355002228205, ’reg___path_smooth’: 3.9920905035989747}

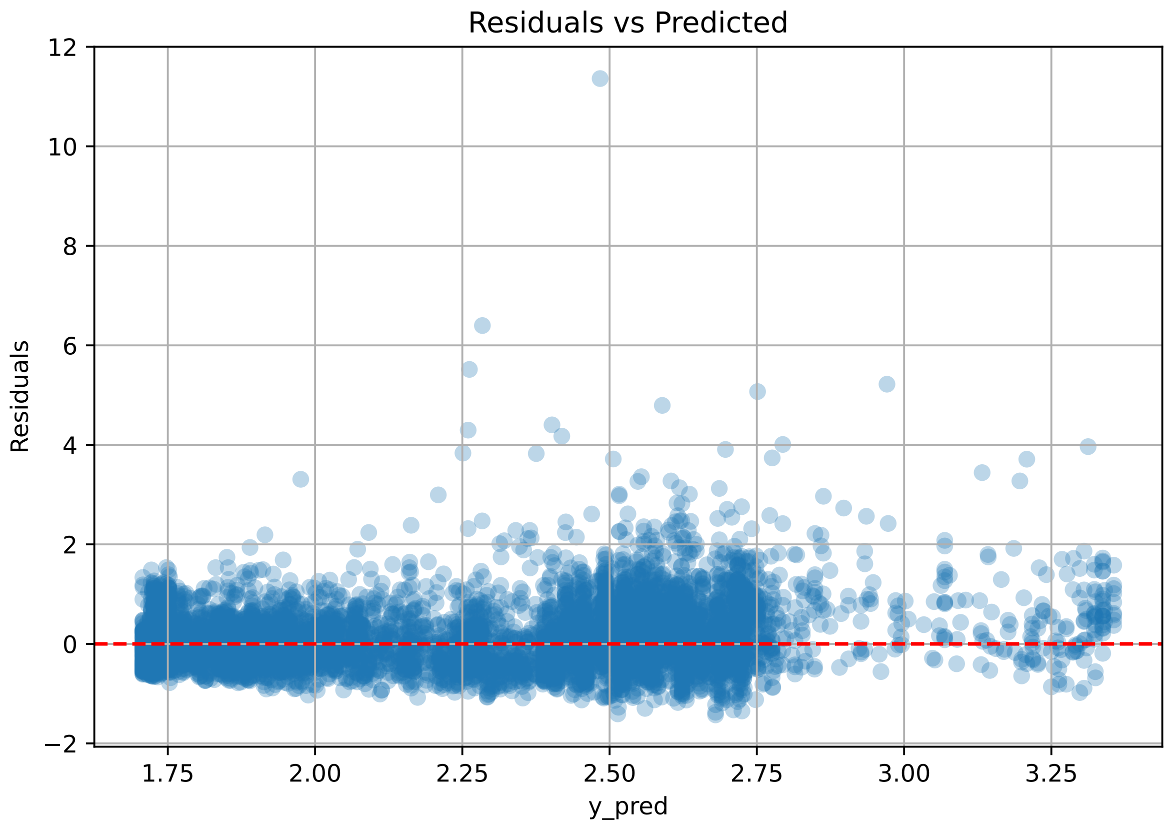
Rank 9: {’reg___n_estimators’: 800, ’reg___max_depth’: 3, ’reg___learning_rate’: 0.057148105910447916, ’reg___num_leaves’: 22, ’reg___subsample’: 0.8028435125732856, 'reg___colsample_bytree': 0.7122439752642935, 'reg___min_child_samples': 34, 'reg___reg_alpha': 7.59988319588502, 'reg___reg_lambda': 21.88132179612233, 'reg___min_split_gain': 1.2830896210545633, 'reg___path_smooth': 3.906522965678149}

Rank 10: {'reg___n_estimators': 700, 'reg___max_depth': 4, 'reg___learning_rate': 0.0463659512760143, 'reg___num_leaves': 44, 'reg___subsample': 0.626272126177956, 'reg___colsample_bytree': 0.5635531850363787, 'reg___min_child_samples': 37, 'reg___reg_alpha': 7.77952655766117, 'reg___reg_lambda': 13.257414395063849, 'reg___min_split_gain': 3.5352524401057424, 'reg___path_smooth': 1.6237391273094182}

2.4.6 Scatter



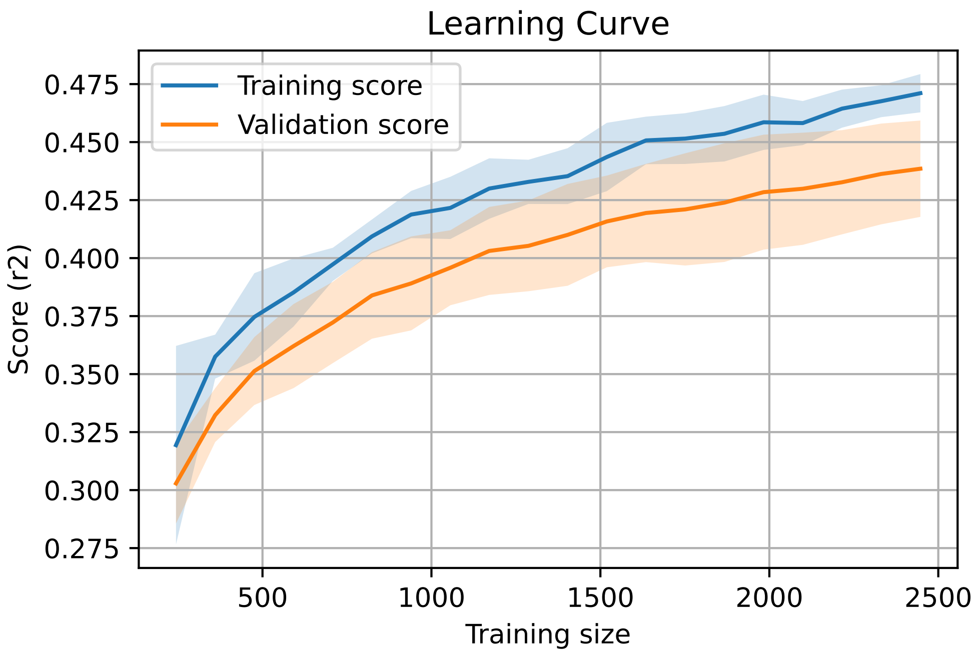
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2025_Sep

Len(group) : 4996, **Len(training)** : 3498,

Len(training_real) : 2449, **Len(test_of_training)** : 1049, **Len(test)** : 1498



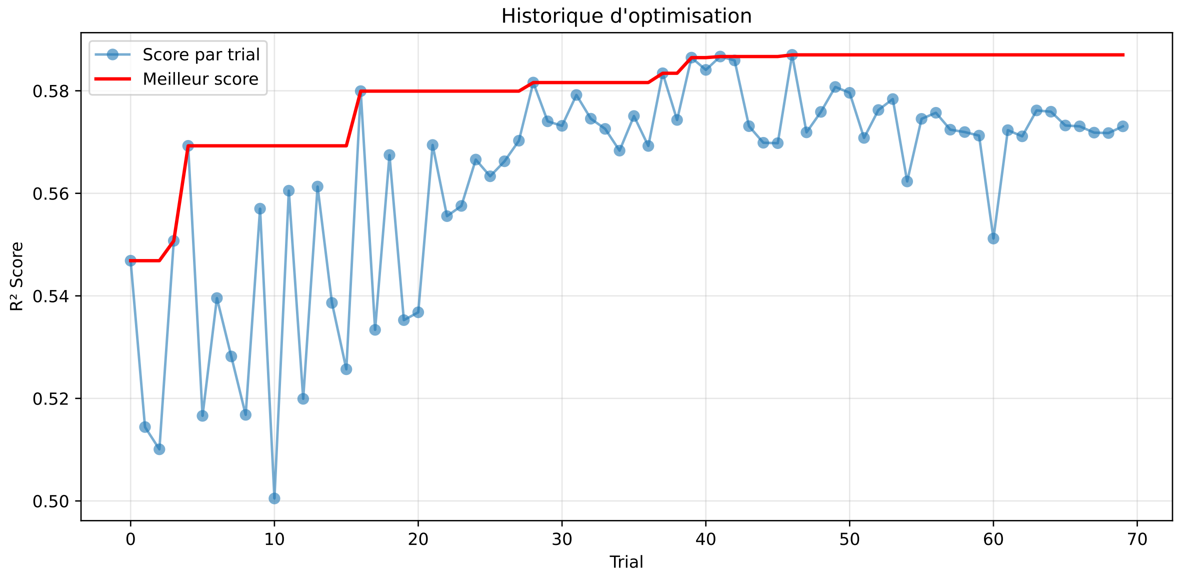
2.5 Model : CatBoost

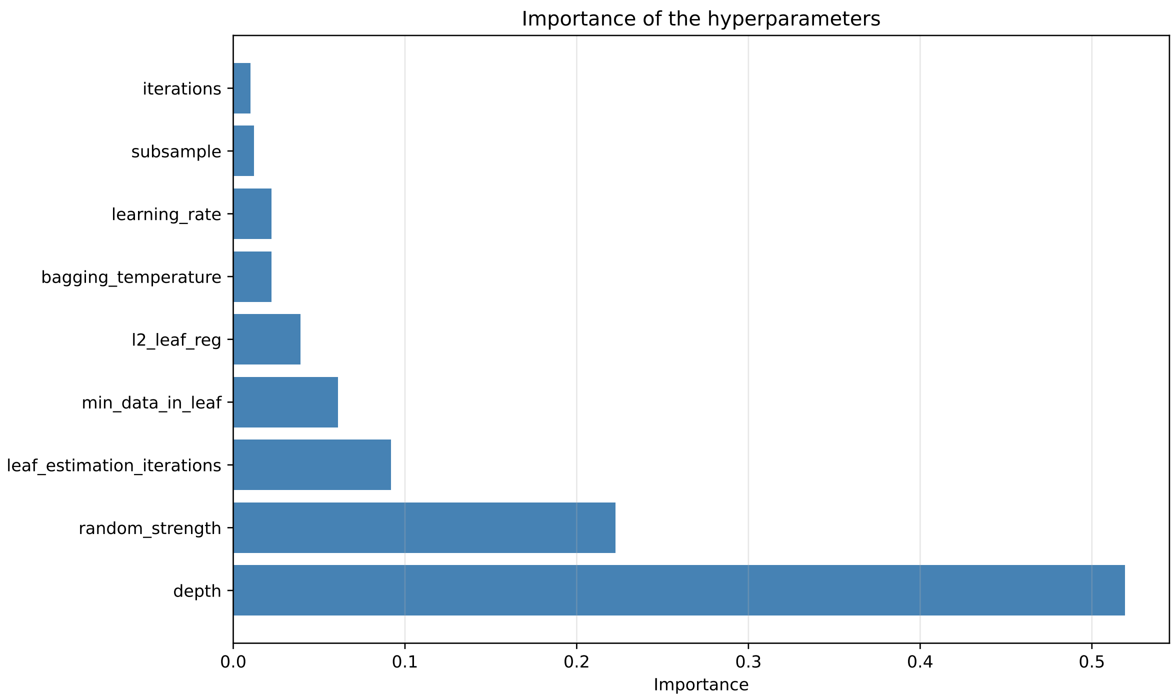
2.5.1 Gridsearch — Group: 2025_Jun

Len(group) : 3172, **Len(training)** : 2221, **Len(test)** : 951

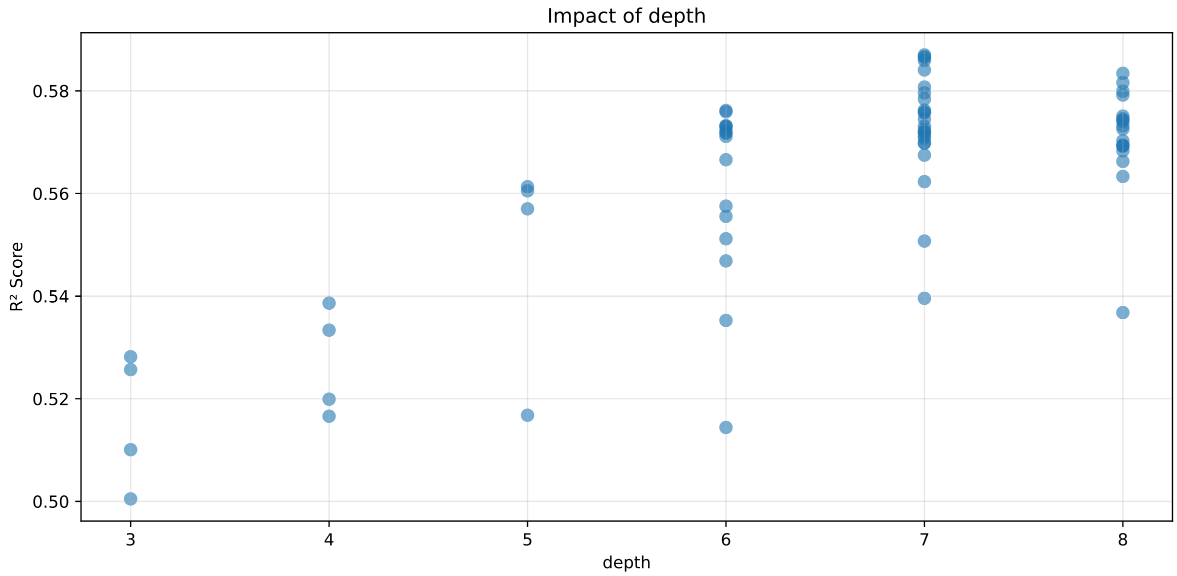
Distribution y_train vs y_test: y_train: mean=2.661, std=0.770, min=1.305, max=13.845 y_test: mean=2.621, std=0.704, min=1.253, max=5.627

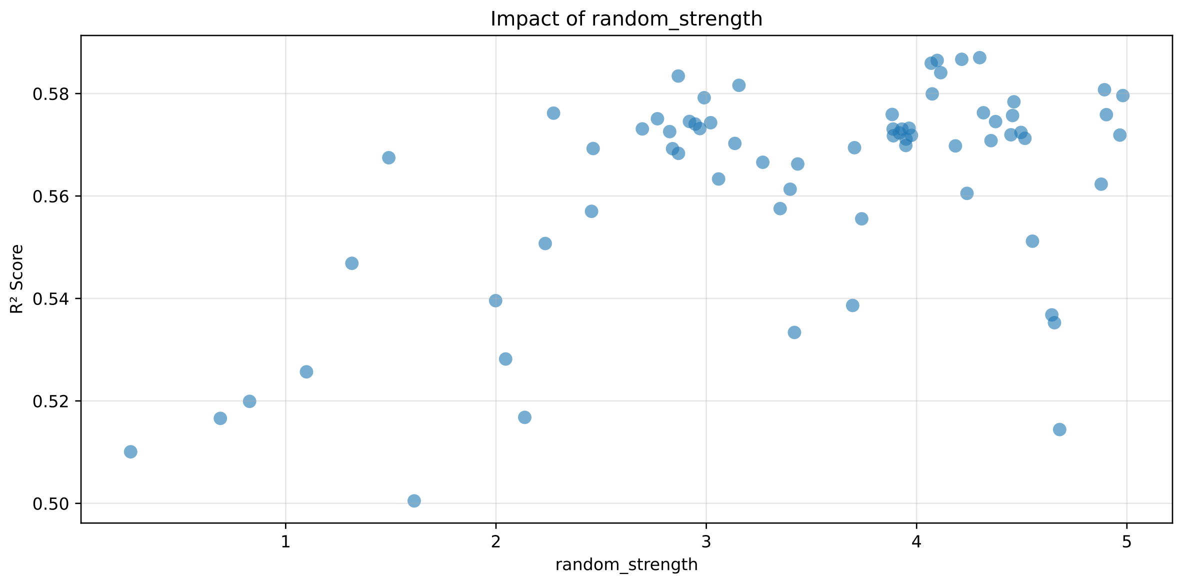
==================== TOP Importance FEATURES ==================== feature importance r_NH3 22.395019 r_umidity 22.048641 r_temperature 19.165452 r_CO2 18.510638 r_THI 17.880250

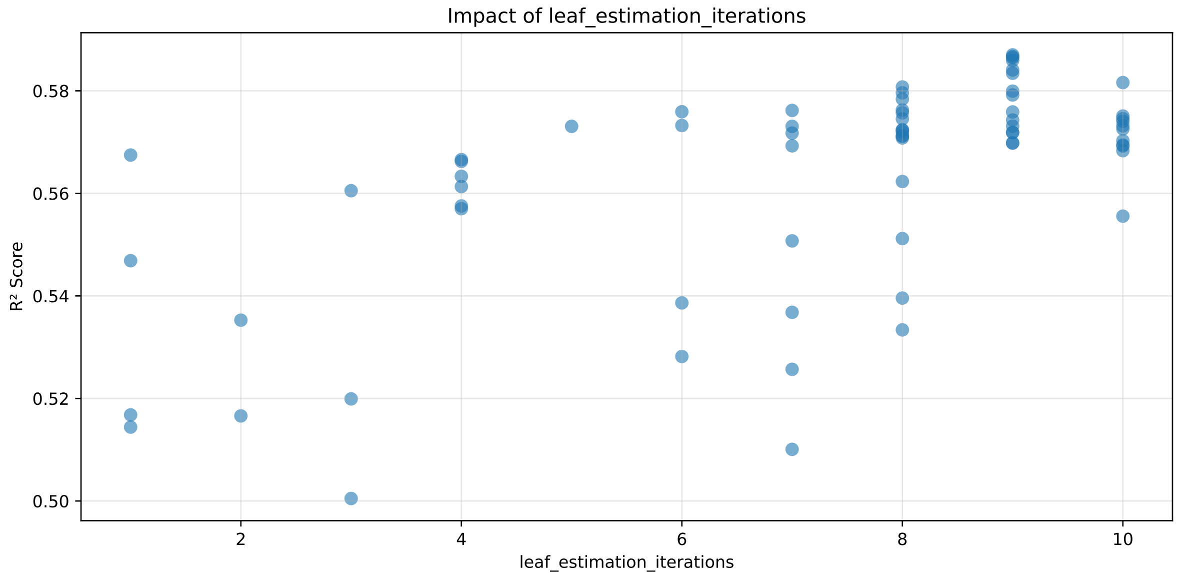


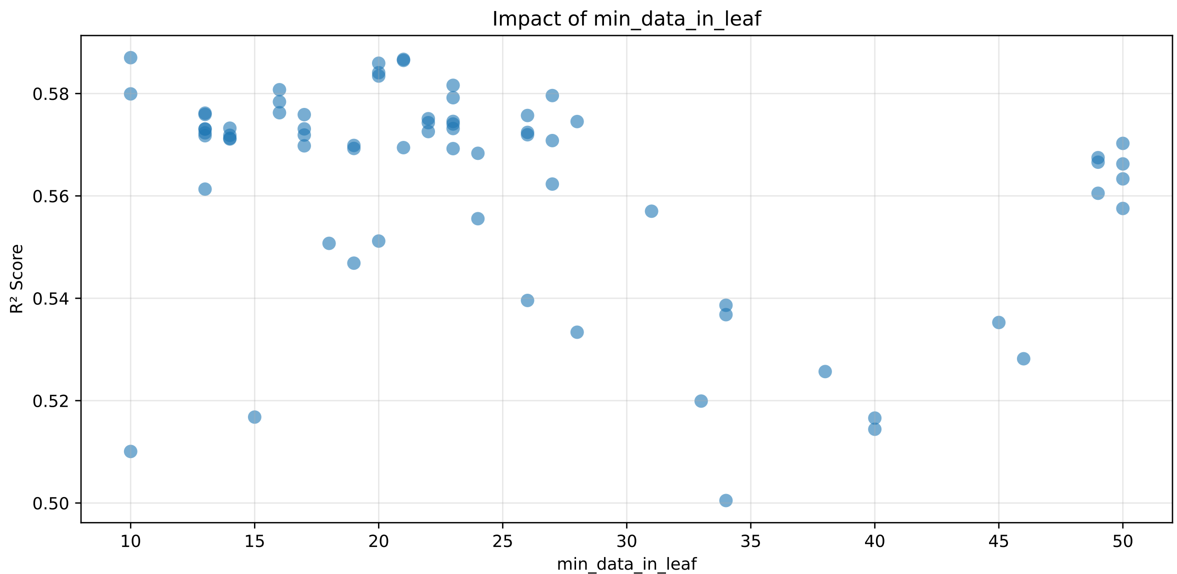


Paramètres avec >5% d'importance : ['depth', 'random_strength', 'leaf_estimation_iterations', 'min_data_in_leaf']


Impact of depth
R² Score
0.50
0.52
0.54
0.56
0.58
3
4
5
6
7
8
depth


Impact of random_strength
R² Score
0.50
0.52
0.54
0.56
0.58
1
2
3
4
5
random_strength



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C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```
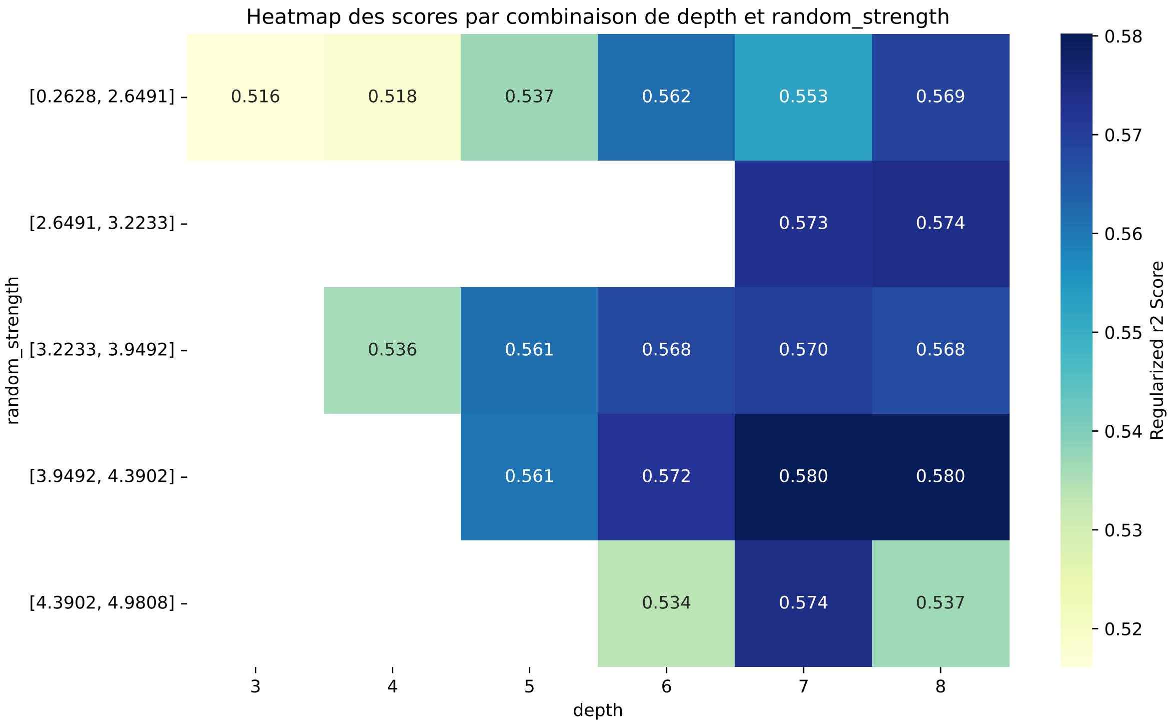



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5481	0.5005	0.5188	0.0321	0.3577	0.5353	0.0349	1.0696	1.0951
2	0.599	0.5353	0.5513	0.032	0.3439	0.5662	0.0309	1.0578	1.1192
3	0.5758	0.5168	0.5396	0.0312	0.3398	0.5636	0.0468	1.0905	1.1142
4	0.5742	0.5144	0.5378	0.0328	0.335	0.5649	0.0505	1.0982	1.1162
5	0.6285	0.5396	0.5564	0.0322	0.2727	0.5631	0.0236	1.0437	1.1649
6	0.6052	0.5166	0.5405	0.0287	0.2507	0.5592	0.0426	1.0825	1.1716
7	0.6356	0.5368	0.5555	0.0351	0.2404	0.5644	0.0277	1.0515	1.1841
8	0.6571	0.5512	0.561	0.0326	0.2335	0.5698	0.0186	1.0338	1.1921
9	0.6691	0.5469	0.5687	0.0301	0.1801	0.5863	0.0394	1.072	1.2236
10	0.6352	0.5199	0.5344	0.0248	0.1742	0.5574	0.0375	1.0721	1.2217
---	---	---	---	---	---	---	---	---	---
70	0.989	0.5703	0.5629	0.0231	-0.686	0.5952	0.025	1.0438	1.7343

Hyperparameters:

Rank 1: {’reg___iterations’: 600, ’reg___depth’: 3, ’reg___learning_rate’: 0.020645394658656637, ’reg___l2_leaf_reg’: 38.035634698417915, ’reg___bagging_temperature’: 0.19476670543760677, ’reg___random_strength’: 1.6111709530341856, ’reg___subsample’: 0.5292571711114443, ’reg___min_data_in_leaf’: 34, ’reg___leaf_estimation_iterations’: 3}

Rank 2: {’reg___iterations’: 600, ’reg___depth’: 6, ’reg___learning_rate’: 0.017536541596944726,

’reg__l2_leaf_reg’: 28.175256318686014, ’reg__bagging_temperature’: 4.71311309405983, ’reg__random_strength’: 4.655548828858429, ’reg__subsample’: 0.6892465407183084, ’reg__min_data_in_leaf’: 45, ’reg__leaf_estimation_iterations’: 2}

Rank 3: {’reg__iterations’: 700, ’reg__depth’: 5, ’reg__learning_rate’: 0.02589659962922482, ’reg__l2_leaf_reg’: 47.81364334446059, ’reg__bagging_temperature’: 3.3186313530823988, ’reg__random_strength’: 2.136729164091506, ’reg__subsample’: 0.641539764043184, ’reg__min_data_in_leaf’: 15, ’reg__leaf_estimation_iterations’: 1}

Rank 4: {’reg__iterations’: 400, ’reg__depth’: 6, ’reg__learning_rate’: 0.037992355079723907, ’reg__l2_leaf_reg’: 30.239976623625516, ’reg__bagging_temperature’: 0.058669494940533085, ’reg__random_strength’: 4.68026502043531, ’reg__subsample’: 0.8148511240531334, ’reg__min_data_in_leaf’: 40, ’reg__leaf_estimation_iterations’: 1}

Rank 5: {’reg__iterations’: 500, ’reg__depth’: 7, ’reg__learning_rate’: 0.011563219111309284, ’reg__l2_leaf_reg’: 30.151476232312138, ’reg__bagging_temperature’: 0.9802645203381988, ’reg__random_strength’: 1.99836403854964, ’reg__subsample’: 0.8343565302890563, ’reg__min_data_in_leaf’: 26, ’reg__leaf_estimation_iterations’: 8}

Rank 6: {’reg__iterations’: 800, ’reg__depth’: 4, ’reg__learning_rate’: 0.01448813111596352, ’reg__l2_leaf_reg’: 23.550642806211208, ’reg__bagging_temperature’: 0.013217643810891633, ’reg__random_strength’: 0.6894347911177678, ’reg__subsample’: 0.7713578789406945, ’reg__min_data_in_leaf’: 40, ’reg__leaf_estimation_iterations’: 2}

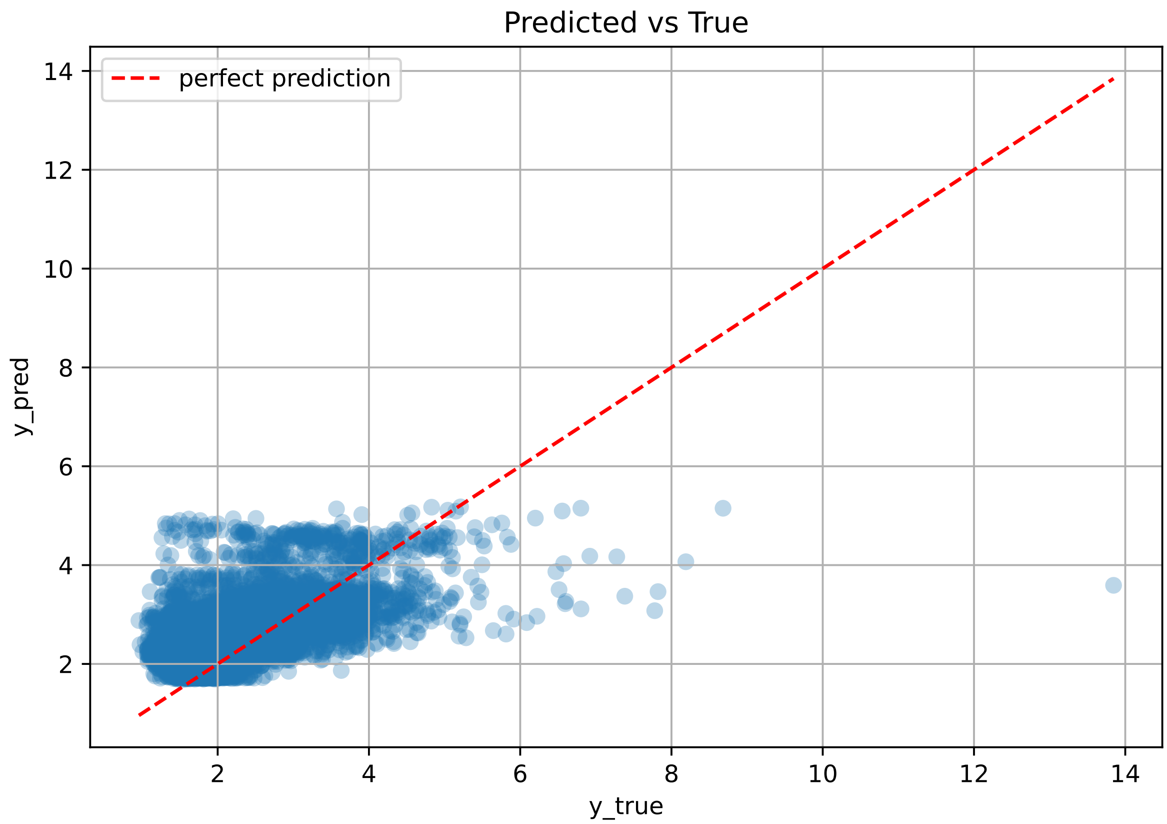
Rank 7: {’reg__iterations’: 500, ’reg__depth’: 8, ’reg__learning_rate’: 0.01123912462365434, ’reg__l2_leaf_reg’: 16.739257219364696, ’reg__bagging_temperature’: 0.485309279754752, ’reg__random_strength’: 4.643024447511373, ’reg__subsample’: 0.7652805882462834, ’reg__min_data_in_leaf’: 34, ’reg__leaf_estimation_iterations’: 7}

Rank 8: {’reg__iterations’: 300, ’reg__depth’: 6, ’reg__learning_rate’: 0.038033244788847516, ’reg__l2_leaf_reg’: 47.32199315897943, ’reg__bagging_temperature’: 0.5542354516734357, ’reg__random_strength’: 4.550895336560329, ’reg__subsample’: 0.5993089301463792, ’reg__min_data_in_leaf’: 20, ’reg__leaf_estimation_iterations’: 8}

Rank 9: {’reg__iterations’: 900, ’reg__depth’: 6, ’reg__learning_rate’: 0.032616151968889846, ’reg__l2_leaf_reg’: 46.19780523475378, ’reg__bagging_temperature’: 3.755792253814396, ’reg__random_strength’: 1.314872788550876, ’reg__subsample’: 0.5920141380793889, ’reg__min_data_in_leaf’: 19, ’reg__leaf_estimation_iterations’: 1}

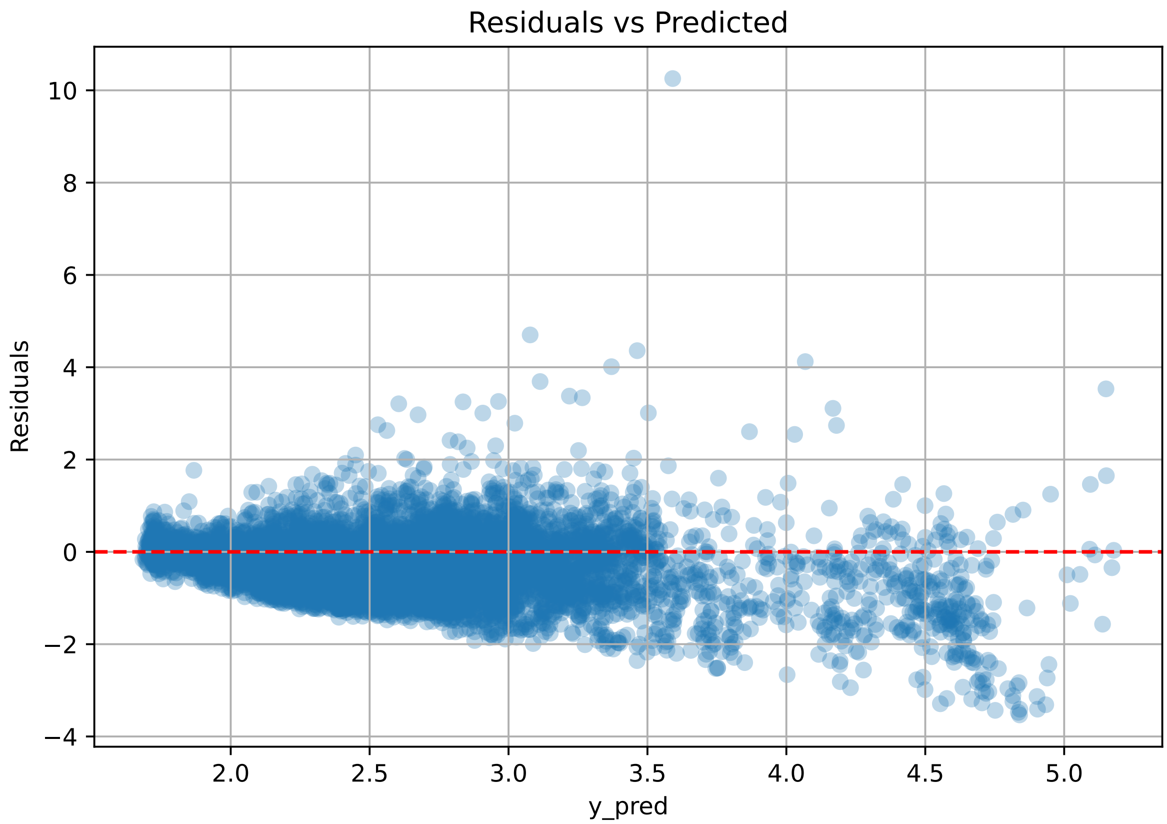
Rank 10: {’reg__iterations’: 200, ’reg__depth’: 4, ’reg__learning_rate’: 0.04822298665742902, ’reg__l2_leaf_reg’: 10.893836907494494, ’reg__bagging_temperature’: 1.900370905092486, ’reg__random_strength’: 0.828431692100145, ’reg__subsample’: 0.8550766257998406, ’reg__min_data_in_leaf’: 33, ’reg__leaf_estimation_iterations’: 3}

2.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

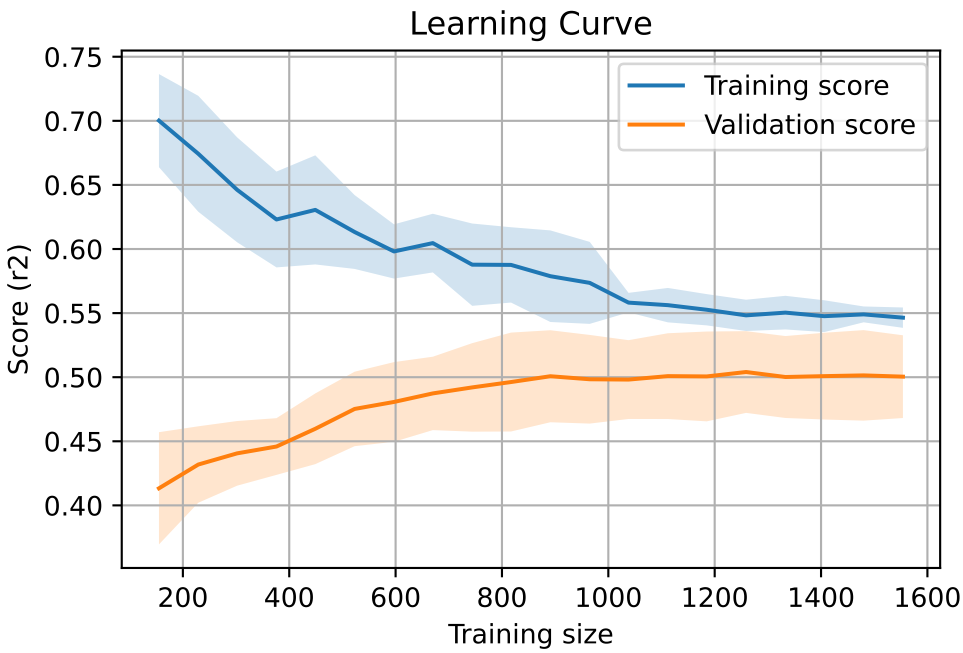
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 2025_Jun

$\textbf{Len(group)}$: 3172, $\textbf{Len(training)}$: 2221,

$\textbf{Len(training\_real)}$: 1555, $\textbf{Len(test\_of\_training)}$: 666, $\textbf{Len(test)}$: 951

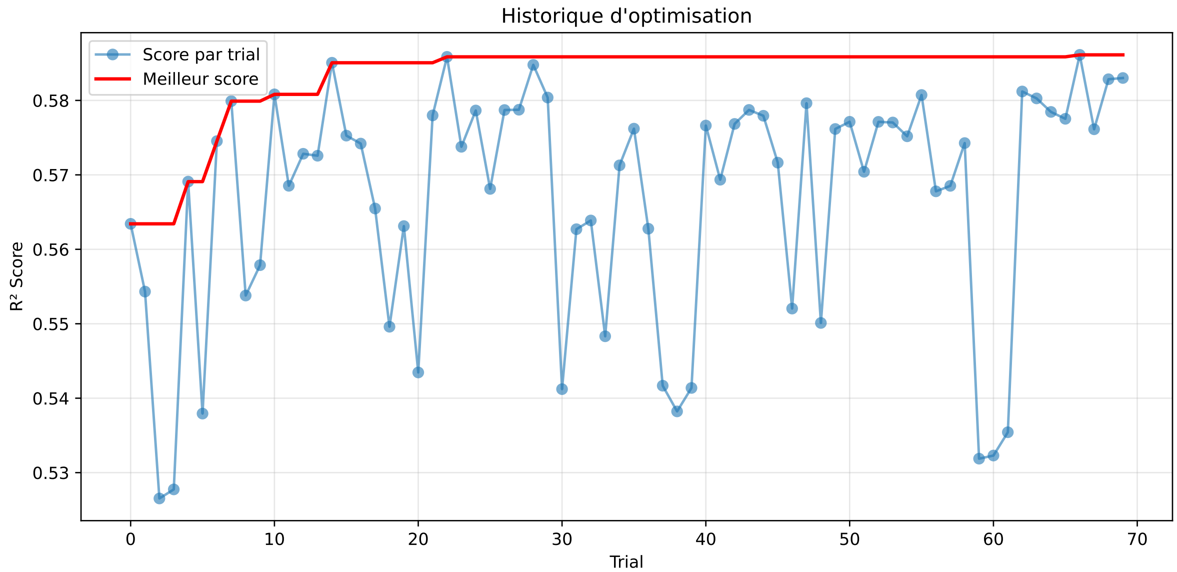


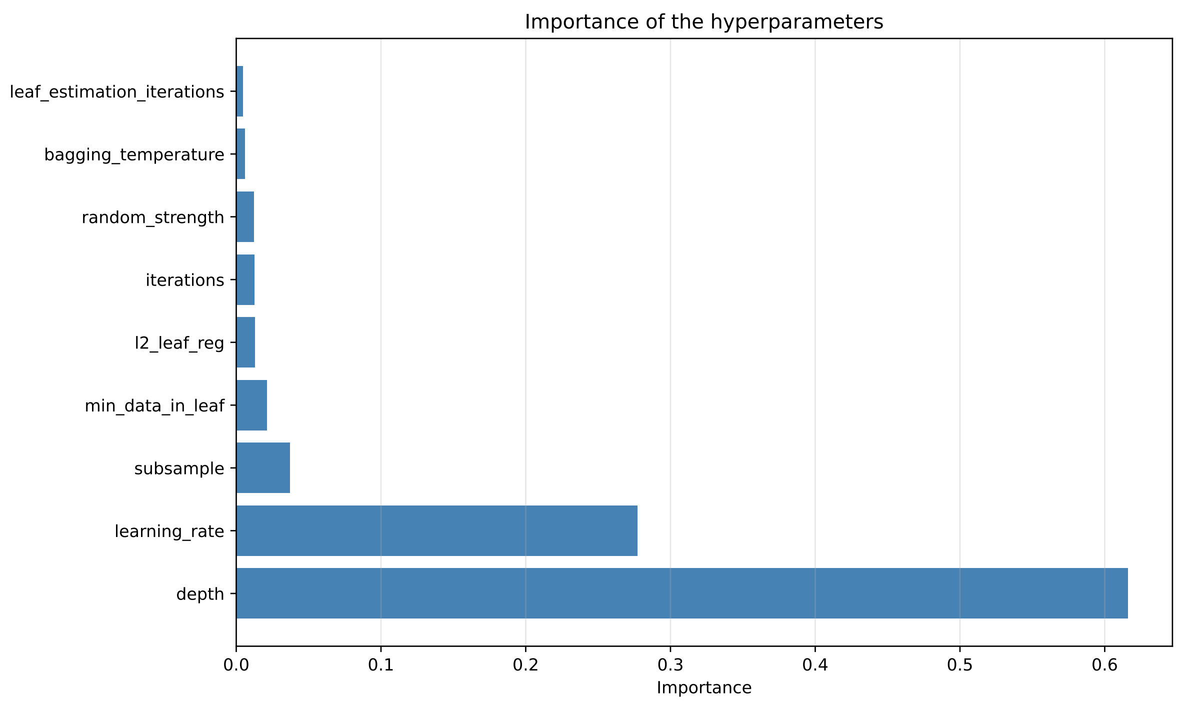
2.5.5 Gridsearch — Group: 2025_Sep

Len(group) : 4996, **Len(training)** : 3498, **Len(test)** : 1498

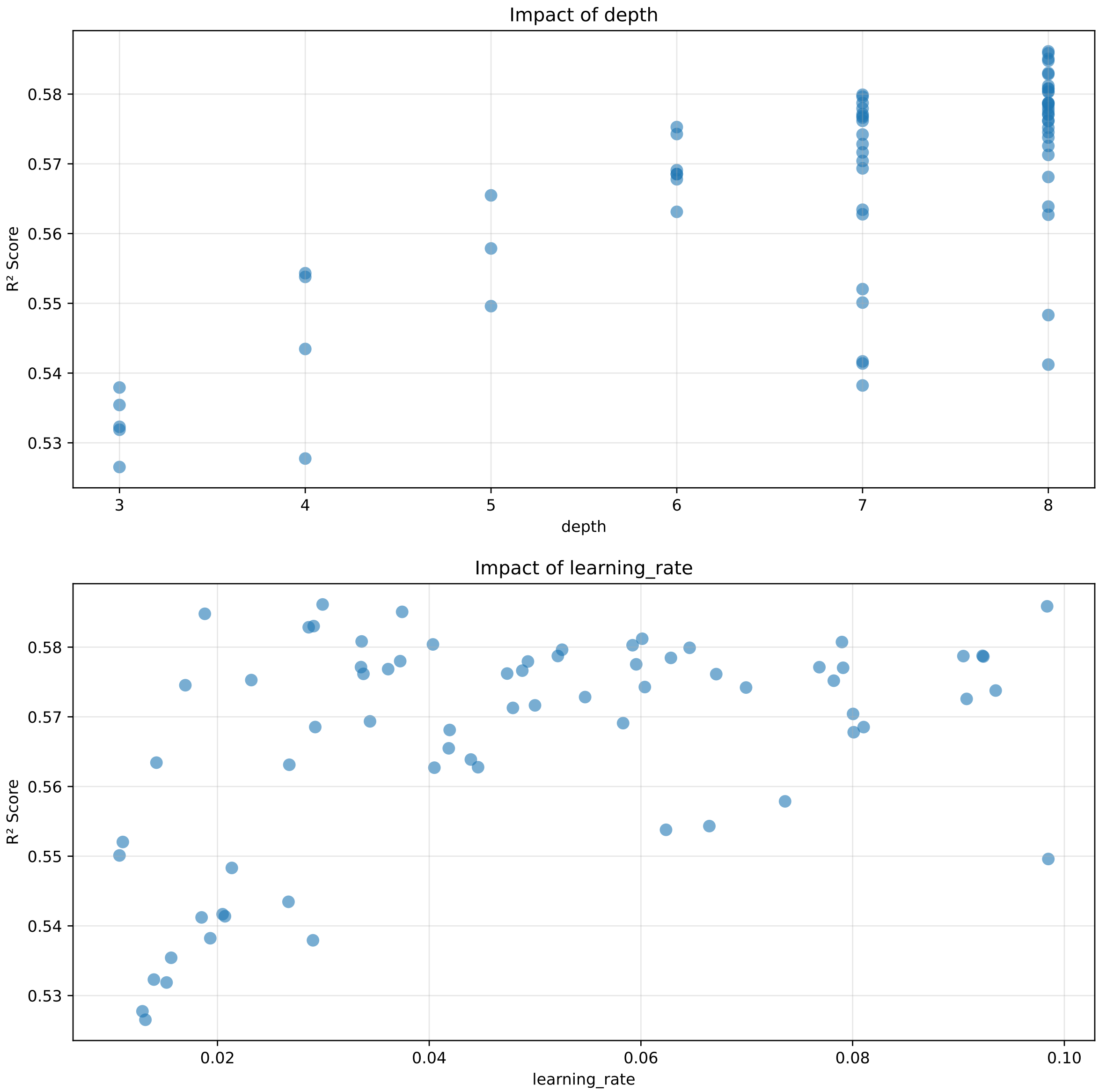
Distribution y_train vs y_test: y_train: mean=2.160, std=0.756, min=0.973, max=8.190 y_test: mean=2.160, std=0.777, min=0.959, max=7.779

==================== TOP Importance FEATURES ==================== feature importance r_umidity 35.605972 r_NH3 27.681787 r_CO2 15.837477 r_THI 10.572138 r_temperature 10.302625

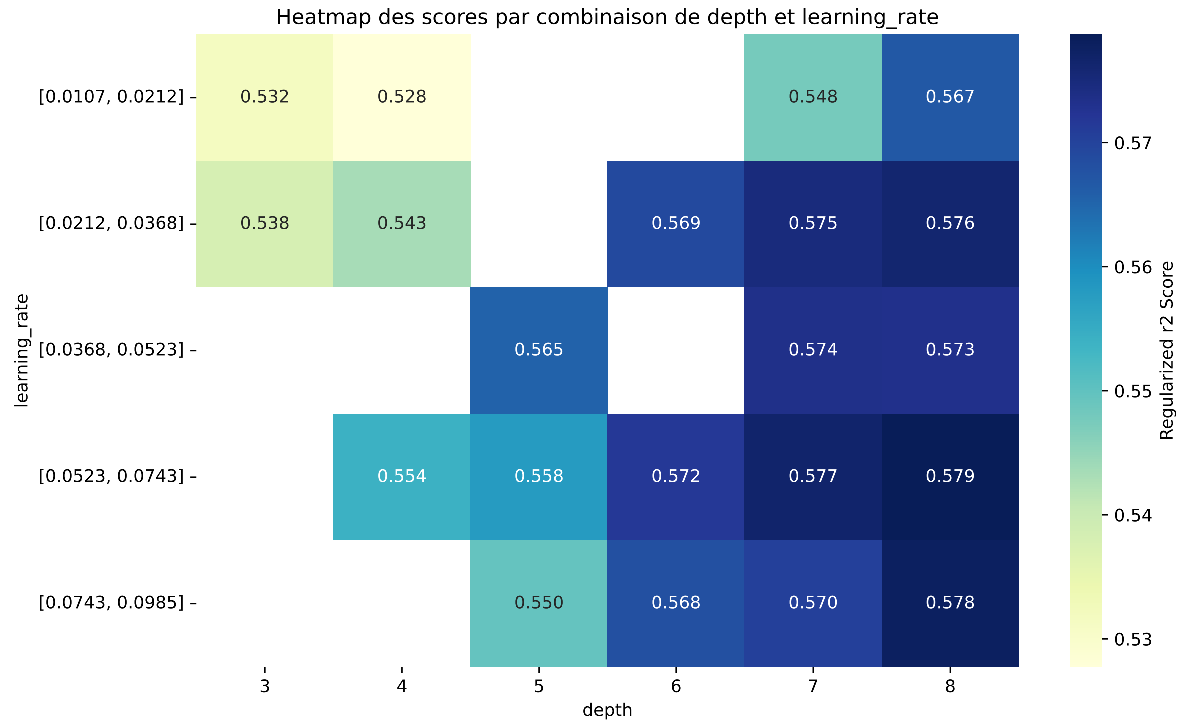




Paramètres avec >5% d'importance : ['depth', 'learning_rate']




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5891	0.5277	0.5218	0.0153	0.3438	0.5232	-0.0045	0.9914	1.1162
2	0.6	0.5265	0.5182	0.0162	0.3062	0.5206	-0.0059	0.9887	1.1395
3	0.6159	0.5319	0.5237	0.0147	0.2799	0.5262	-0.0057	0.9893	1.1579
4	0.6194	0.5323	0.524	0.0149	0.271	0.5272	-0.0051	0.9905	1.1636
5	0.6286	0.5354	0.5274	0.014	0.2559	0.5306	-0.0048	0.991	1.174
6	0.6376	0.5412	0.536	0.0158	0.252	0.5409	-0.0003	0.9994	1.1781
7	0.648	0.5483	0.5429	0.0147	0.2493	0.5507	0.0024	1.0043	1.1818
8	0.6349	0.5382	0.5297	0.0149	0.2483	0.5345	-0.0037	0.9931	1.1796
9	0.6402	0.5417	0.5314	0.0134	0.2462	0.5382	-0.0035	0.9936	1.1819
10	0.6354	0.5379	0.5305	0.0135	0.2455	0.5345	-0.0034	0.9936	1.1812
---	---	---	---	---	---	---	---	---	---
70	0.9596	0.5726	0.5644	0.0274	-0.5884	0.5851	0.0126	1.022	1.6759

Hyperparameters:

Rank 1: {'reg___iterations': 700, 'reg___depth': 4, 'reg___learning_rate': 0.01291092180451708, 'reg___l2_leaf_reg': 36.9418100688893, 'reg___bagging_temperature': 3.7389827118439163, 'reg___random_strength': 4.923620304981766, 'reg___subsample': 0.5711865811771292, 'reg___min_data_in_leaf': 19, 'reg___leaf_estimation_iterations': 9}

Rank 2: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.013190687869253287,

’reg__l2_leaf_reg’: 31.03323581500452, ’reg__bagging_temperature’: 3.7031924056721937, ’reg__random_strength’: 1.505735567397234, ’reg__subsample’: 0.8979524496127389, ’reg__min_data_in_leaf’: 11, ’reg__leaf_estimation_iterations’: 6}

Rank 3: {’reg__iterations’: 900, ’reg__depth’: 3, ’reg__learning_rate’: 0.015189421593494484, ’reg__l2_leaf_reg’: 12.629060419067061, ’reg__bagging_temperature’: 1.1110180284250024, ’reg__random_strength’: 3.0039487455824836, ’reg__subsample’: 0.5494936496126231, ’reg__min_data_in_leaf’: 12, ’reg__leaf_estimation_iterations’: 6}

Rank 4: {’reg__iterations’: 1000, ’reg__depth’: 3, ’reg__learning_rate’: 0.013991193174614296, ’reg__l2_leaf_reg’: 12.733856557011638, ’reg__bagging_temperature’: 3.4406175385104776, ’reg__random_strength’: 2.8244068435502108, ’reg__subsample’: 0.5534688537575058, ’reg__min_data_in_leaf’: 25, ’reg__leaf_estimation_iterations’: 9}

Rank 5: {’reg__iterations’: 1000, ’reg__depth’: 3, ’reg__learning_rate’: 0.01562312839363973, ’reg__l2_leaf_reg’: 12.517771185590686, ’reg__bagging_temperature’: 1.0867736771284828, ’reg__random_strength’: 3.0330898070045382, ’reg__subsample’: 0.6011012645939136, ’reg__min_data_in_leaf’: 34, ’reg__leaf_estimation_iterations’: 6}

Rank 6: {’reg__iterations’: 200, ’reg__depth’: 8, ’reg__learning_rate’: 0.018498111998290677, ’reg__l2_leaf_reg’: 17.783767234045424, ’reg__bagging_temperature’: 1.6381721547194772, ’reg__random_strength’: 1.8567140930257067, ’reg__subsample’: 0.72011857057304, ’reg__min_data_in_leaf’: 22, ’reg__leaf_estimation_iterations’: 10}

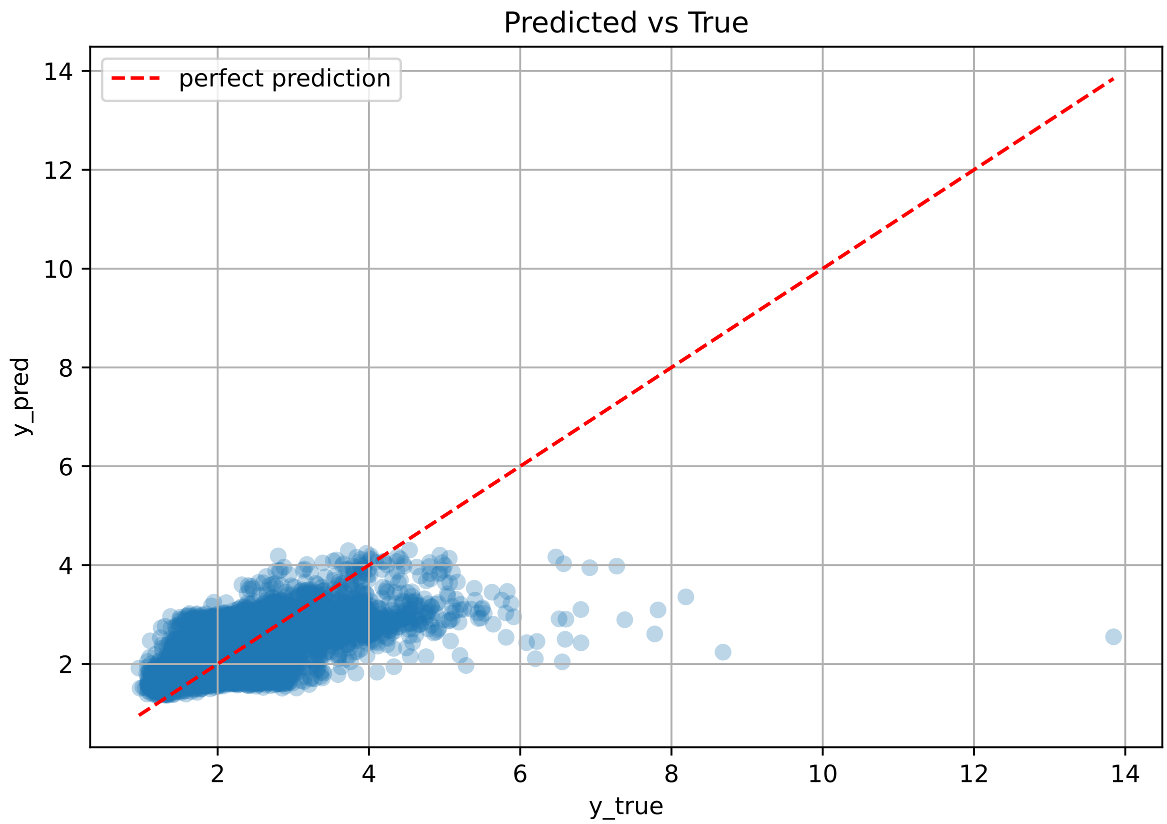
Rank 7: {’reg__iterations’: 200, ’reg__depth’: 8, ’reg__learning_rate’: 0.021352406215596288, ’reg__l2_leaf_reg’: 18.718162956801756, ’reg__bagging_temperature’: 1.9405061188335924, ’reg__random_strength’: 1.8300982817706353, ’reg__subsample’: 0.7531242296856223, ’reg__min_data_in_leaf’: 22, ’reg__leaf_estimation_iterations’: 8}

Rank 8: {’reg__iterations’: 200, ’reg__depth’: 7, ’reg__learning_rate’: 0.01931739899927768, ’reg__l2_leaf_reg’: 1.992674000798023, ’reg__bagging_temperature’: 1.8078578756538892, ’reg__random_strength’: 1.8943116549172623, ’reg__subsample’: 0.7400888817166058, ’reg__min_data_in_leaf’: 15, ’reg__leaf_estimation_iterations’: 8}

Rank 9: {’reg__iterations’: 200, ’reg__depth’: 7, ’reg__learning_rate’: 0.020479870127970864, ’reg__l2_leaf_reg’: 2.349912139054041, ’reg__bagging_temperature’: 2.839657608410097, ’reg__random_strength’: 1.8830678282519162, ’reg__subsample’: 0.7491723853772537, ’reg__min_data_in_leaf’: 15, ’reg__leaf_estimation_iterations’: 8}

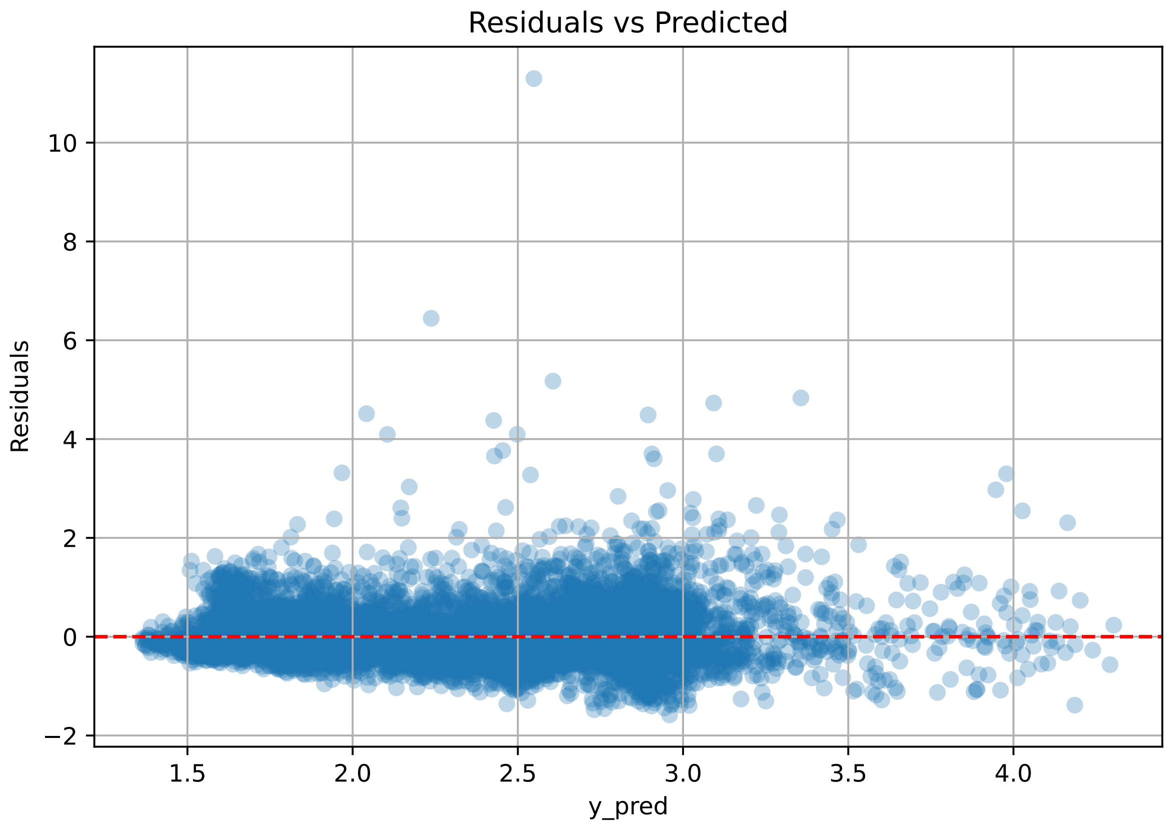
Rank 10: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.02901867354233179, ’reg__l2_leaf_reg’: 11.771335264659179, ’reg__bagging_temperature’: 4.556903396534492, ’reg__random_strength’: 4.4479189409903315, ’reg__subsample’: 0.5782763682056387, ’reg__min_data_in_leaf’: 27, ’reg__leaf_estimation_iterations’: 2}

2.5.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

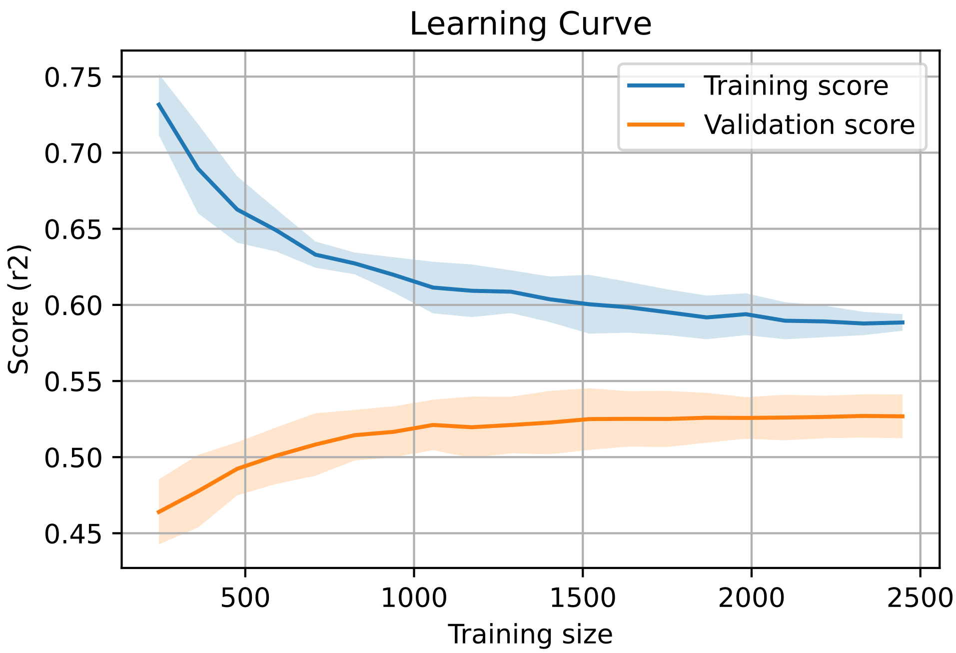
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2025_Sep

Len(group) : 4996, **Len(training)** : 3498,

Len(training_real) : 2449, **Len(test_of_training)** : 1049, **Len(test)** : 1498



2.6 Sumup-table

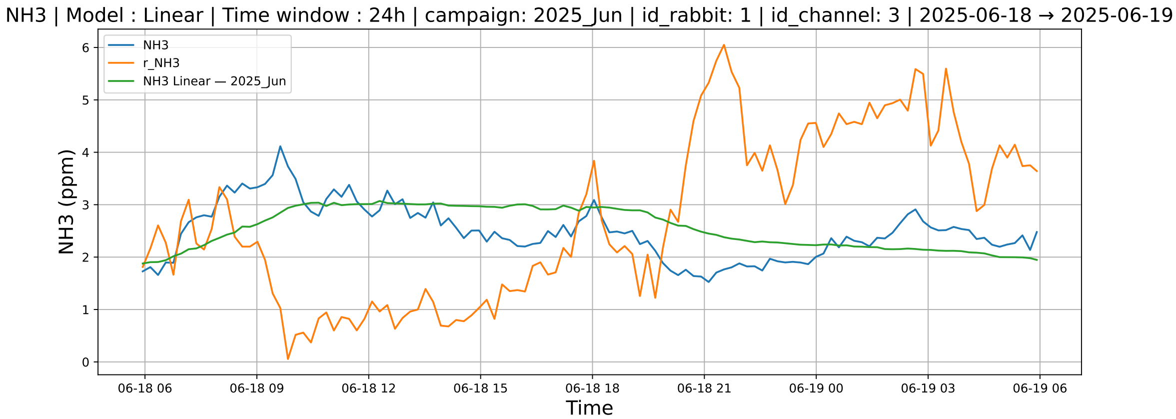
model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	2025_Jun	2221	951	3172	0.223	0.229	0.018	0.25	0.252	0.023	1.1	0.972	nan
Linear	2025_Sep	3498	1498	4996	0.371	0.386	0.02	0.349	0.349	-0.037	0.904	0.961	nan
Linear	MEAN	5719	2449	8168	0.297	0.308	0.019	0.3	0.301	-0.007	1.002	0.966	nan
LGBM	2025_Jun	2221	951	3172	0.456	0.436	0.039	0.482	0.511	0.076	1.174	1.046	0.4356
LGBM	2025_Sep	3498	1498	4996	0.469	0.437	0.021	0.426	0.441	0.004	1.009	1.072	0.3421
LGBM	MEAN	5719	2449	8168	0.462	0.436	0.03	0.454	0.476	0.04	1.092	1.059	0.389
CatBoost	2025_Jun	2221	951	3172	0.548	0.5	0.032	0.519	0.535	0.035	1.07	1.095	0.3577
CatBoost	2025_Sep	3498	1498	4996	0.589	0.528	0.015	0.522	0.523	-0.004	0.991	1.116	0.3438
CatBoost	MEAN	5719	2449	8168	0.569	0.514	0.024	0.52	0.529	0.015	1.03	1.106	0.351

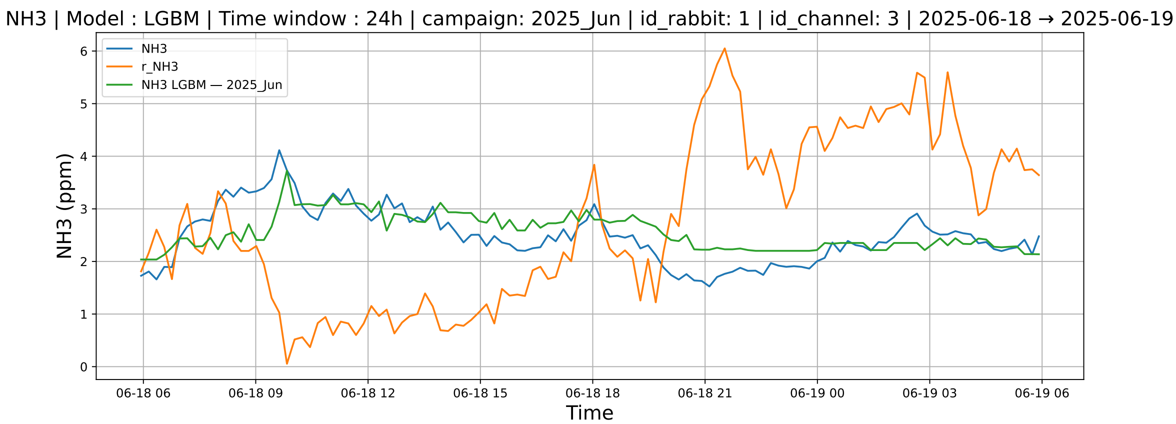
3 Plot

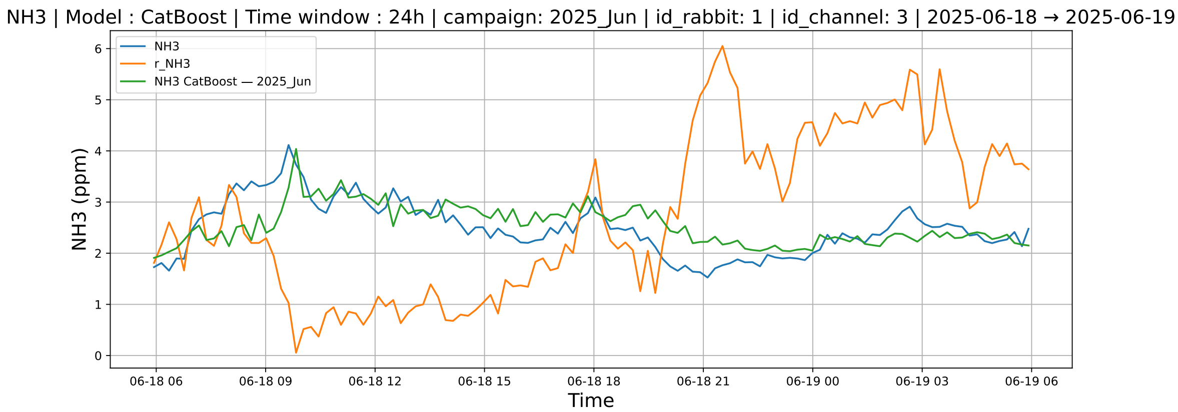
3.1 Standalone plots

3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

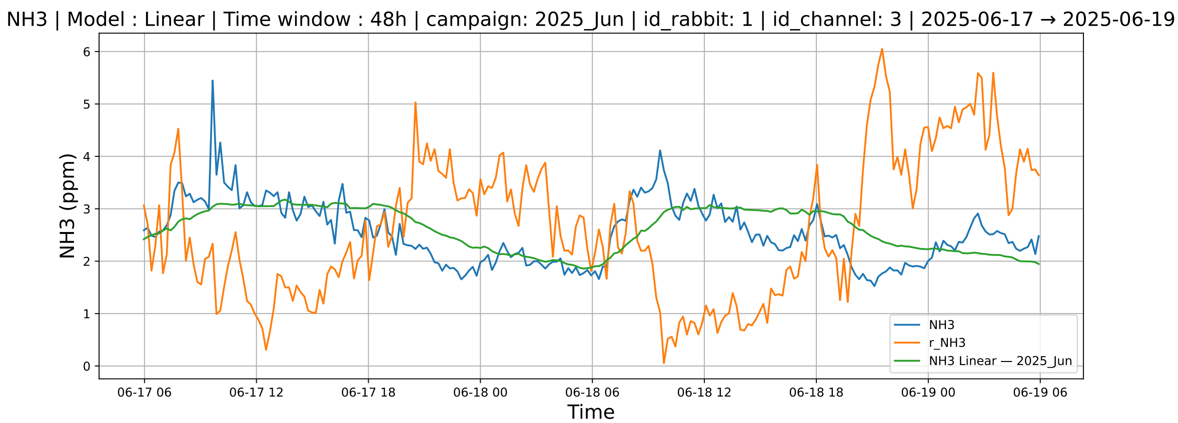
3.1.1.1 Time window : 24

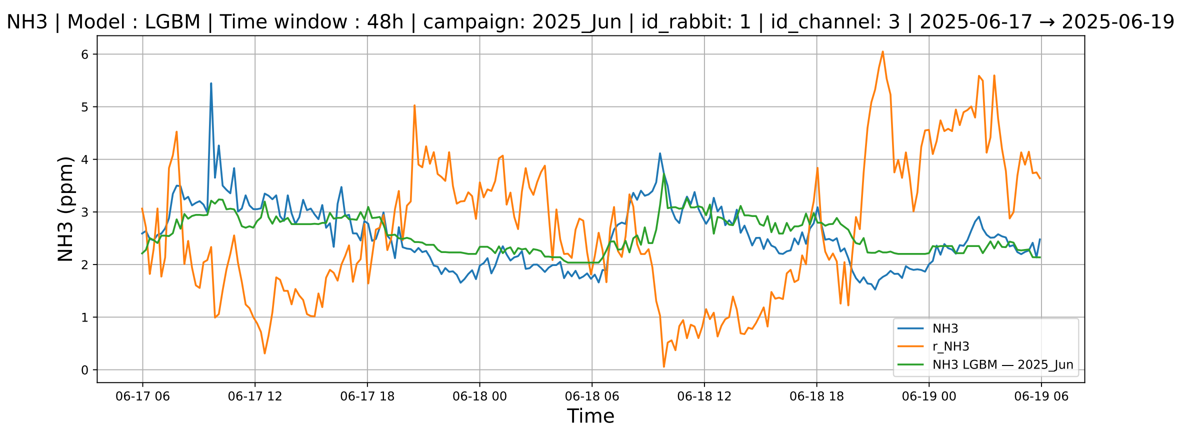




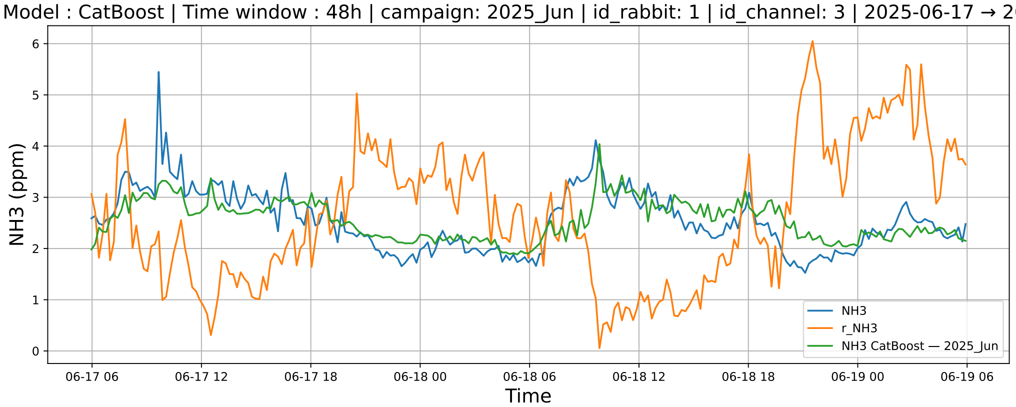


3.1.1.2 Time window : 48



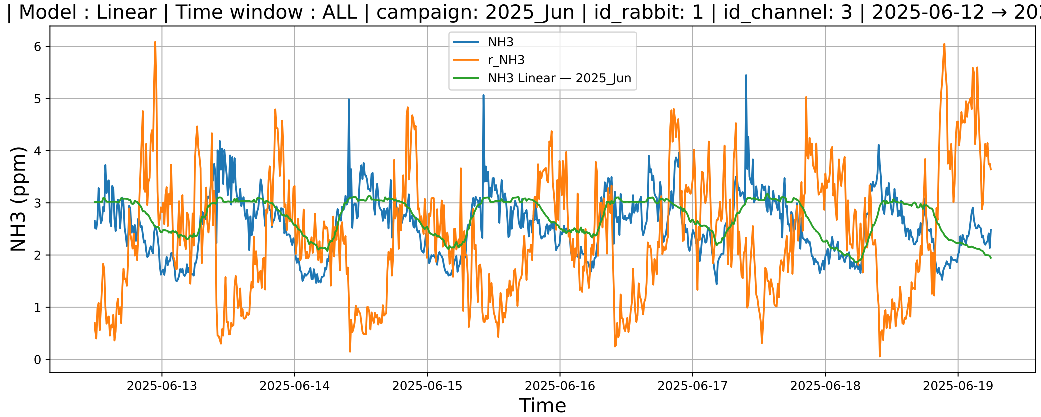


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

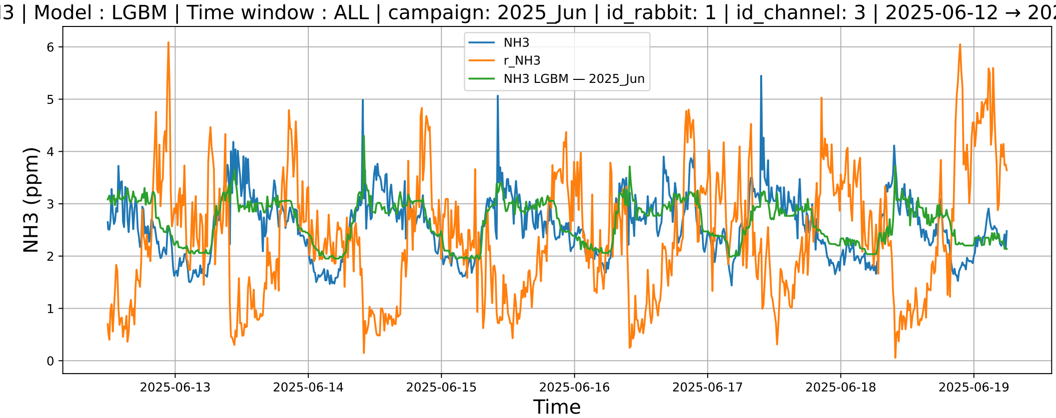


3.1.1.3 Time window : ALL

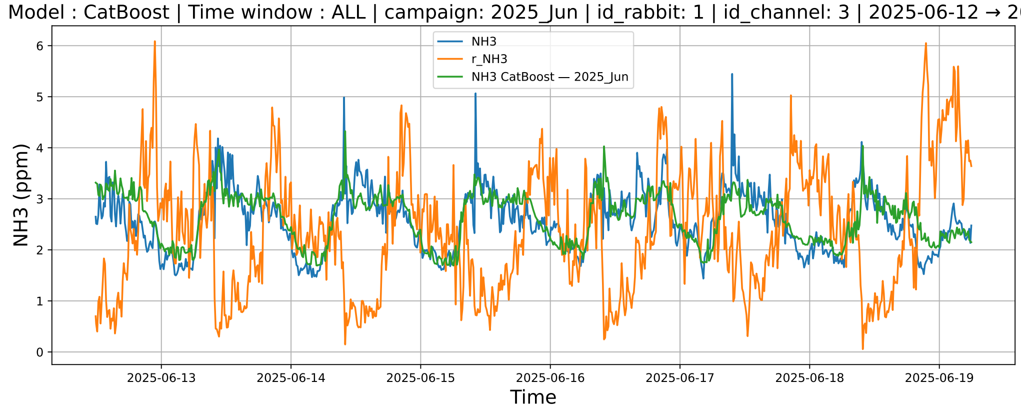
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



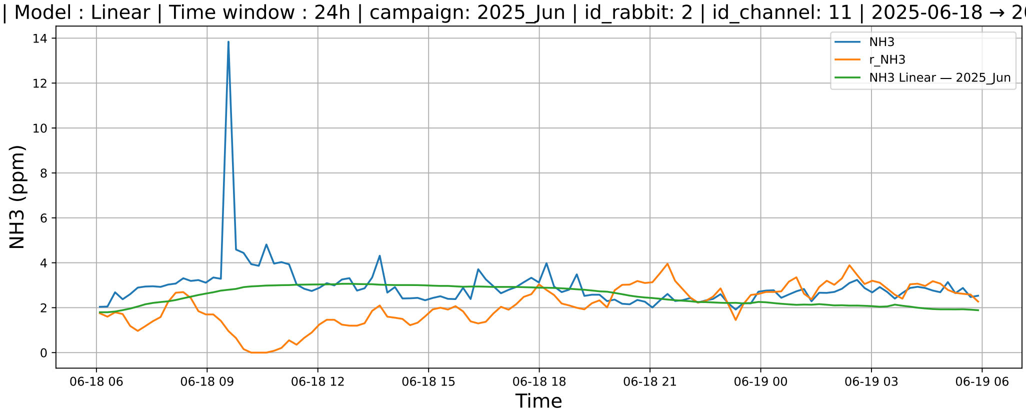
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



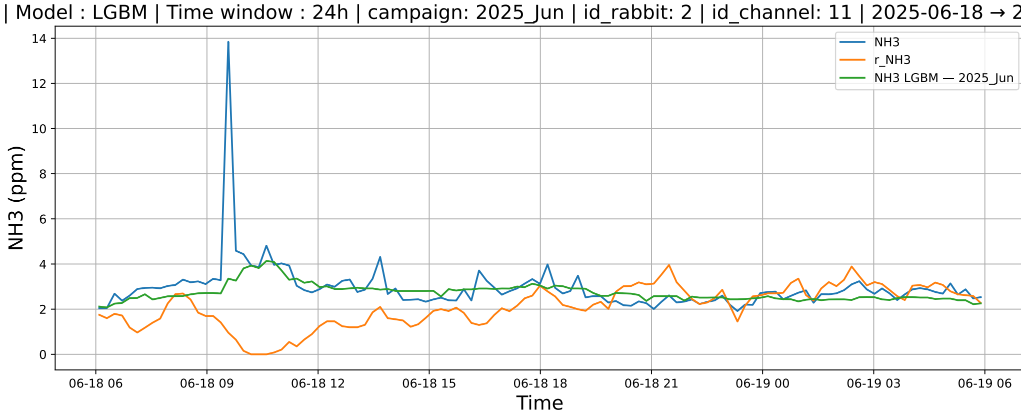
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

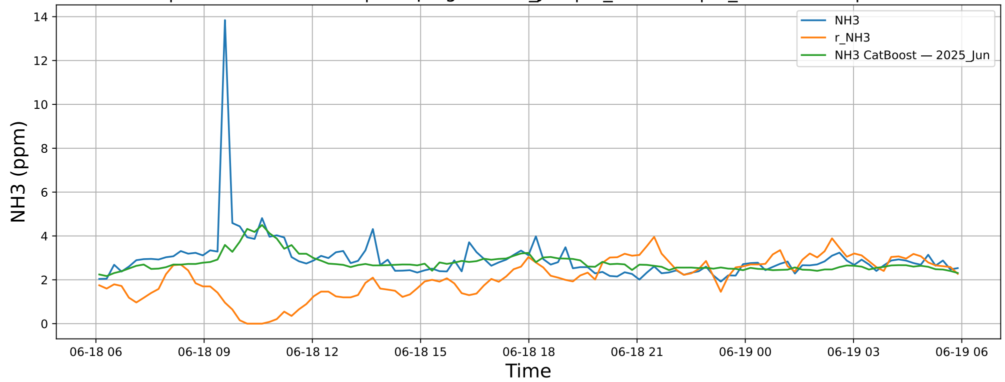
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

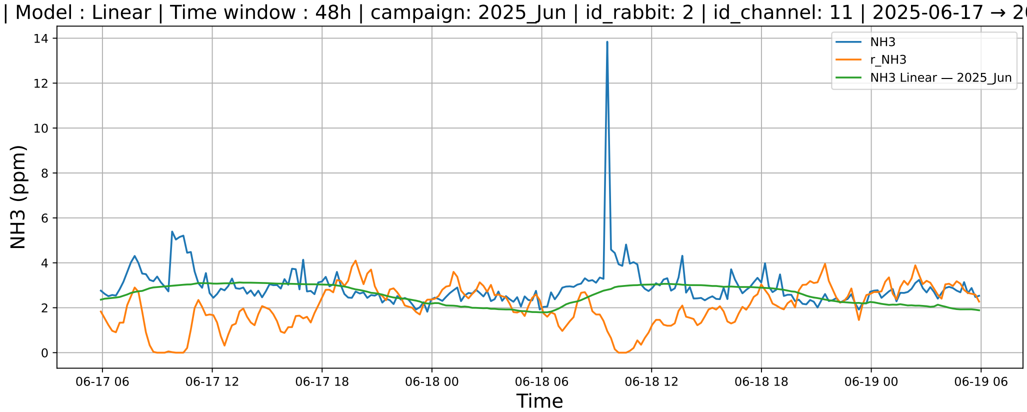


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

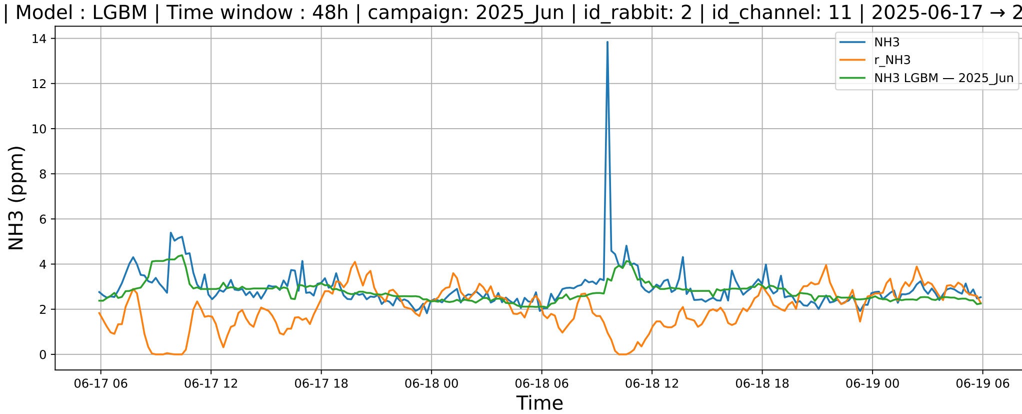


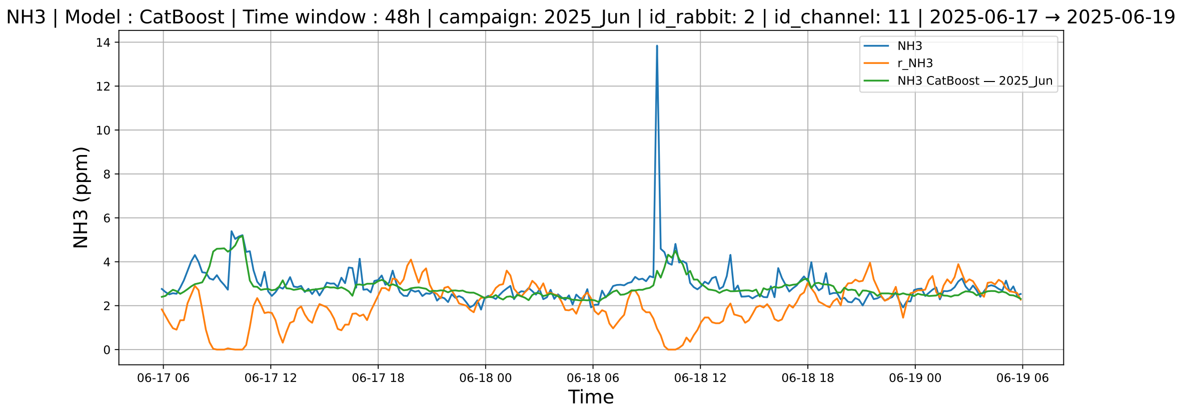
3.1.2.2 Time window : 48

NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

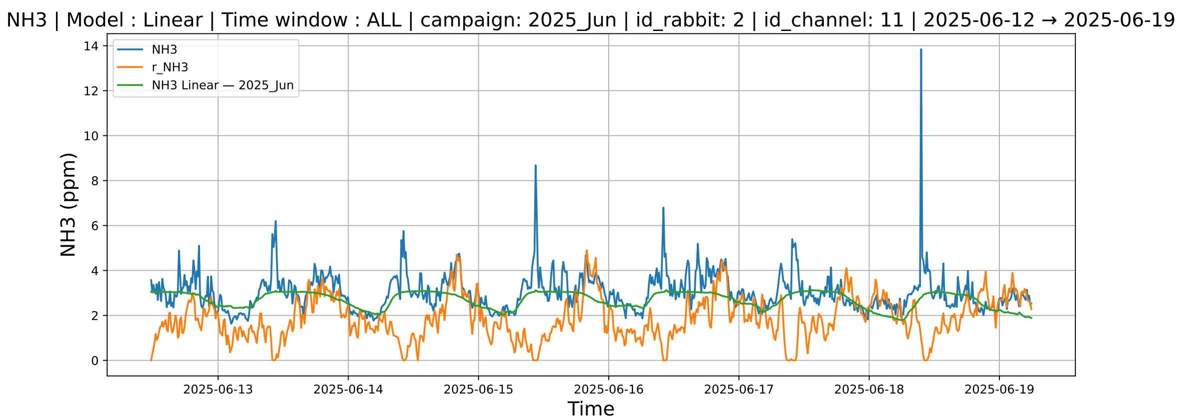


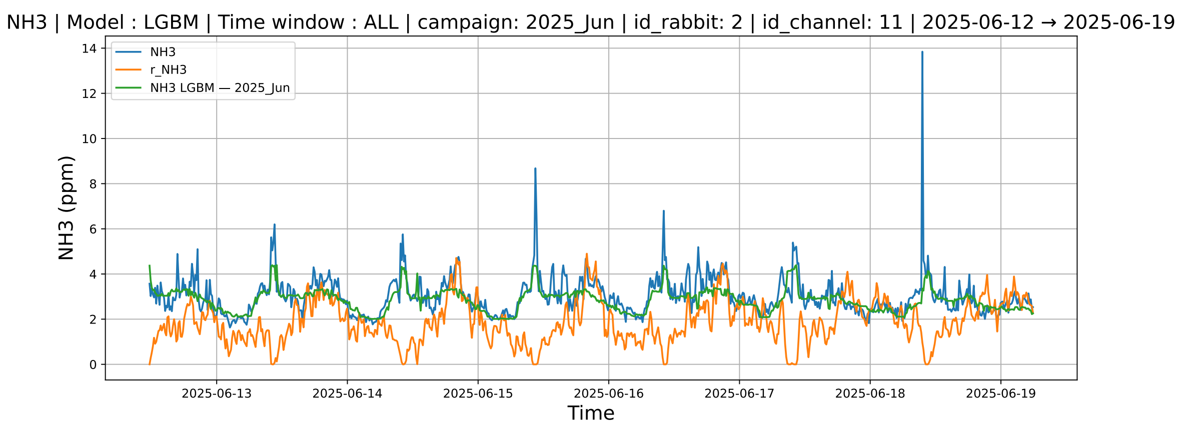
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



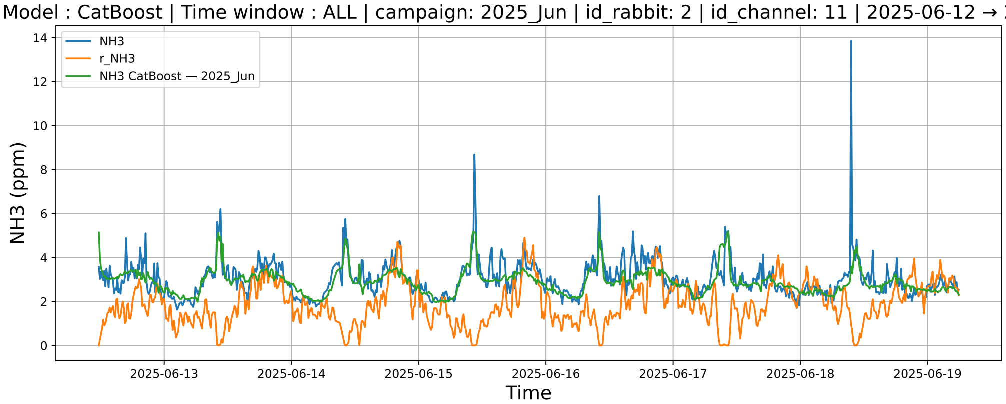


3.1.2.3 Time window : ALL





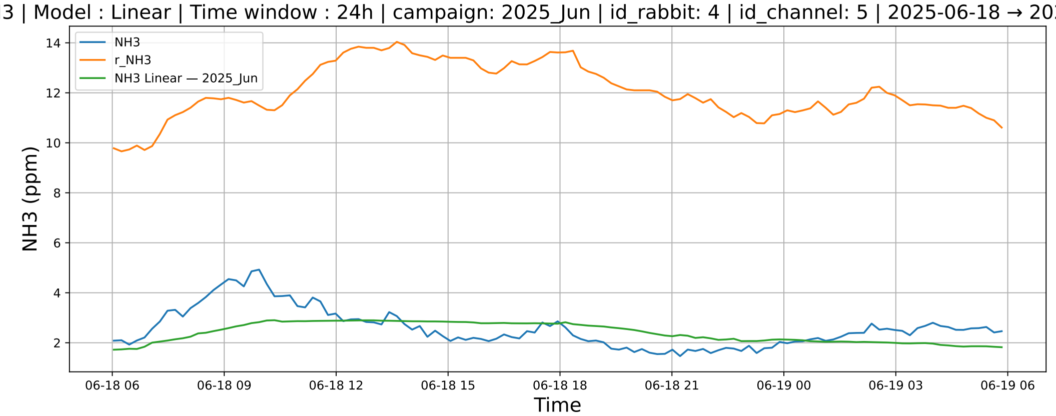
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



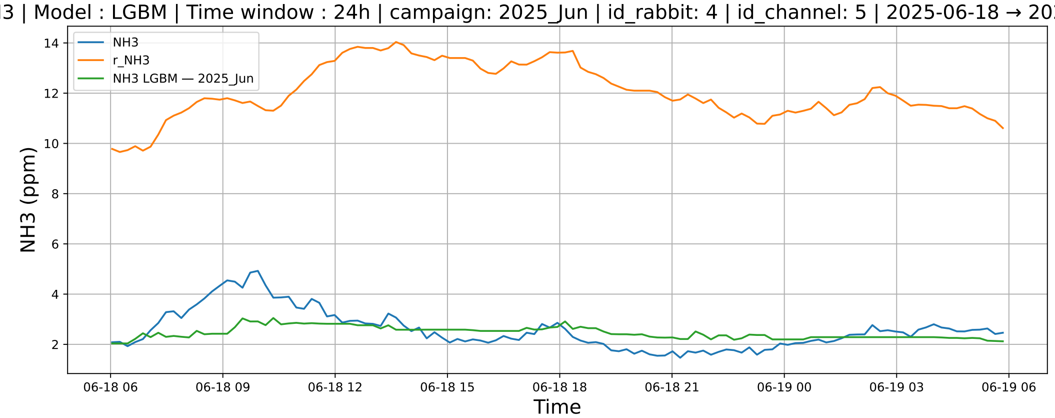
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

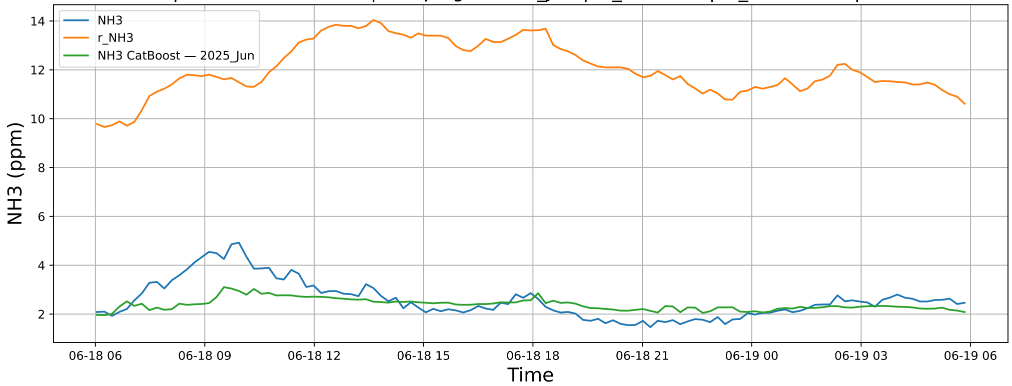
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

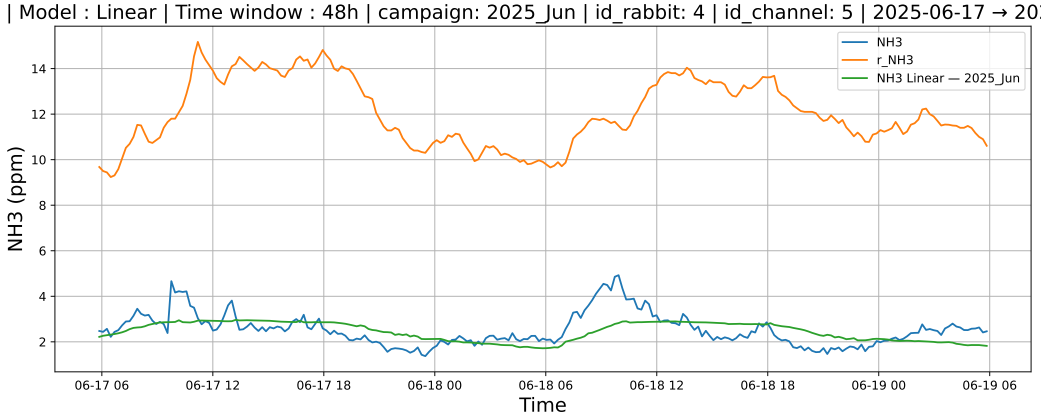


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

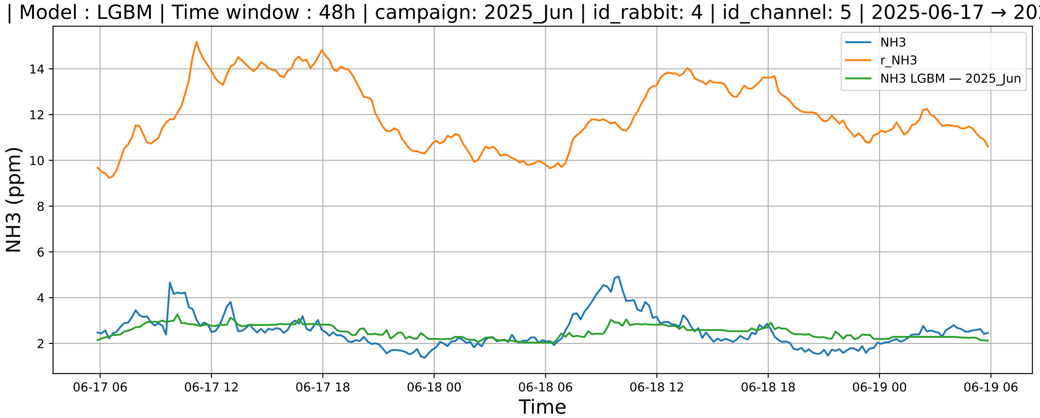


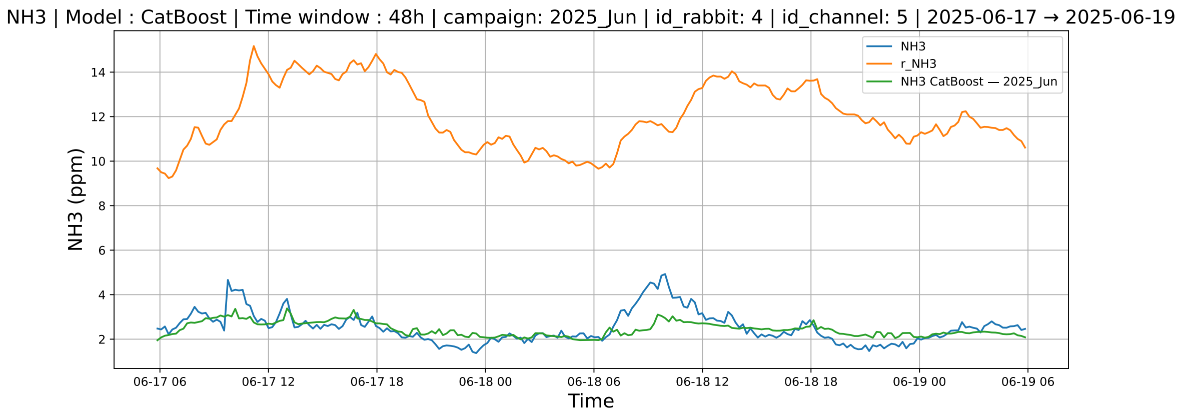
3.1.3.2 Time window : 48

NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

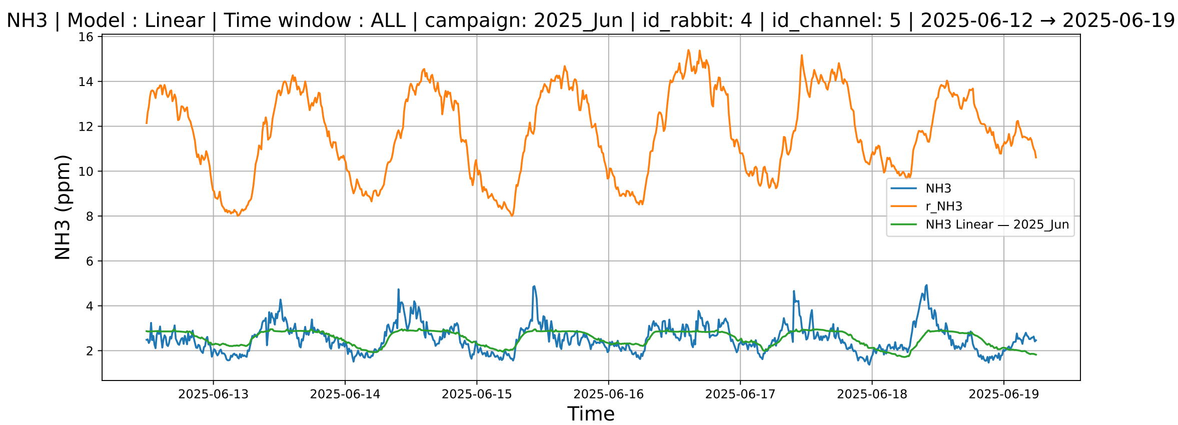


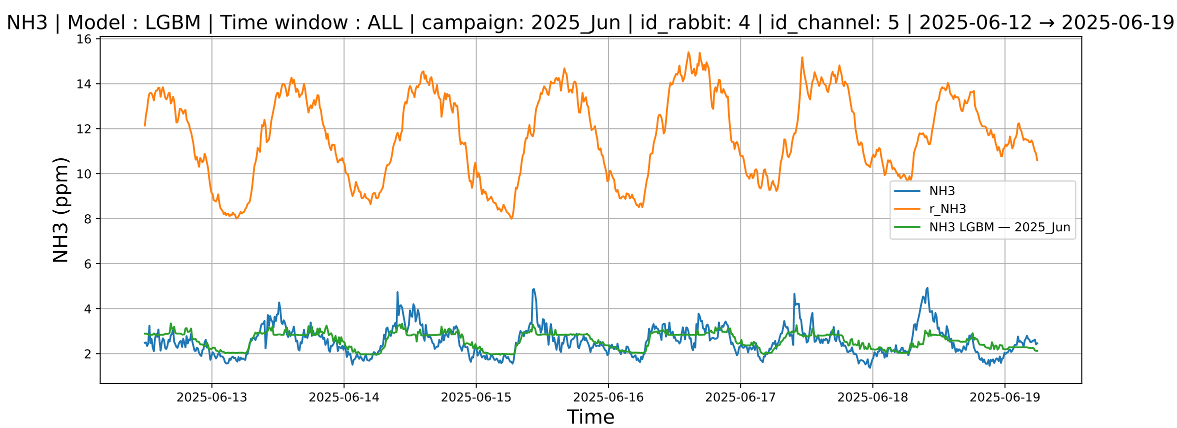
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



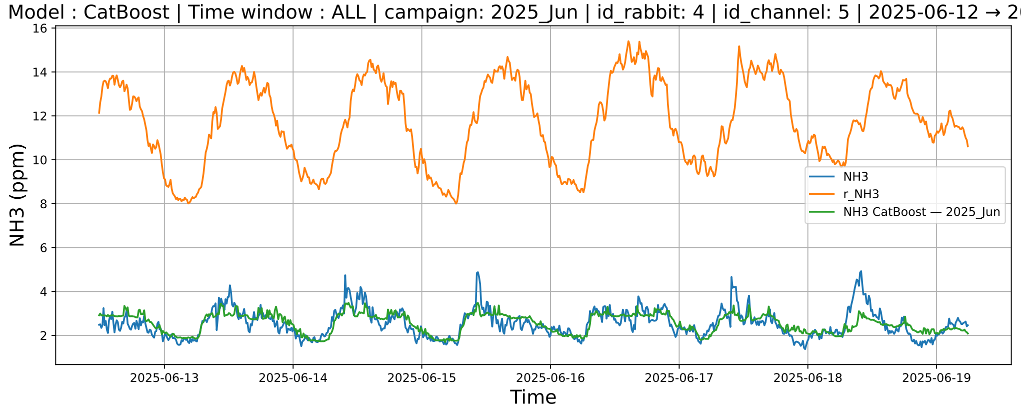


3.1.3.3 Time window : ALL





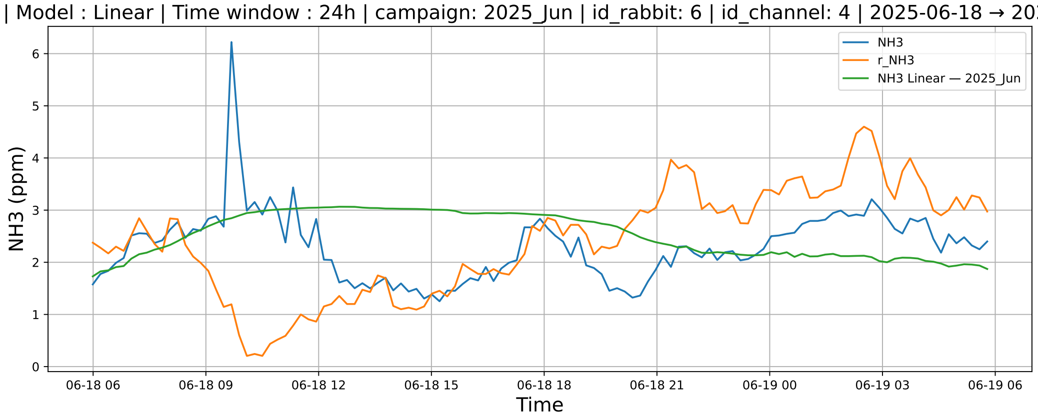
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



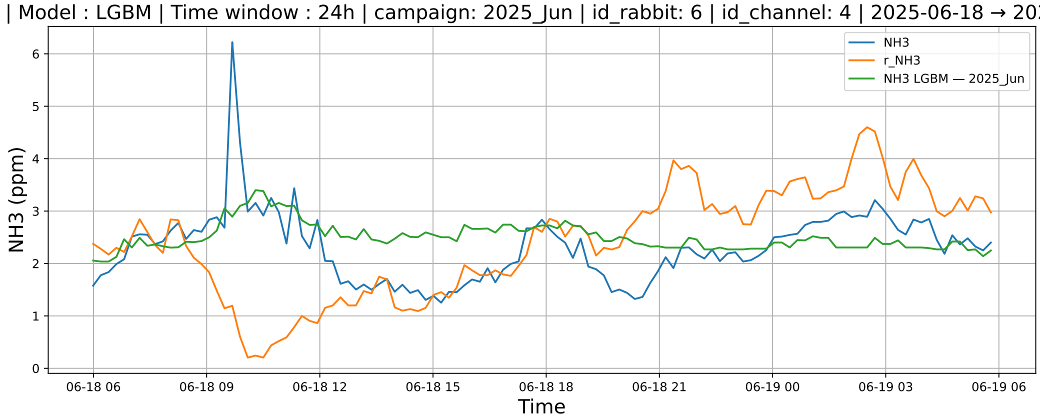
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

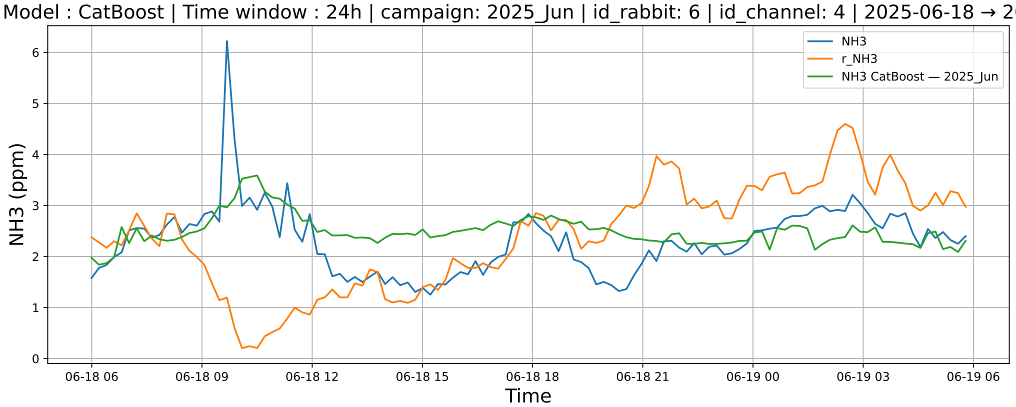
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

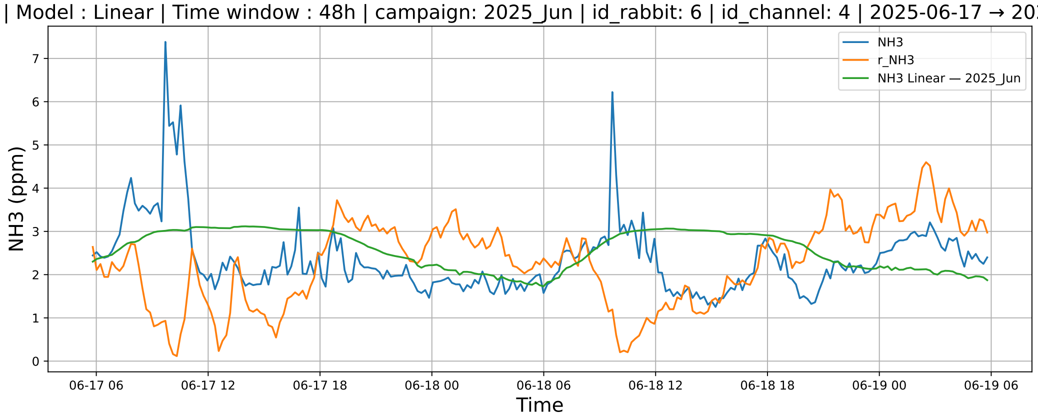


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

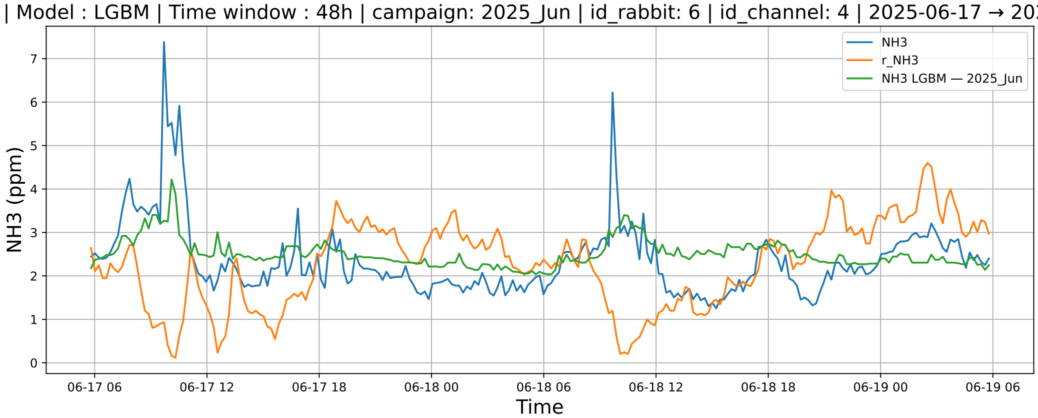


3.1.4.2 Time window : 48

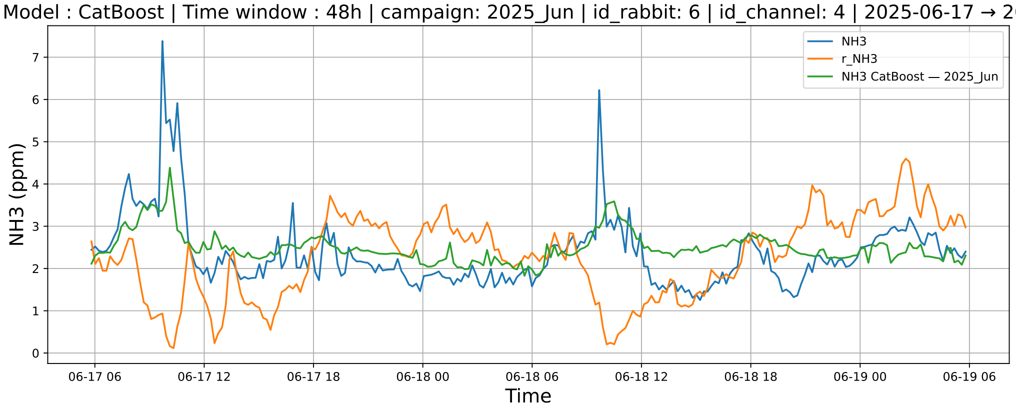
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

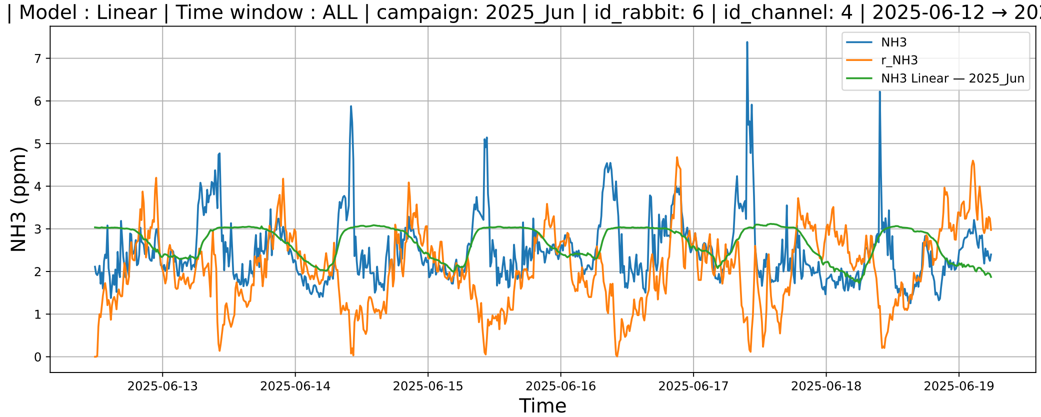


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

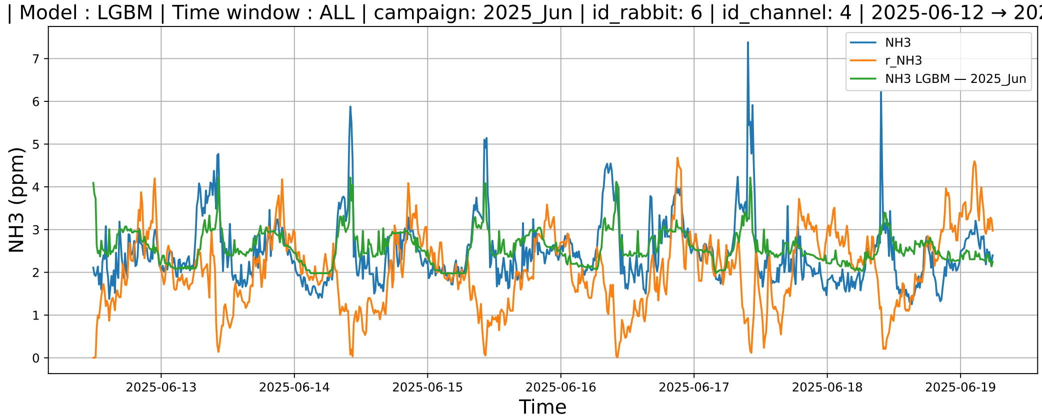


3.1.4.3 Time window : ALL

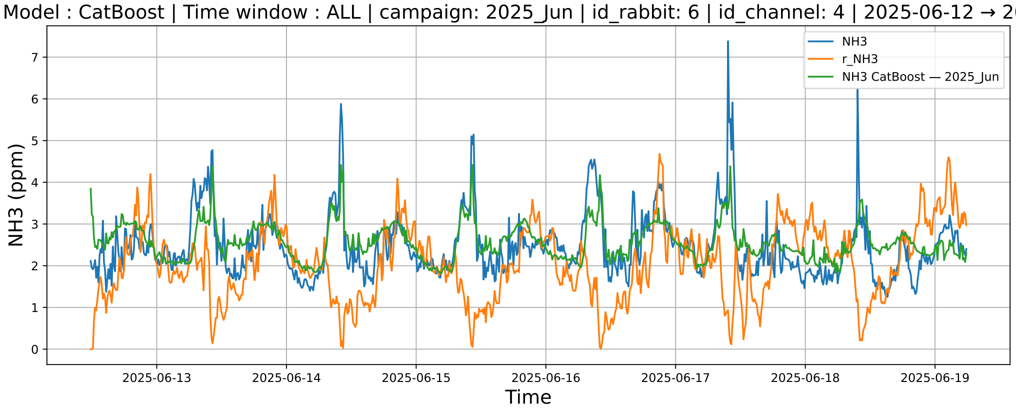
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



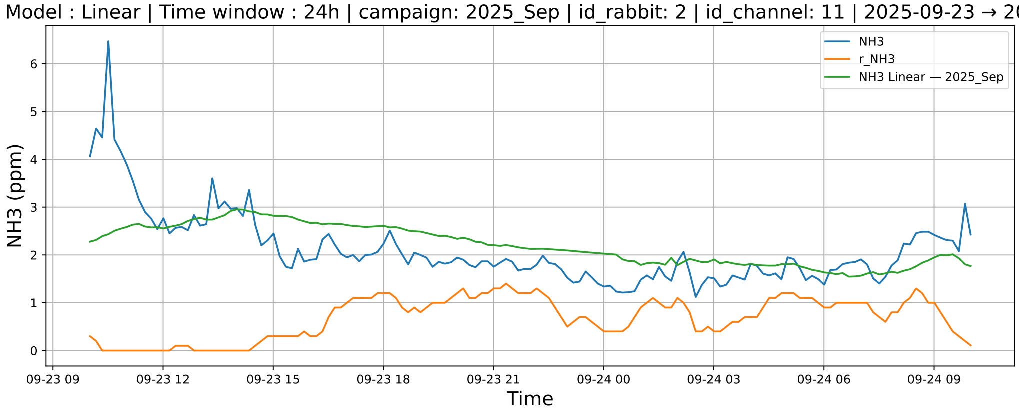
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



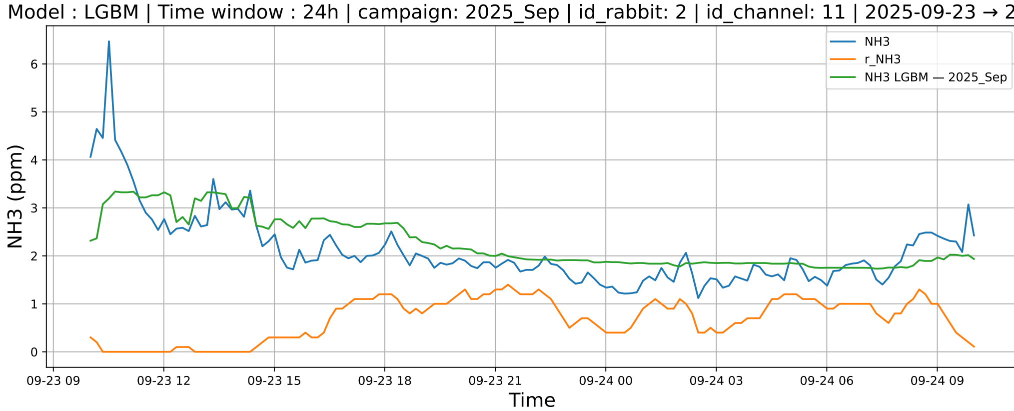
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

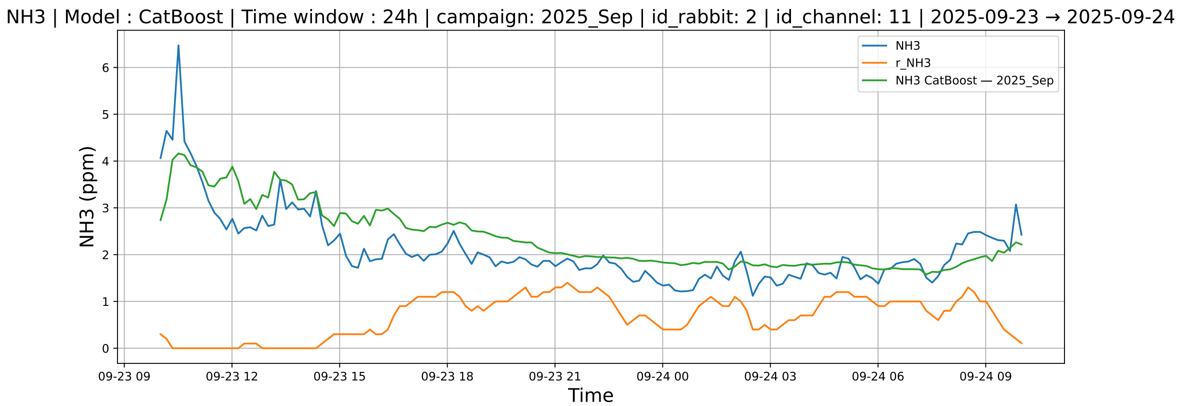
3.1.5.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

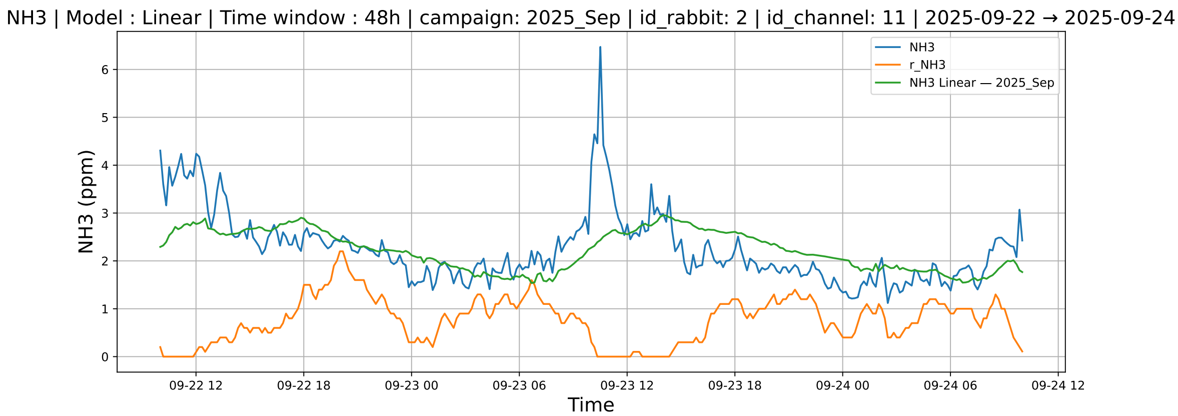


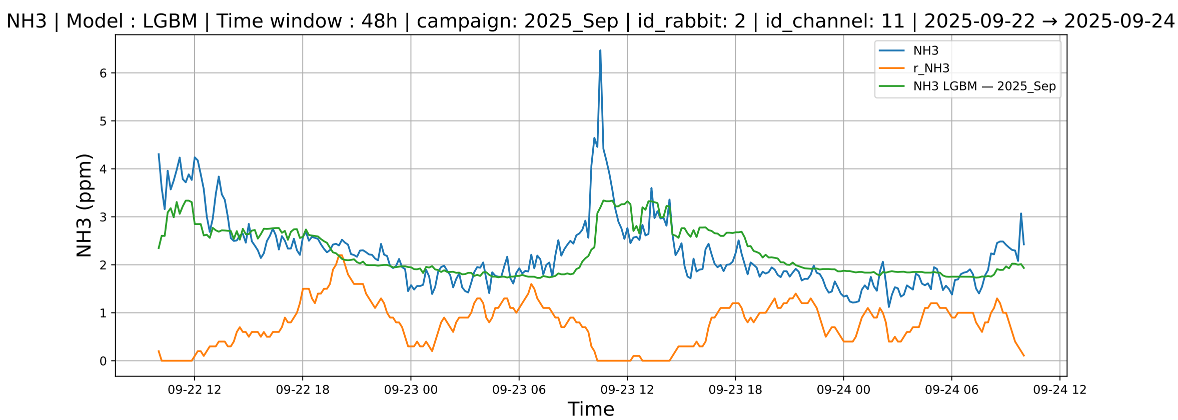
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



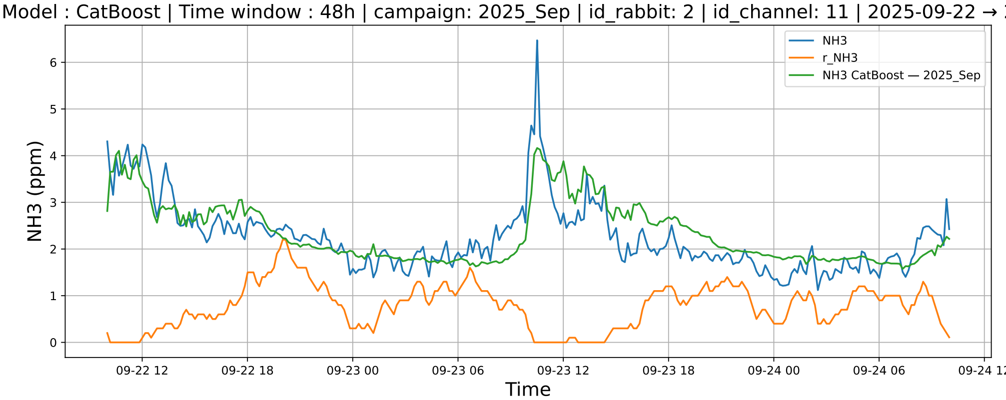


3.1.5.2 Time window : 48



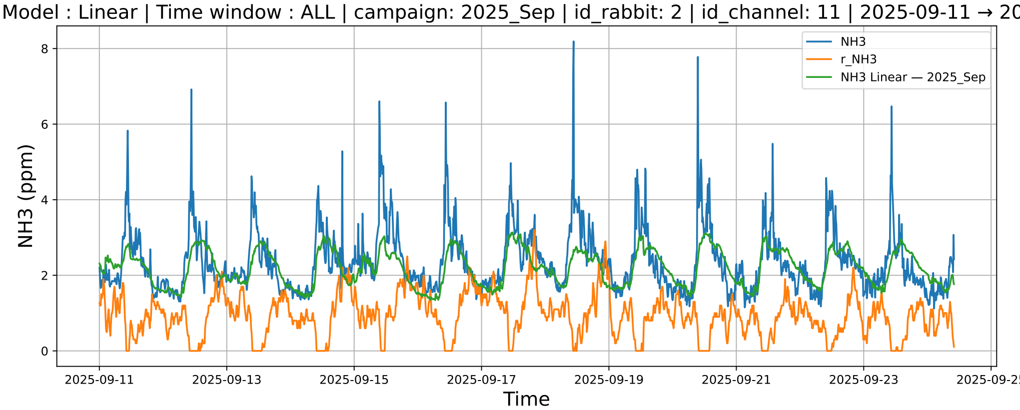


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

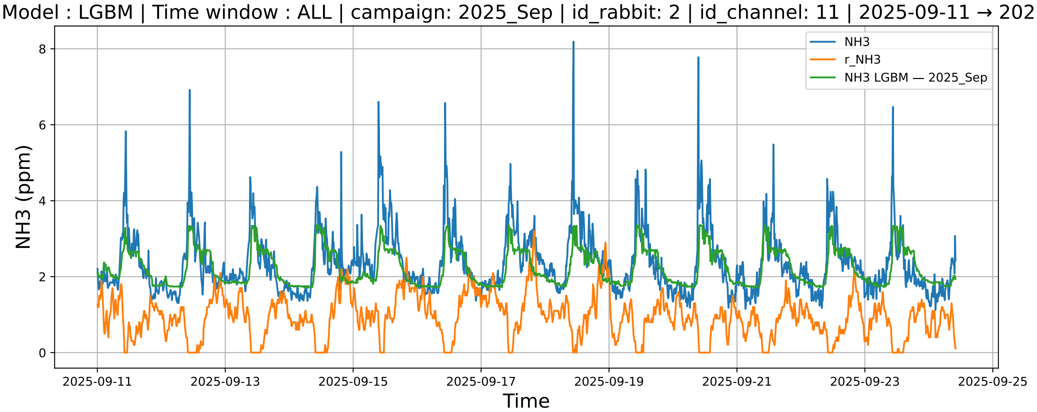


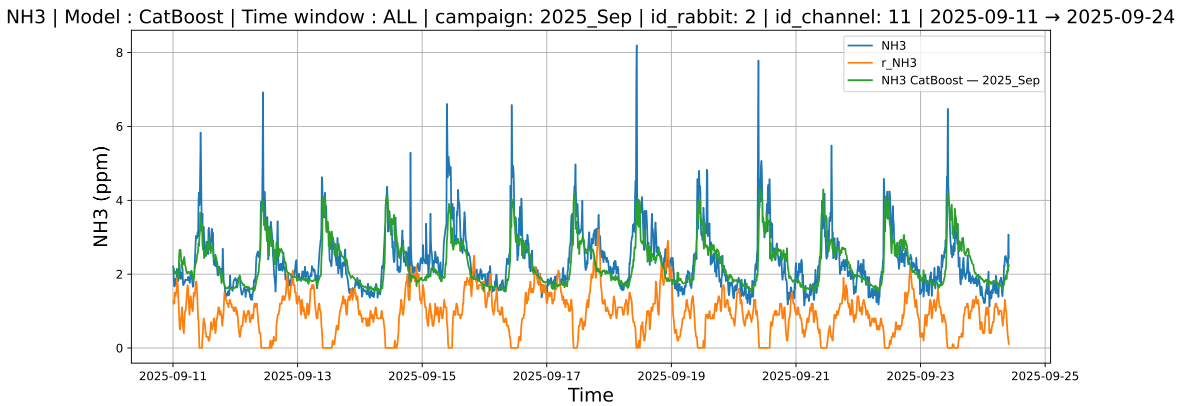
3.1.5.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



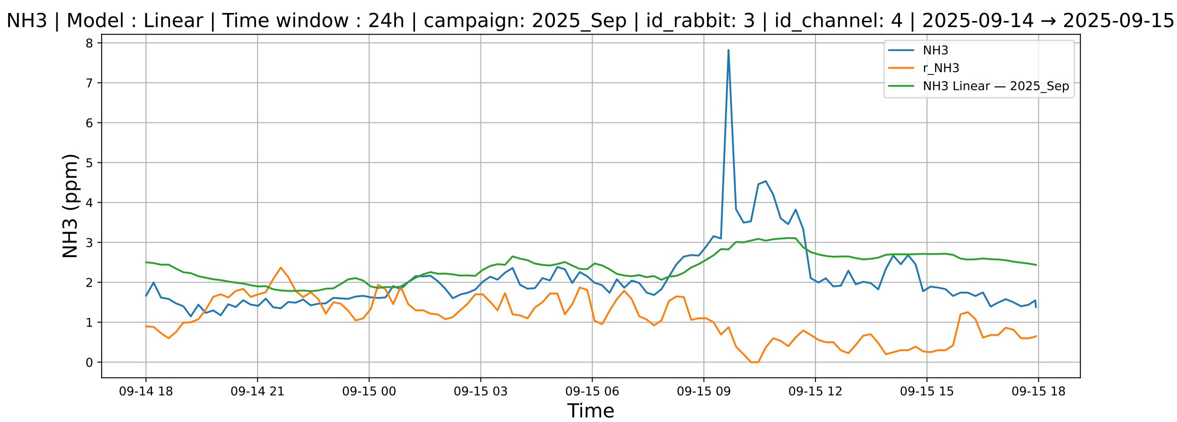
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

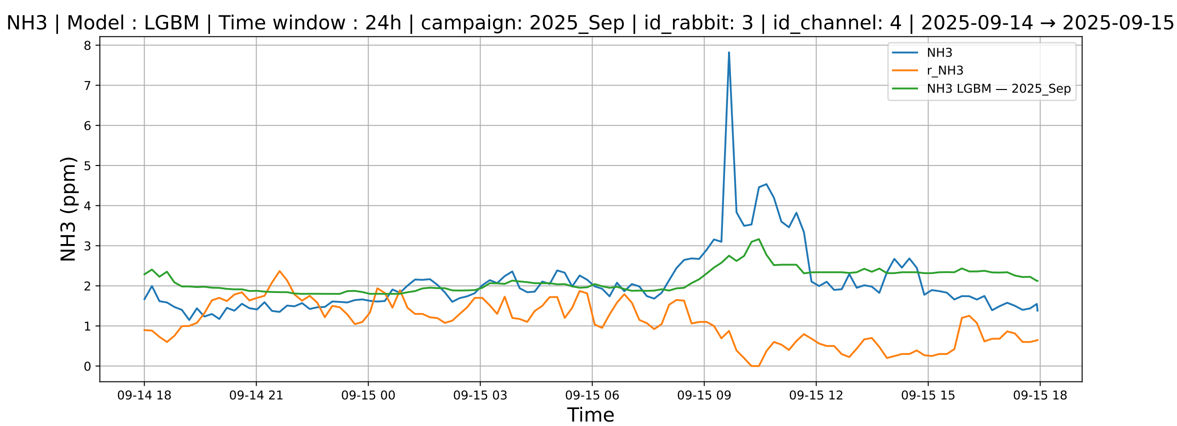




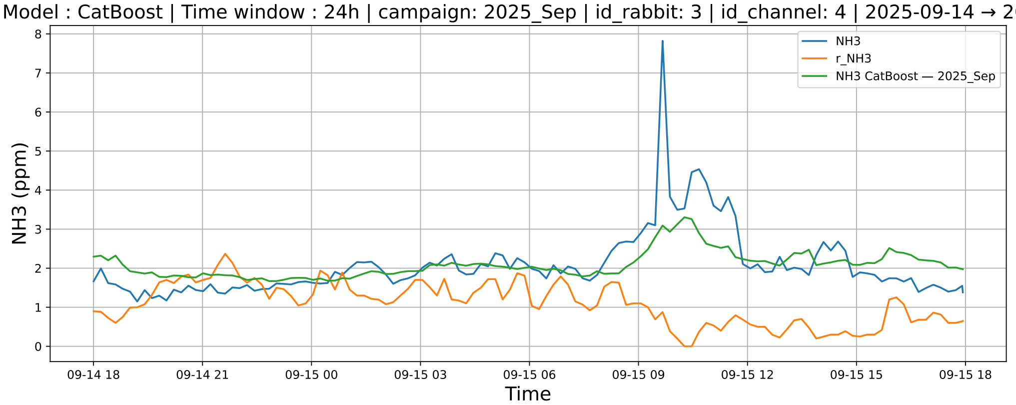
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24



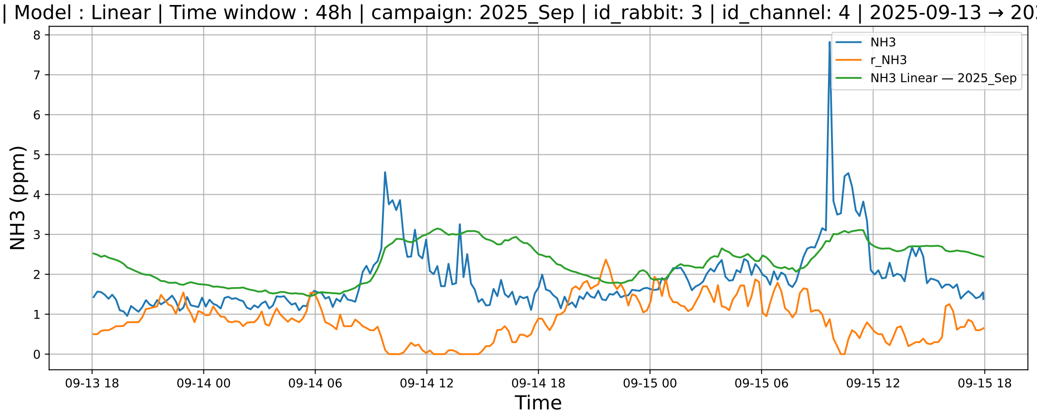


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

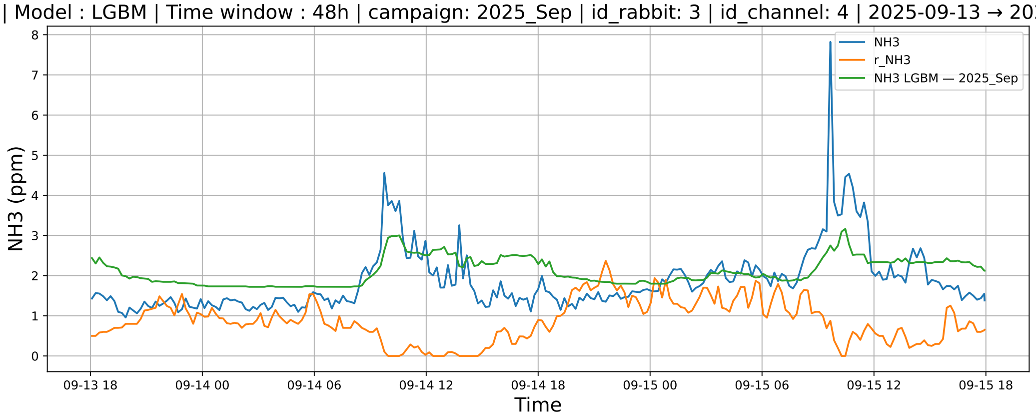


3.1.6.2 Time window : 48

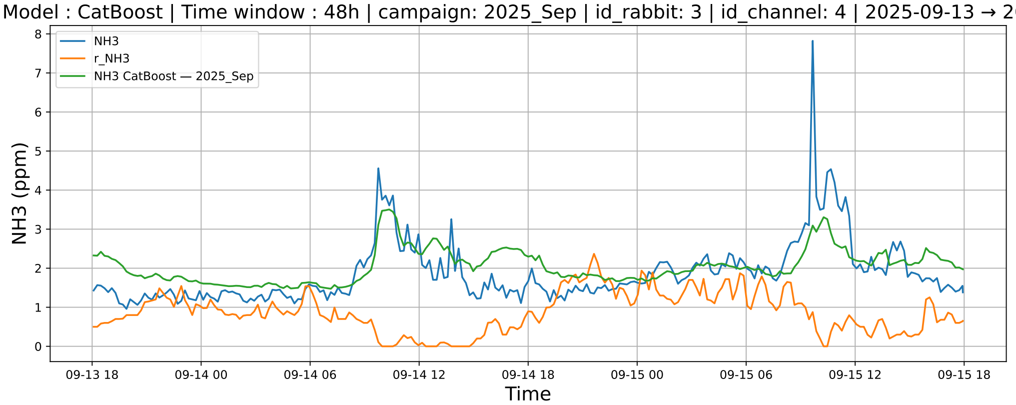
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

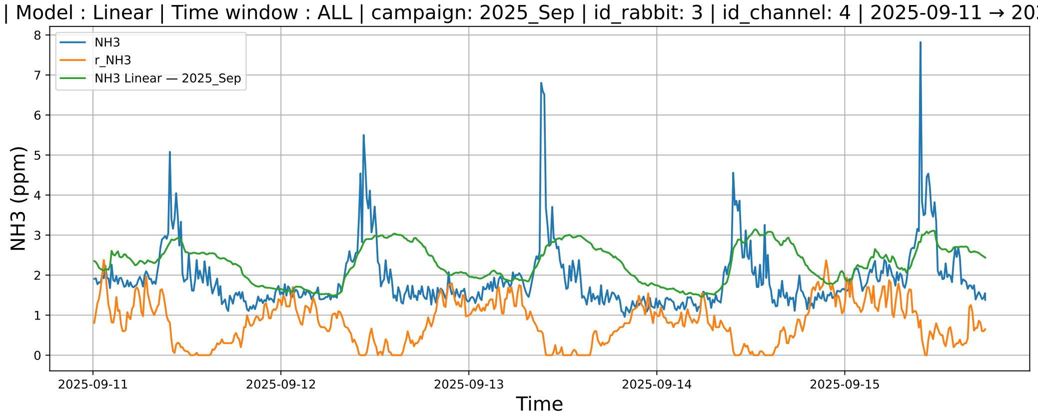


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

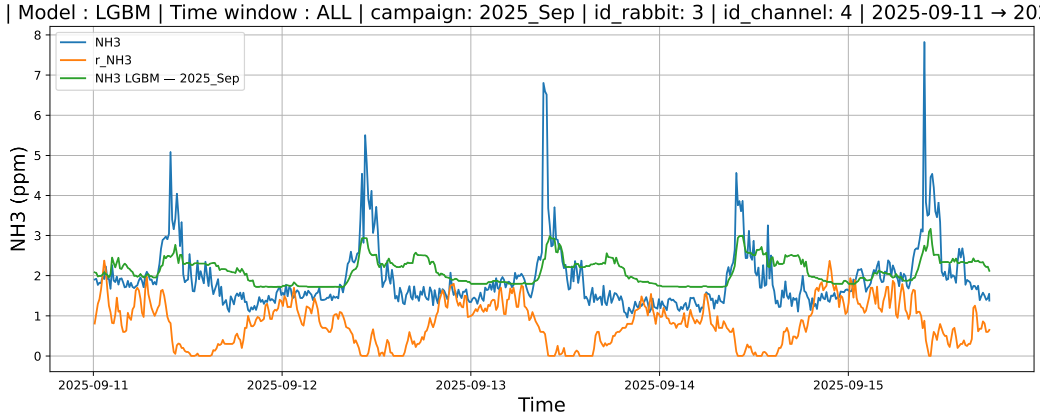


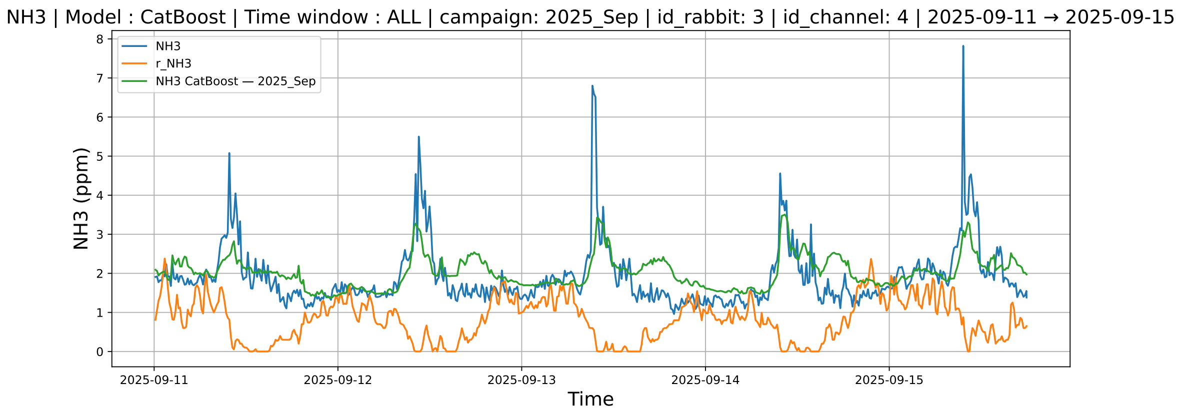
3.1.6.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



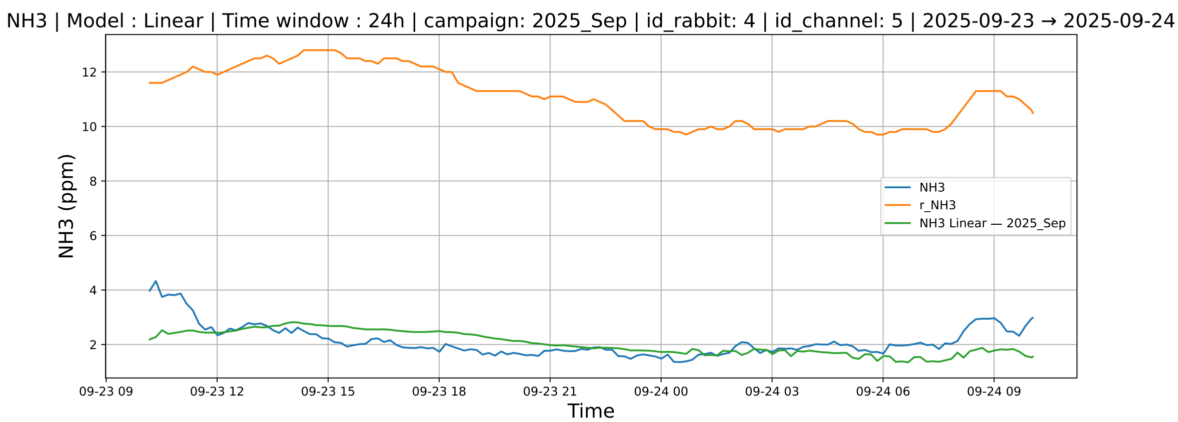
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

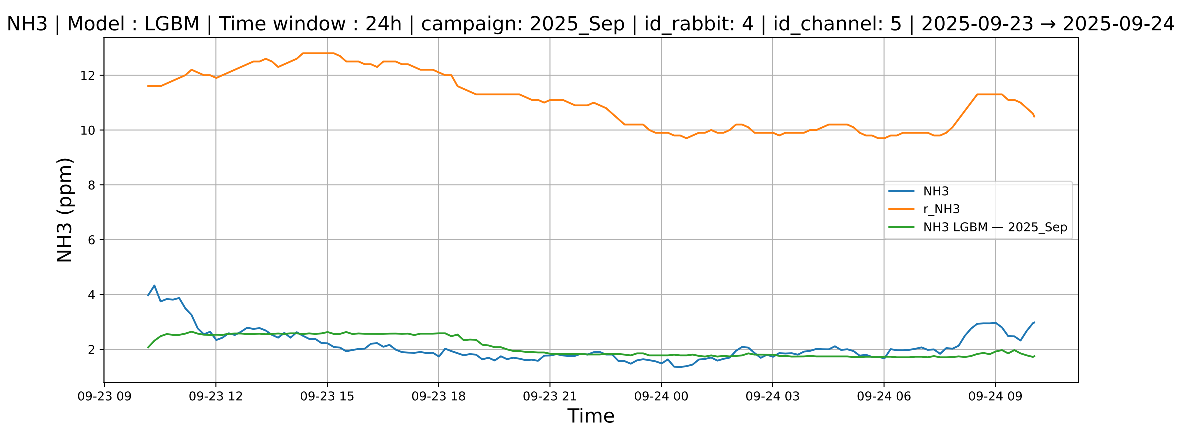




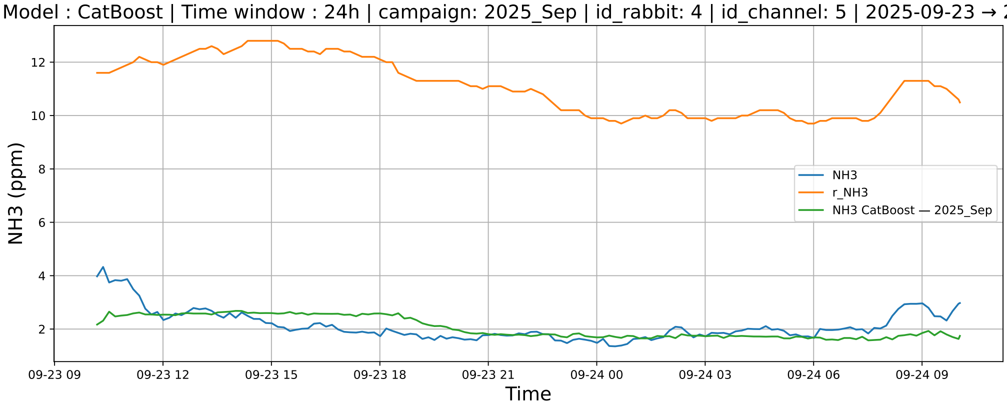
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24



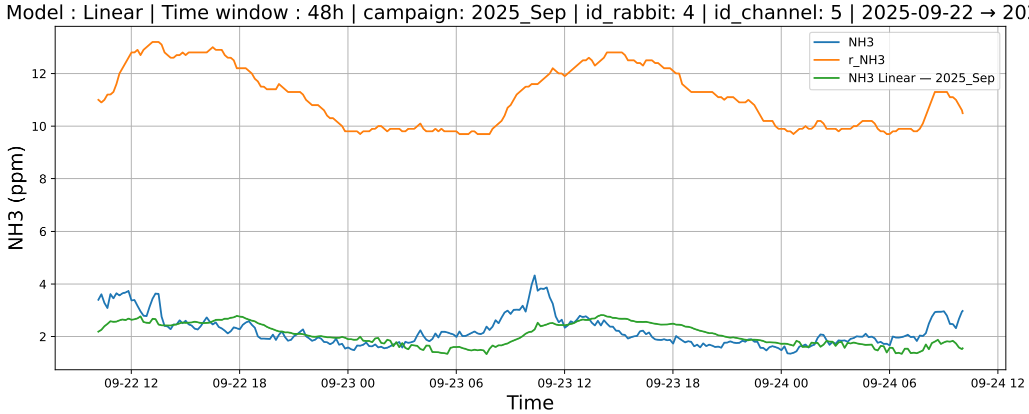


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

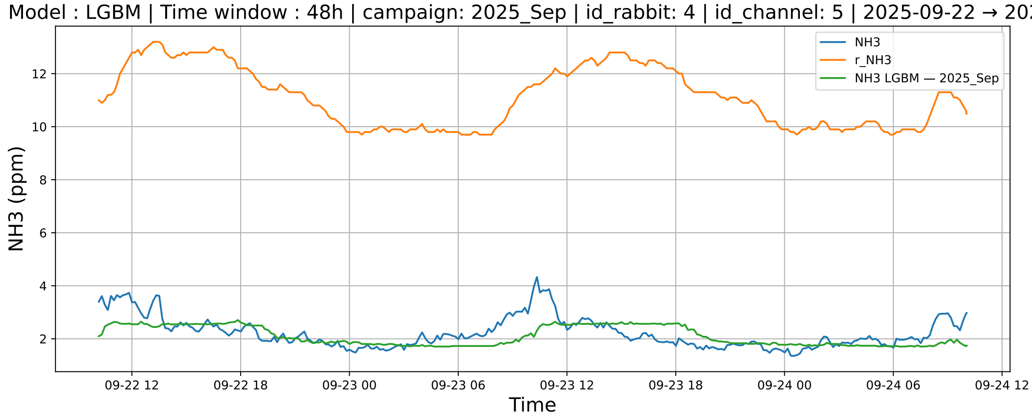


3.1.7.2 Time window : 48

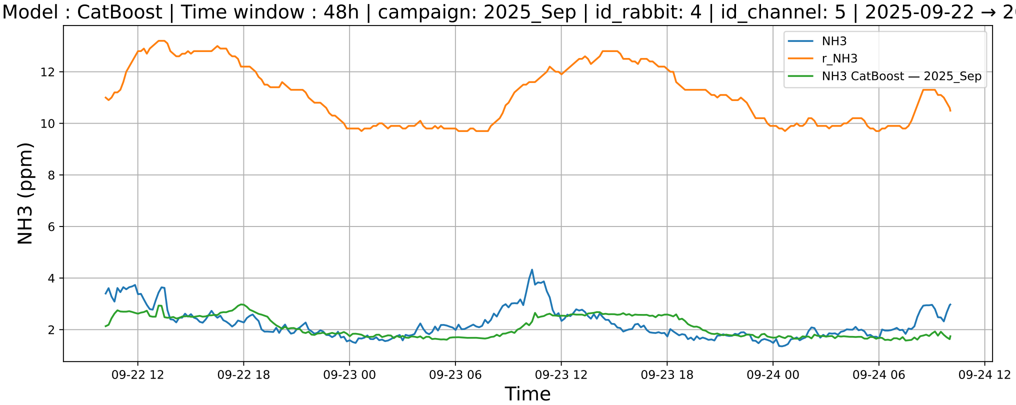
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

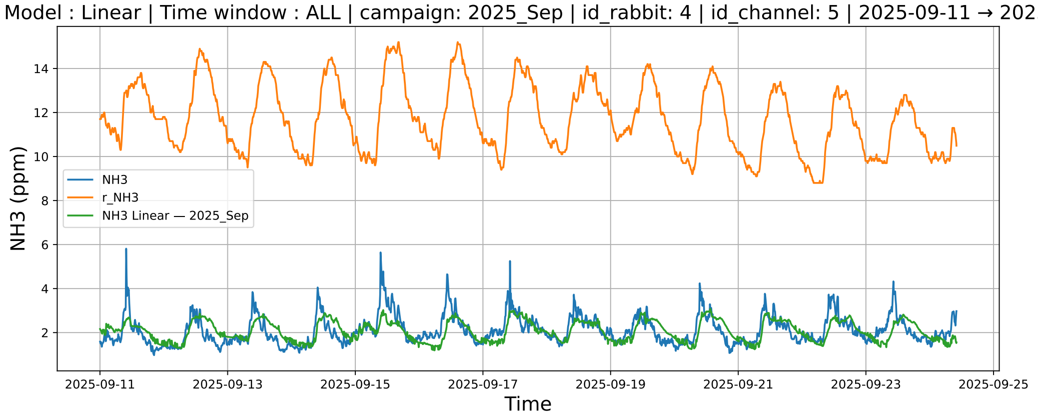


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

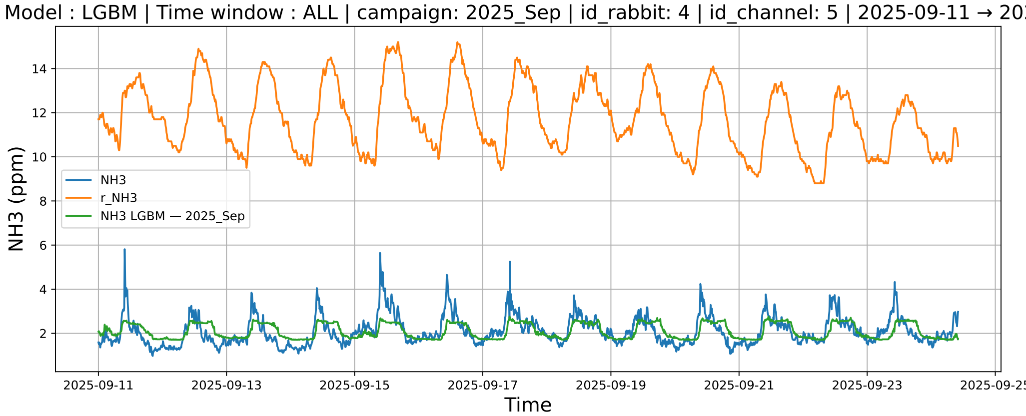


3.1.7.3 Time window : ALL

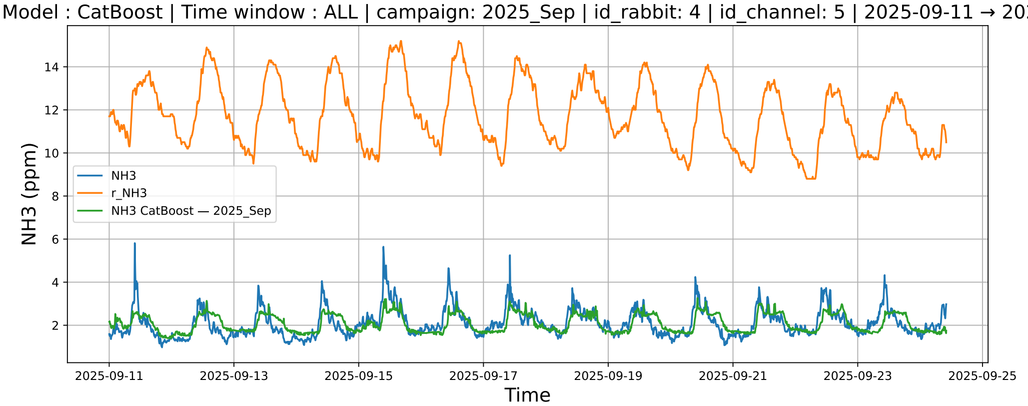
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



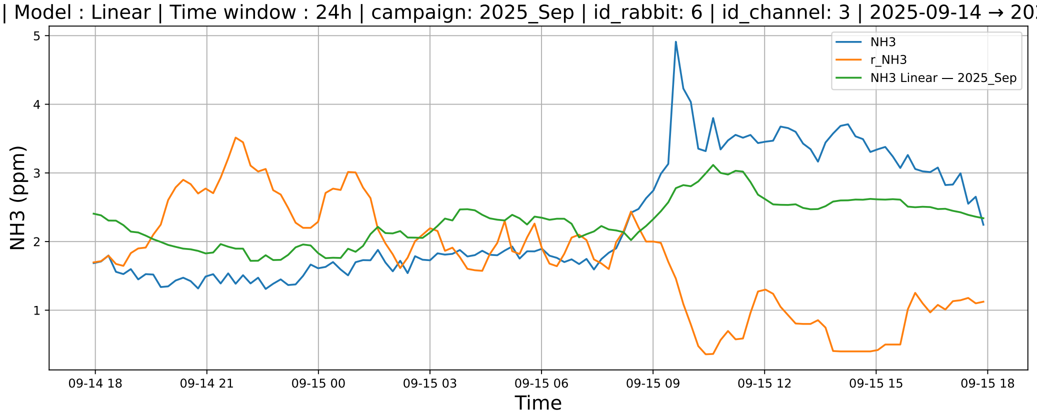
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



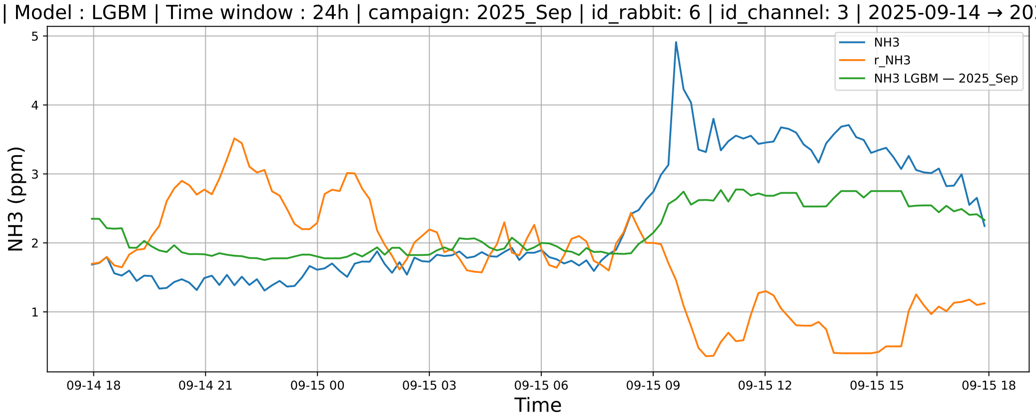
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

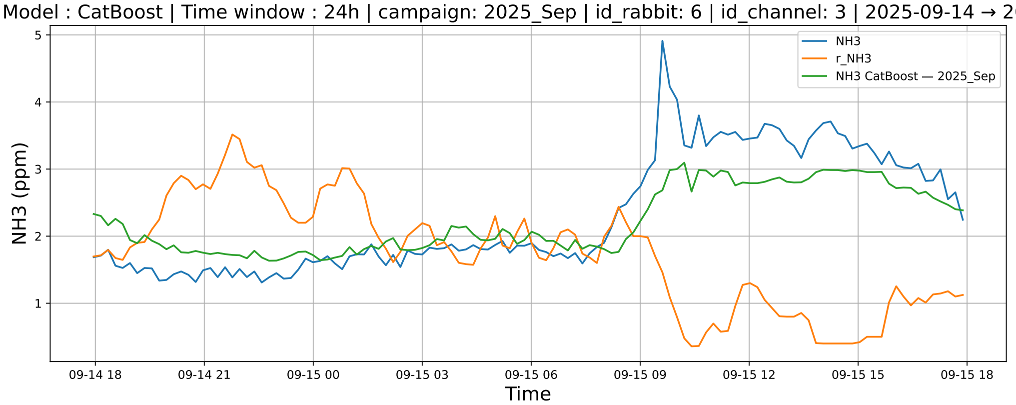
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

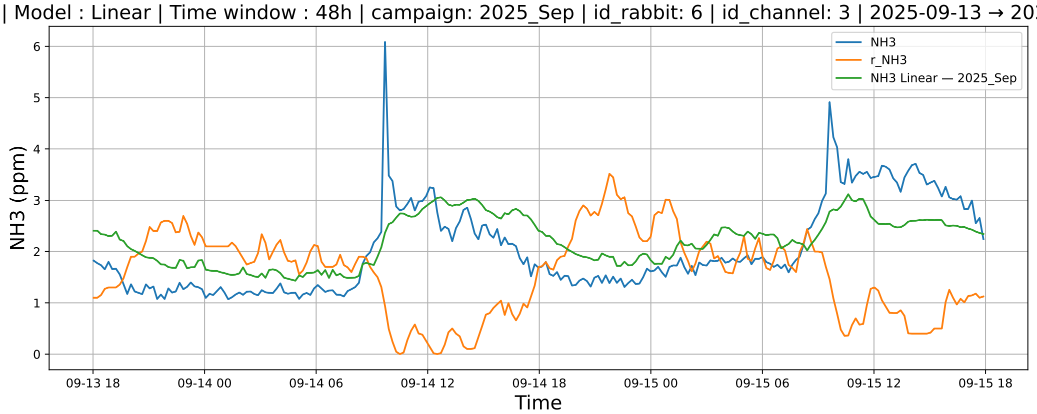


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

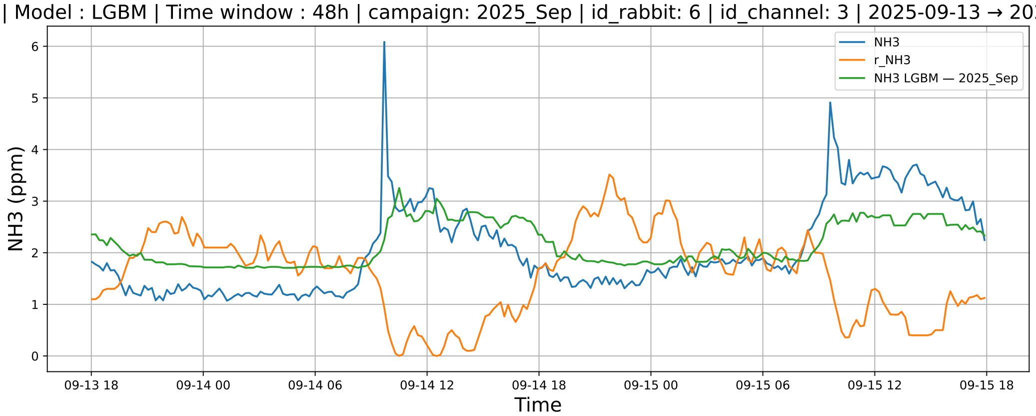


3.1.8.2 Time window : 48

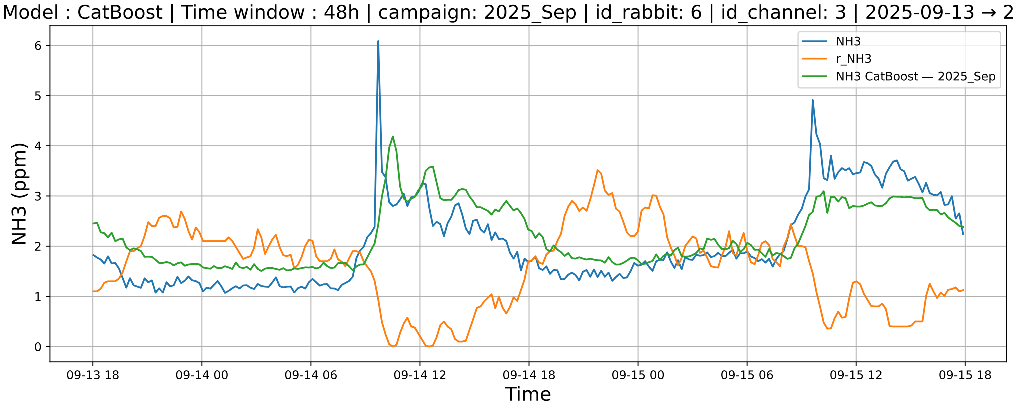
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

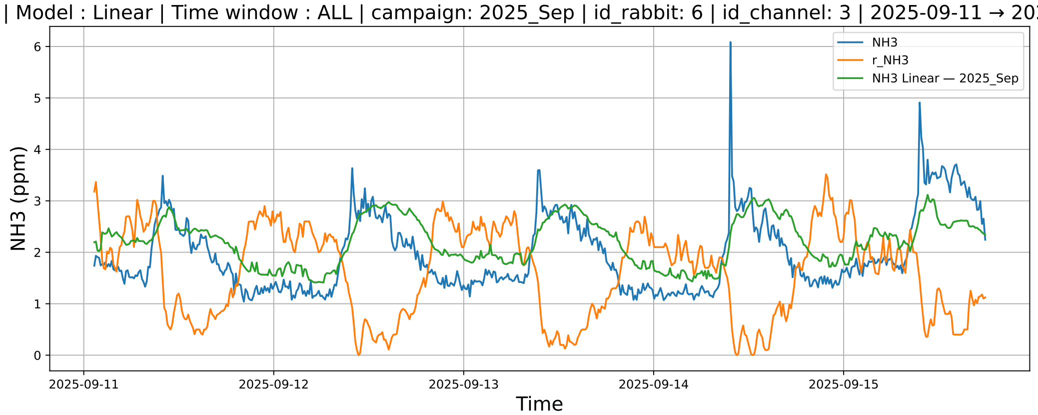


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

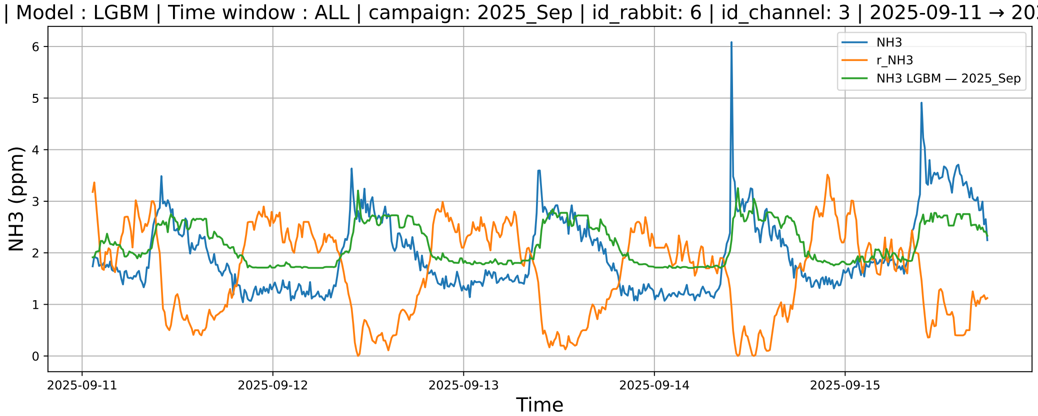


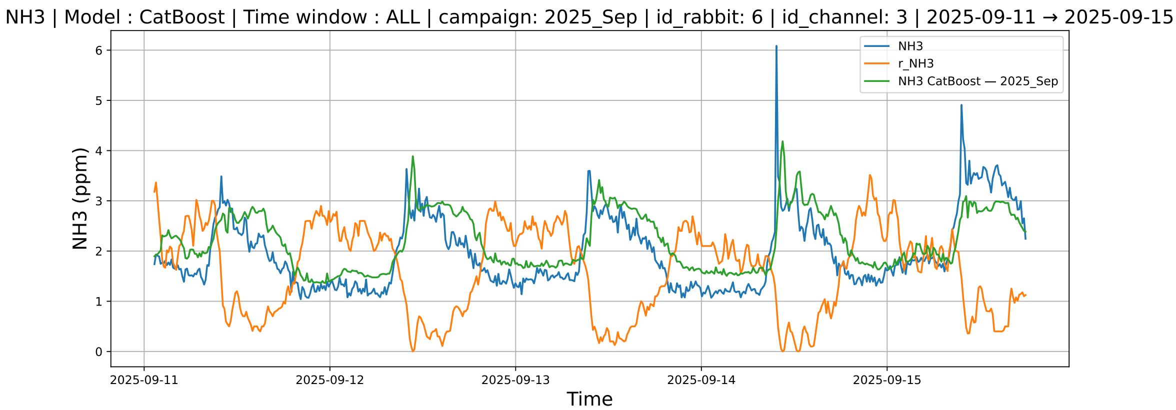
3.1.8.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

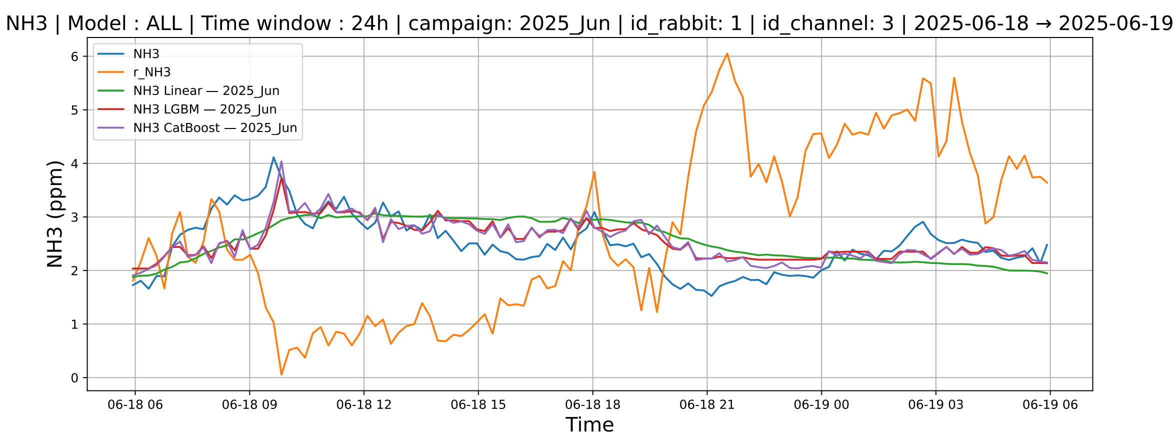




3.2 Compare plots

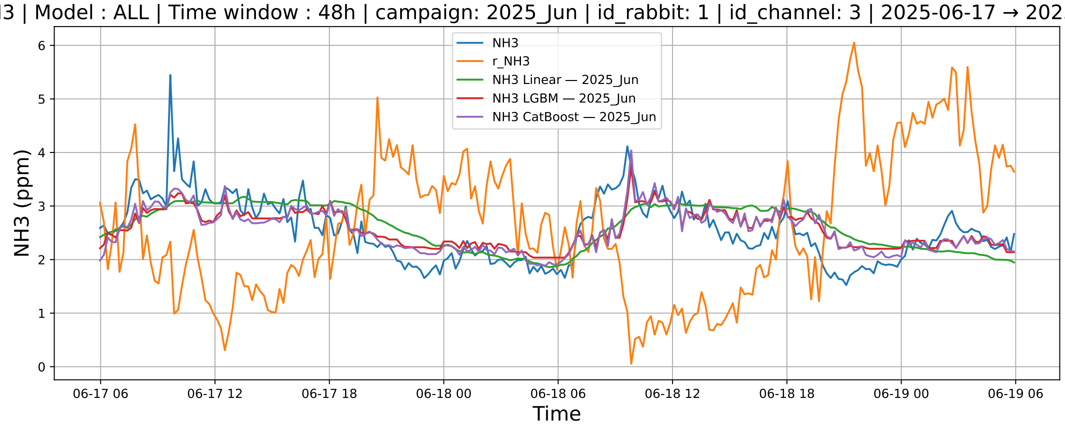
3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.2.1.1 Time window : 24



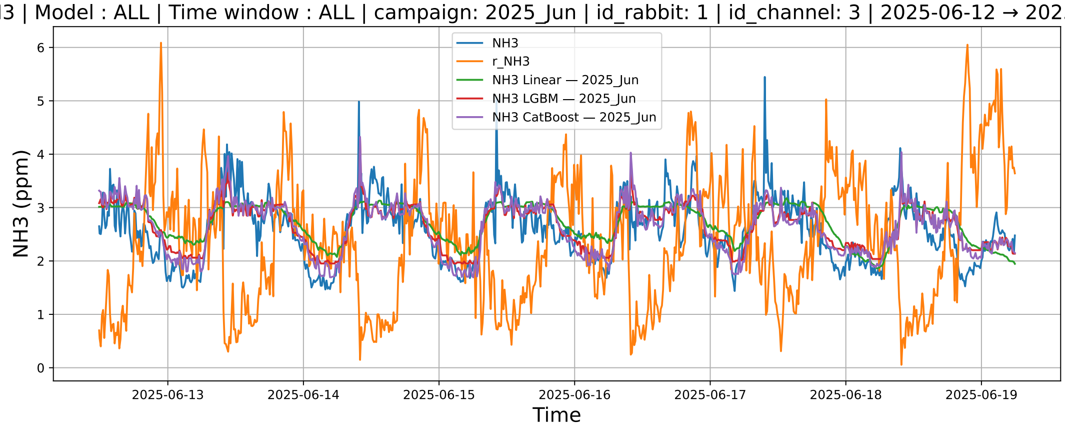
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



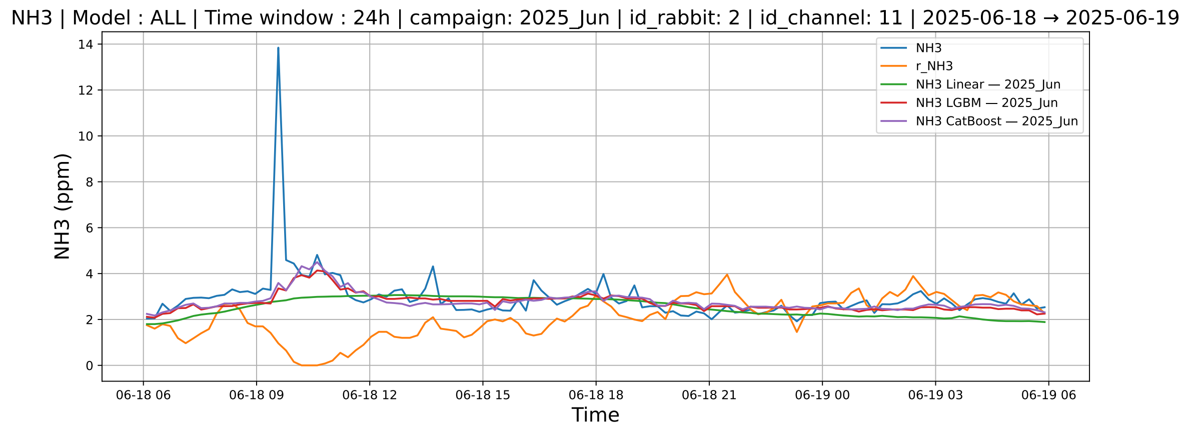
3.2.1.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

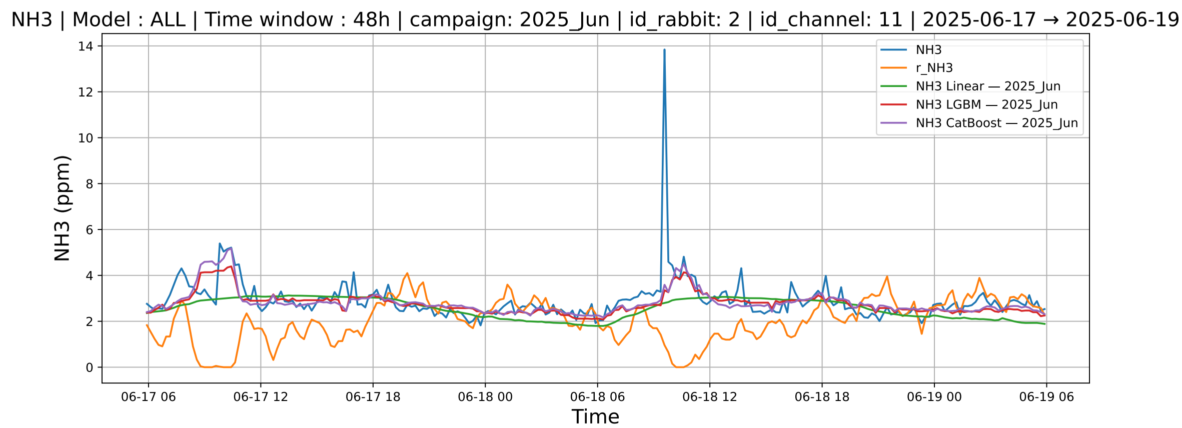


3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

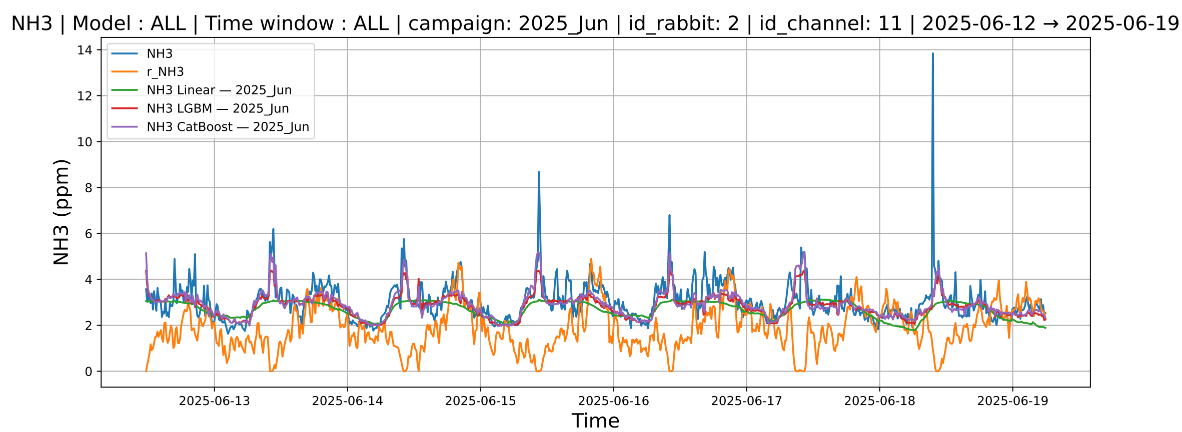
3.2.2.1 Time window : 24



3.2.2.2 Time window : 48



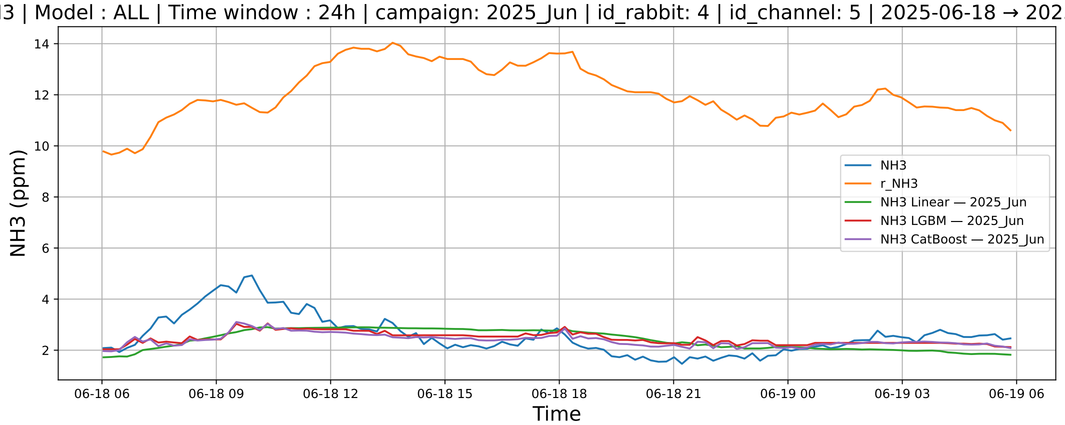
3.2.2.3 Time window : ALL



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

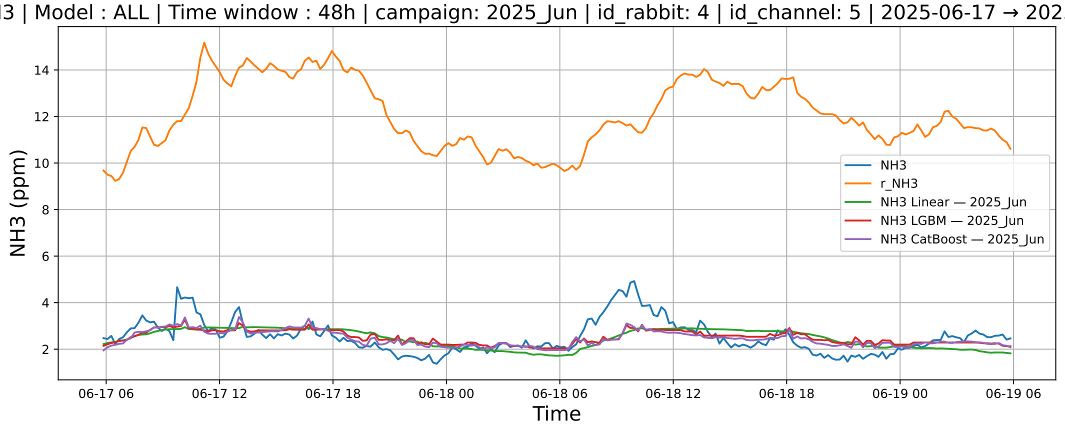
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

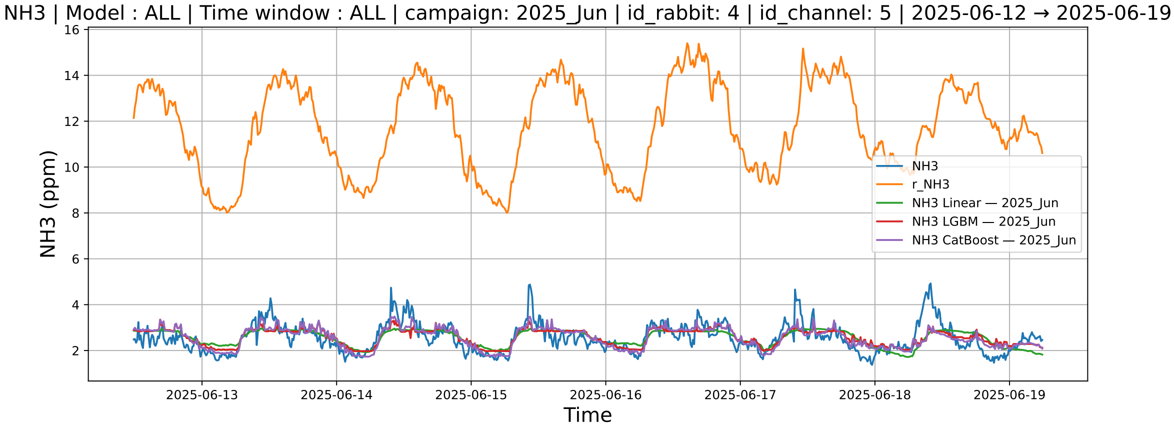


3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

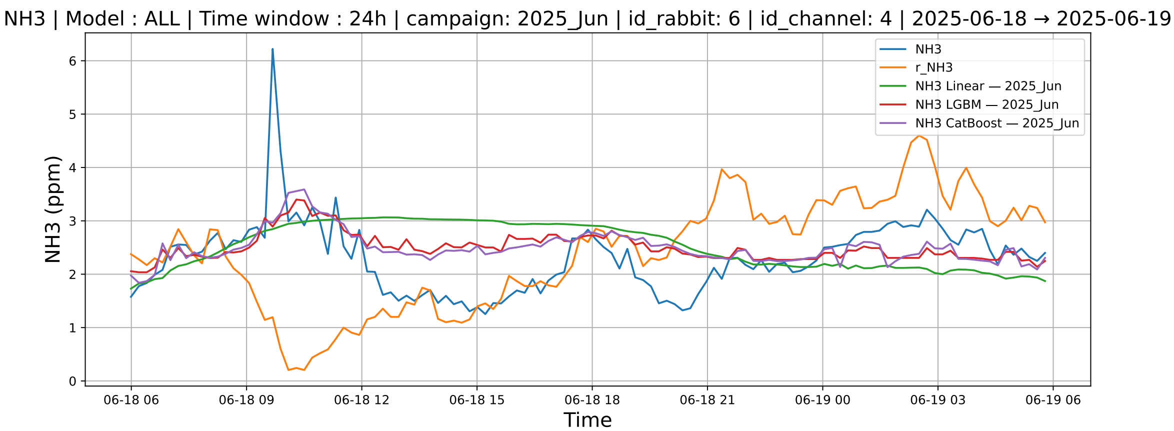


3.2.3.3 Time window : ALL

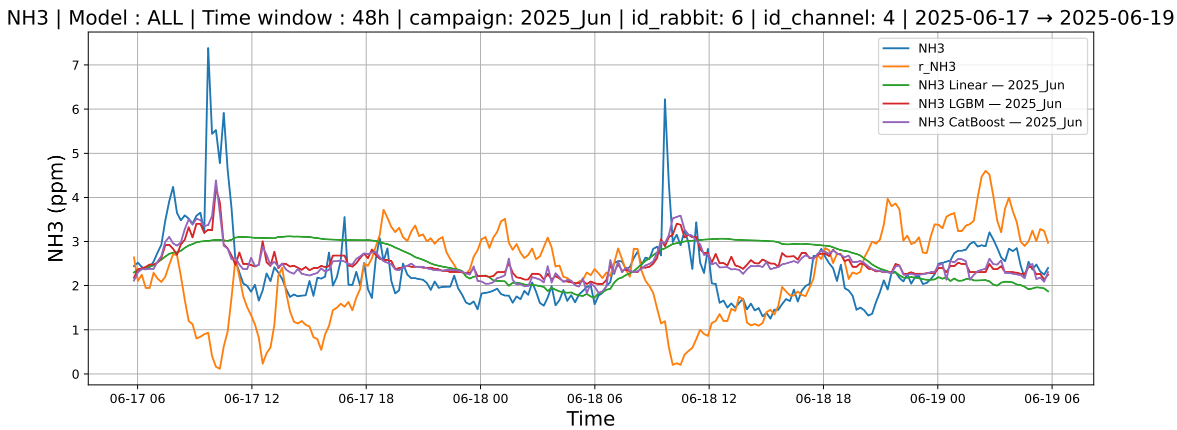


3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

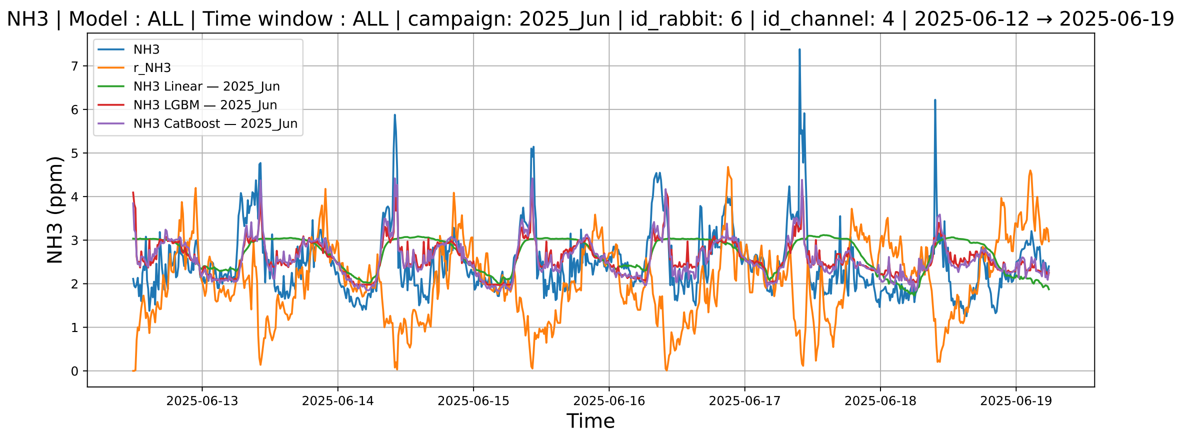
3.2.4.1 Time window : 24



3.2.4.2 Time window : 48



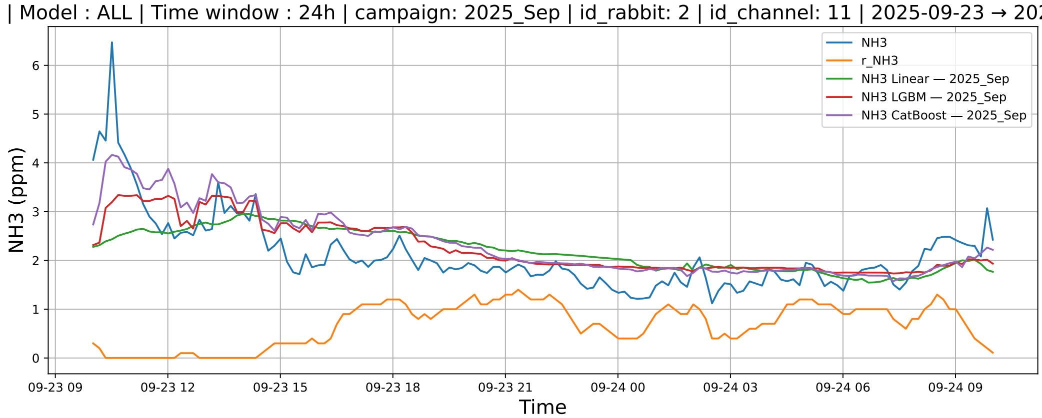
3.2.4.3 Time window : ALL



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

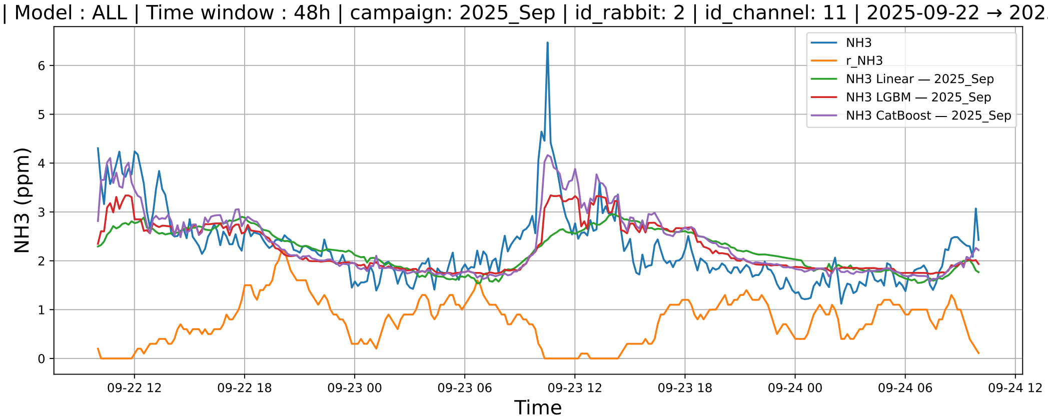
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



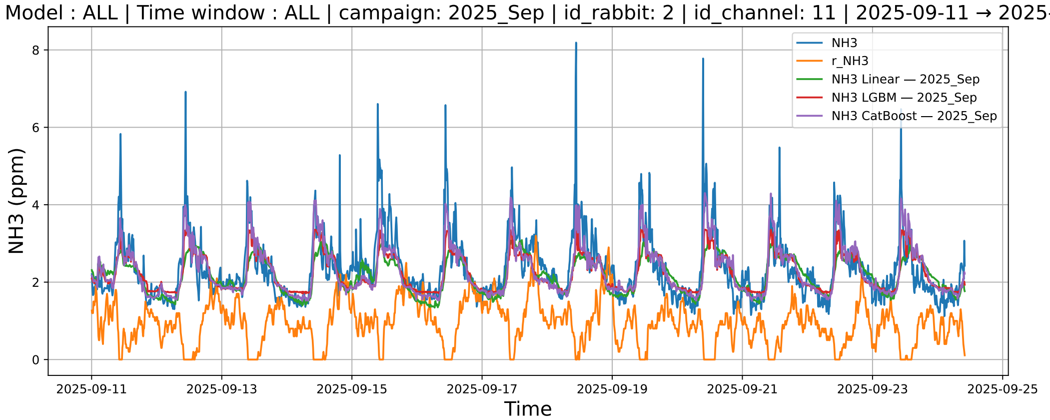
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



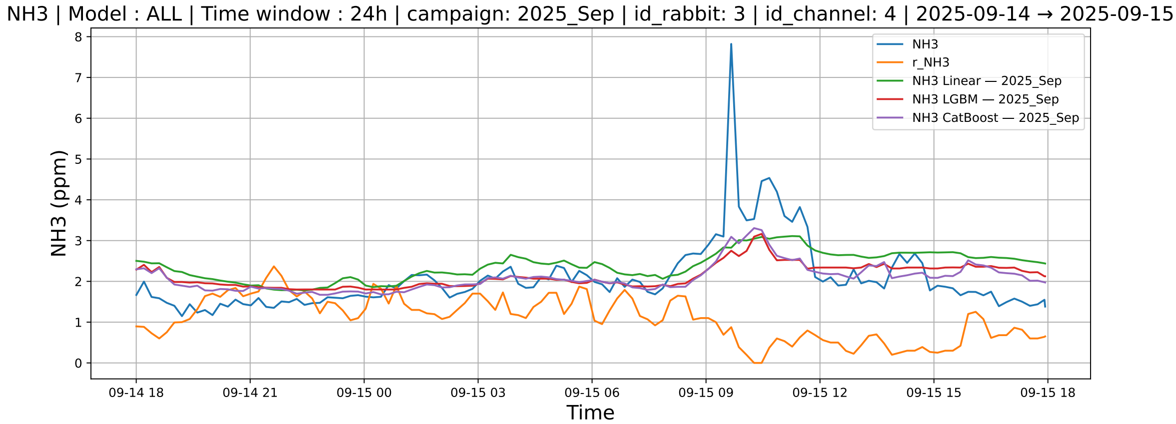
3.2.5.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

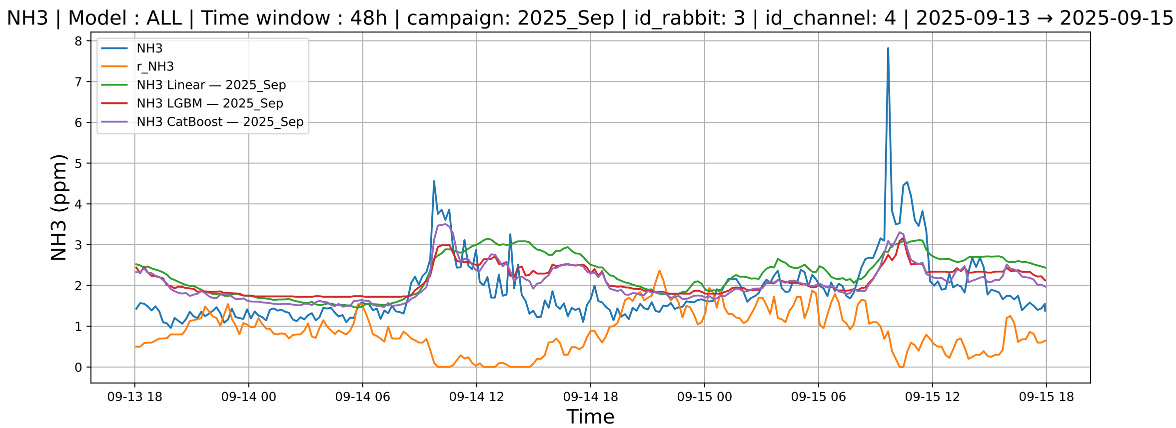


3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

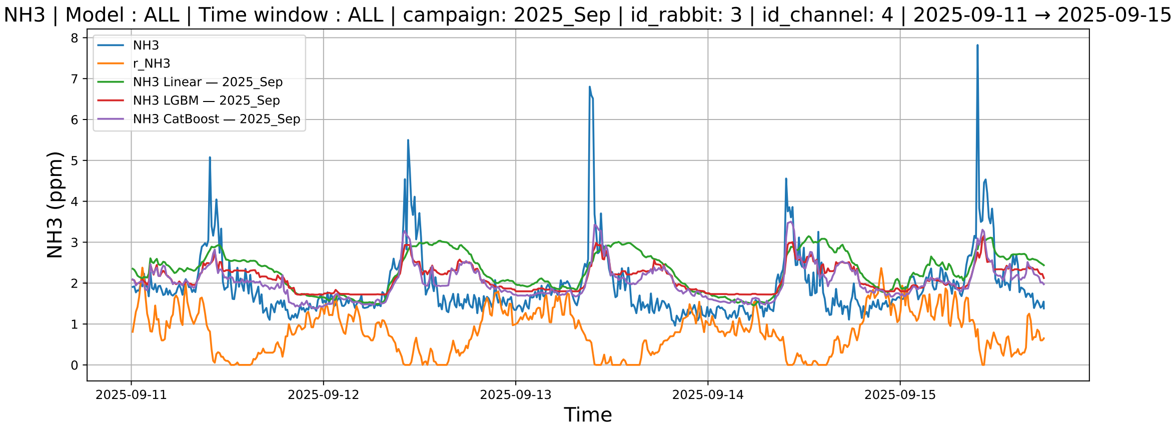
3.2.6.1 Time window : 24



3.2.6.2 Time window : 48

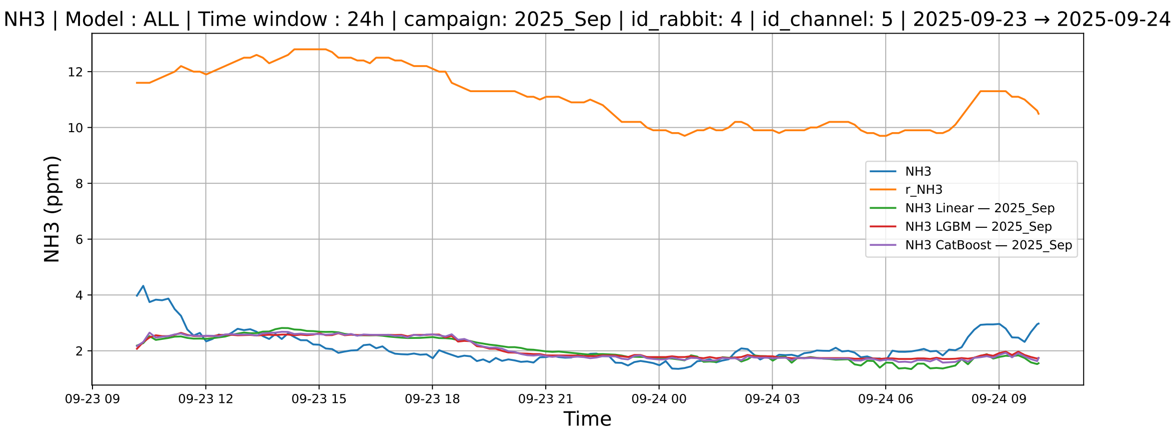


3.2.6.3 Time window : ALL

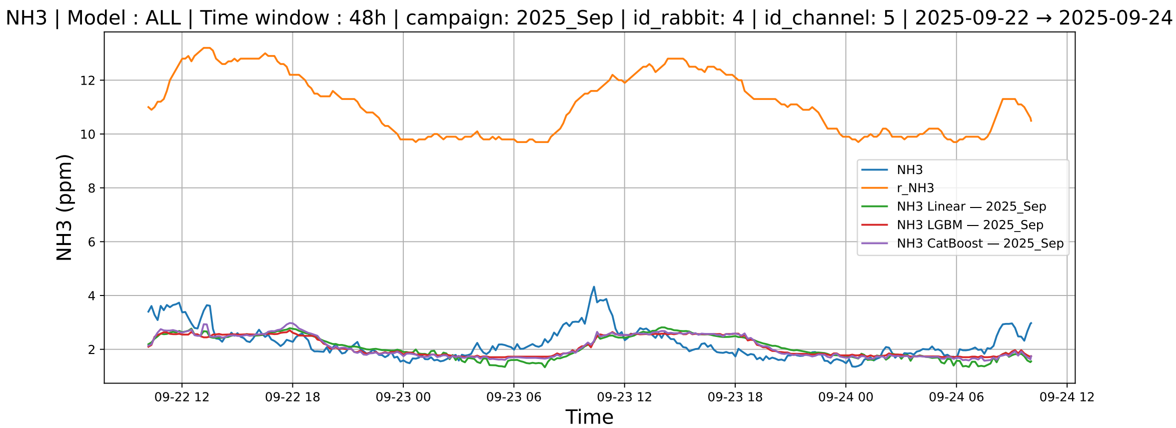


3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

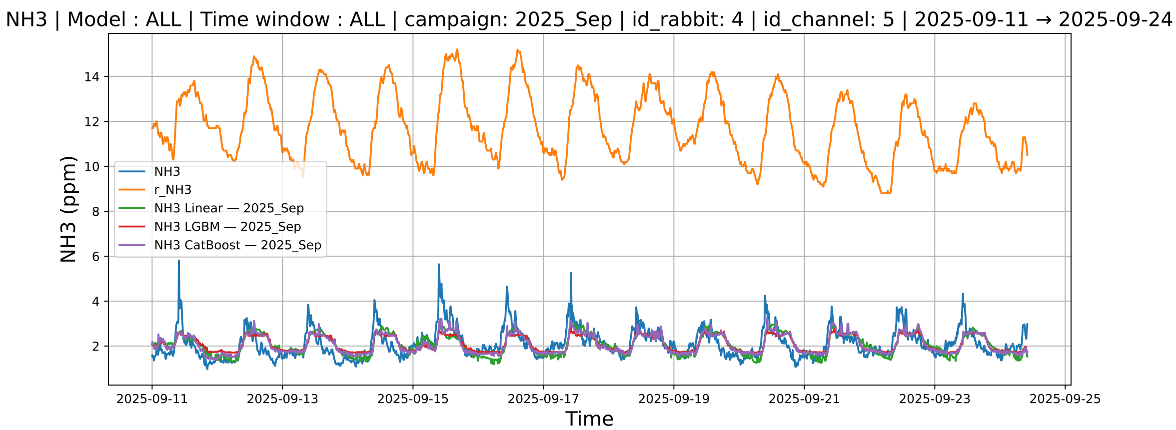
3.2.7.1 Time window : 24



3.2.7.2 Time window : 48



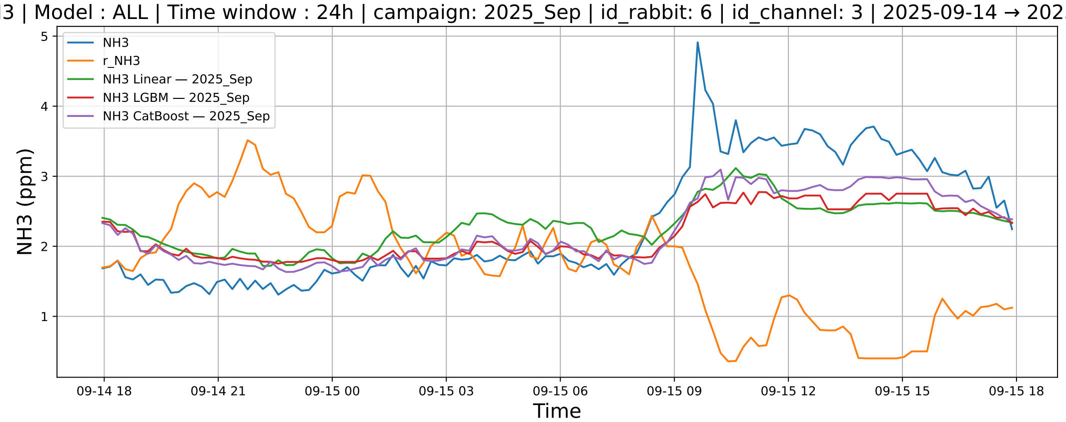
3.2.7.3 Time window : ALL



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

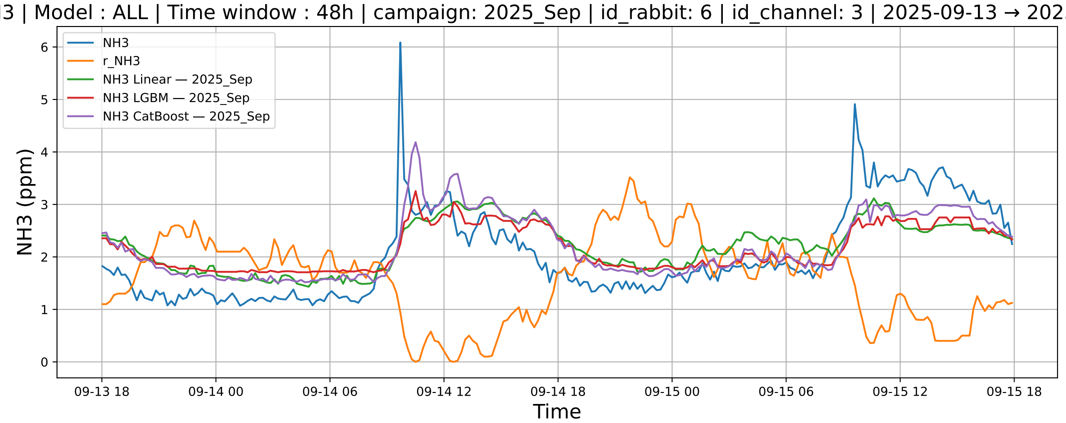
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15